Model-Agnostic Interpretation of Machine Learning via Average Marginal Effects

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Abstract

We propose a framework for interpreting any supervised machine learning model through average marginal effects, the standard inferential tool in the econometrics of nonlinear models. The estimand is the window average marginal effect — the average centered finite difference of the regression function at a fixed, interpretable step — which is well defined for learners with no differentiability anywhere and recovers the classical average derivative as the window shrinks. Because the functional is linear in the regression function, it admits an exact Riesz representation obtained by a change of variables, with no boundary terms and no density derivatives, and the resulting orthogonal score has estimation bias exactly equal to a product of two L^2 errors. A cross-fitted debiased estimator is therefore root-n consistent and asymptotically normal whenever the learner and a Riesz-regression estimate of the representer converge faster than $n^{-1/4}$ in L^2 — rather than at the parametric rate in a function-space norm — attains the semiparametric efficiency bound, and returns a coefficient table directly analogous to parametric regression output without resampling; the classical generalized linear model result is nested as a special case, and binary contrasts arise from the same representation with inverse propensity weights. The score is doubly robust:

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when the representer is known or parametrically estimable — designed experiments, randomized treatments, simulation designs — inference remains valid with no consistency requirement on the learner, covering the fixed-complexity settings in which off-the-shelf learners converge to pseudo-true limits. A pairing identity characterizes, apparently for the first time, what the marginal effect of a tree-based model measures: the jump measure of the fitted surface integrated against a locally averaged covariate density. Gradient consistency results for neural networks in Sobolev spaces and for boosted trees and Mondrian forests in the space of functions of bounded variation delineate when uncorrected plug-in effects are already consistent. Experiments on three datasets and simulations illustrate the framework and display the regularization-bias mechanism that the orthogonal correction removes.

1 Introduction

The rise of flexible machine learning methods has created a persistent tension in empirical research. Models such as random forests (Breiman, 2001a), gradient boosted trees (Chen and Guestrin, 2016), and deep neural networks (Hastie et al., 2009) routinely achieve predictive accuracy that parametric models cannot match. Yet the standard tools of statistical inference — coefficient estimates, standard errors, hypothesis tests — apply naturally only to parametric models whose functional form is specified in advance. Breiman (2001b) framed this as a conflict between two cultures of statistical modeling: one oriented toward interpretable data-generating models, the other toward accurate algorithmic prediction. The conventional resolution has been to treat accuracy and interpretability as opposing ends of a spectrum, accepting reduced interpretability as the price of improved prediction. This paper argues that the tradeoff is not fundamental. It is an artifact of requiring parametric assumptions to make interpretation rigorous, and it disappears once that requirement is dropped.

The machine learning literature has produced a rich set of post-hoc interpretability methods that attempt to explain black-box model predictions without parametric assumptions.

Partial dependence plots (Friedman, 2001) visualize the marginal relationship between a feature and the predicted outcome by averaging over the empirical distribution of other features. LIME (Ribeiro et al., 2016) fits a local linear approximation to the prediction function in the neighborhood of individual observations. SHAP (Lundberg and Lee, 2017) assigns each feature a contribution to each prediction based on Shapley values from cooperative game theory, providing a unified framework that nests several earlier methods. These tools are widely used in practice and are surveyed comprehensively in Molnar (2025) and Doshi-Velez and Kim (2017). However, they share a fundamental limitation: they produce point estimates with no associated sampling distributions. A SHAP value, a PDP slope, and a LIME coefficient are descriptive quantities — there is no standard error, no confidence interval, and no basis for a hypothesis test. A researcher cannot determine whether a SHAP value is statistically distinguishable from zero, or whether two features have significantly different effects, or whether the effect of a variable differs across model specifications. For applied researchers who need to draw defensible conclusions from their models, this is a serious constraint.

Two prior contributions are closest in spirit to the present paper. Scholbeck et al. (2024) propose forward marginal effects for nonlinear prediction functions as a model-agnostic interpretability method, establishing a general framework for computing feature effects for any supervised learner. Their framework is descriptive rather than inferential: it characterizes how feature effects vary across the covariate space but does not provide standard errors, confidence intervals, or hypothesis tests, and it explicitly argues against summarizing feature effects in a single metric such as the average marginal effect. The present paper takes the opposite position: we argue that the average marginal effect is precisely the right summary quantity, that it has a rigorous population-level interpretation inherited from the econometric literature, and that valid, semiparametrically efficient inference is achievable via a debiased, cross-fitted estimator with an explicit influence function. Löwe et al. (2024) provide the `fmeffects` R package implementing this descriptive framework, demonstrating the practical

demand for marginal-effect-style interpretation of machine learning models, but without the theoretical justification — no population estimand, no efficiency theory, no characterization of when the marginal effects are well-defined — that the present paper provides through the Riesz representation of the effect functional and semiparametric efficiency theory.

The econometric literature has solved exactly this inferential problem for parametric models through average marginal effects (Greene, 2012; Wooldridge, 2010). For a generalized linear model with inverse link function μ and linear predictor $\eta = \mathbf{x}^\top \boldsymbol{\beta}$, the average marginal effect of variable j is $\text{AME}_j = \beta_j \cdot \mathbb{E}[\mu'(\eta)]$ — the regression coefficient scaled by the average derivative of the inverse link, averaged over the empirical distribution of the linear predictor. For the identity link this reduces to the OLS coefficient. For the logistic link it averages the logistic density over the sample. The result is a quantity measured in the natural units of the outcome, interpretable as the average change in the predicted outcome associated with a one-unit change in x_j, with standard errors obtained by the delta method or bootstrap. This is the output of `margins` in R and Stata (Leeper, 2021), and it is the standard form in which marginal effects are reported in applied economics, political science, sociology, and public health. The limitation is that it has been restricted to parametric models: to compute $\partial f / \partial x_j$ analytically, one must know the functional form of f.

A separate literature, largely disjoint from both the interpretability and the applied marginal effects traditions, has studied the population version of this quantity directly. The average derivative $\mathbb{E}[\partial_j f(X)]$ was introduced as an estimand in its own right by Härdle and Stoker (1989) and Powell et al. (1989), its semiparametric efficiency bounds were derived by Newey and Stoker (1993), and the modern debiased machine learning program (Chernozhukov et al., 2018, 2022) shows how to estimate linear functionals of a regression function at the parametric rate using flexible learners as nuisance estimators. The present paper is deliberately positioned at the junction of these literatures, and its delta relative to each is stated in one sentence: the debiasing literature tells one how to estimate $\mathbb{E}[\partial_j f(X)]$ efficiently when the derivative exists and the learner is treated as a black-box nuisance; this

paper establishes what the estimand *is* when the learner is not differentiable — the window effect, with an exact finite-difference Riesz representation requiring no boundary conditions and no density derivatives — and shows that a single orthogonal score then covers smooth and piecewise constant learners, continuous and discrete regressors, within one construction. Conversely, relative to the interpretability literature, the contribution is to replace descriptive scores with an estimand–estimator–efficiency-bound triple of the classical kind.

This paper removes that restriction. We propose a framework for computing average marginal effects for any supervised machine learning model, requiring only that the prediction function $\hat{f}$ can be evaluated at a point. The estimand is the *window average marginal effect*: the expectation of a centered finite difference of the regression function at a fixed step h scaled to the spread of each feature, trimmed away from the boundary of the support. Treating the finite difference as the estimand rather than as a numerical approximation to a derivative is what makes genuine model-agnosticism possible — the window effect is well defined for tree-based models, for which classical derivatives do not exist anywhere — while the classical average derivative is recovered as the $h \to 0$ limit whenever the regression function has one weak derivative, so nothing is given up relative to the econometric tradition. Estimation combines the plug-in average of finite differences with an orthogonal correction term built from an explicit Riesz representer, computed by cross-fitting; standard errors come from the estimated influence function in a single pass, with no resampling required, and a multiplier bootstrap is available at no additional model fits when a resampling distribution is wanted. The result is a coefficient table — estimate, standard error, and significance level — directly analogous to the output of a parametric regression but with no restrictions on the functional form of $\hat{f}$. The framework nests the classical GLM result as a special case: when $\hat{f}$ is a logistic regression, the estimator recovers the classical AME automatically. When $\hat{f}$ is XGBoost or a random forest or a neural network, the same formula applies and produces output on the same scale and with the same interpretation, and for binary regressors the construction specializes to augmented inverse probability weighting with the regressor in the

role of the treatment.

The theoretical contribution proceeds in three parts, which answer in turn the questions of inference, optimality, and interpretation. First, because the window functional is linear in the regression function, a change of variables yields an exact Riesz representation with an explicitly computable representer, bounded whenever the covariate density is — no integration by parts, no boundary terms, and no derivatives of the covariate density — and the bias of the resulting orthogonal moment is exactly a product of two L^2 estimation errors. A cross-fitted estimator built on this score is root-n consistent and asymptotically normal whenever the learner and a Riesz-regression estimate of the representer each converge faster than $n^{-1/4}$ in L^2 — a condition of the kind modern learning theory actually verifies, in place of the parametric-rate function-space convergence that a delta-method treatment would demand and that no growing-complexity learner satisfies. Second, the score is the efficient influence function: the estimator attains the semiparametric efficiency bound for the window effect, whose variance decomposes into the heterogeneity of the effect and a noise term amplified by the representer, making the cost of narrow windows explicit. The score is moreover doubly robust: with a known or parametrically estimated representer — simulation designs, designed experiments, randomized treatments — inference remains valid with no consistency requirement on the learner at all, covering the fixed-complexity defaults under which off-the-shelf learners converge to pseudo-true limits rather than to the truth. Third, a pairing identity characterizes what the marginal effect of a non-differentiable model measures: for any estimator of bounded variation — every tree-based model — the window effect equals the jump measure of the fitted surface integrated against a locally averaged covariate density, an exact geometric description of the population quantity that a forest's coefficient tracks. Supporting this interpretation theory, gradient consistency is established for neural networks (Hornik et al., 1990) in the Sobolev space $W^{1,p}$, via the Rellich–Kondrachov compact embedding theorem (Evans, 2010), and for gradient boosted models with tree base learners and Mondrian forests (Lakshminarayanan et al., 2014; Mourtada et al., 2017) in the

space of functions of bounded variation (Ambrosio et al., 2000), via the BV compactness theorem. For Breiman forests (Breiman, 2001a; Scornet et al., 2015), gradient consistency is conjectured rather than proved, with the obstruction identified precisely; because the inference theory requires gradient consistency for no class, this leaves two Breiman questions of unequal difficulty — valid inference needs only an L^2 rate from the forest, while the gradient question, which bears on interpretation rather than on coverage, remains a function-space problem and remains open.

We validate the framework empirically on three datasets spanning binary classification in social science, binary classification in finance, and continuous regression: the UCI Adult Income dataset (Kohavi, 1996; Dua and Graff, 2019), the UCI Credit Card Default dataset (Yeh, 2009; Dua and Graff, 2019), and the Ames Housing dataset (De Cock, 2011). Across all three applications we compare AMEs from logistic regression, random forests (Breiman, 2001a), XGBoost (Chen and Guestrin, 2016), and neural networks against SHAP values (Lundberg and Lee, 2017) and partial dependence plot slopes (Friedman, 2001). The debiased AMEs agree closely across the four model classes, which is what one expects once every learner is used to estimate the same window effect rather than to report its own slope, while SHAP values answer a different question — additive attribution relative to a baseline rather than a rate of change — and are correspondingly on a different scale and sometimes of opposite sign — a divergence that is most visible in the regression setting where units are dollars and the magnitudes are large enough to make the discrepancy concrete. PDP slopes, being themselves plug-in averages over a single full-sample fit, track the debiased AME where the plug-in is nearly unbiased and fall short of it, usually toward zero, wherever regularization flattens the fitted surface — the regularization bias made directly visible. Because AMEs come with standard errors and SHAP values and PDP slopes do not, the AME framework supports conclusions that the alternatives cannot. The credit default application is the sharpest case, and it is worth stating at the outset. Under the plug-in the four learners put the effect of the payment ratio at -0.431 and -0.440 for the two tree ensembles,

-0.045 for logistic regression, and 0.0009 — effectively nothing — for the neural network: a set of estimates that cannot all be describing the same world, and among which a reader had no principled way to adjudicate. Under the correction those same four become -0.489, -0.584, -0.265 and -0.194 respectively, every one of them negative and substantial. The disagreement was not four models detecting different structure. It was the regularization bias $\mathbb{E}[\alpha_h(\hat{f} - f)]$, which differs across learners because their regularization differs, and which the orthogonal score removes.

An additional finding concerns Breiman random forests, which are the learner the framework is most often asked to carry and the one for which the interpretation theory is least complete. Under the uncorrected plug-in they behave worst of the four classes: the simulation study shows a bias that does not vanish with the sample size but grows with it, from -0.097 at $n = 250$ to -0.091 at $n = 5{,}000$ in the linear regression design, so that intervals shrink at the usual rate around a center that is not moving and coverage collapses to 0.130. The orthogonal correction removes that term. Estimating the same quantity on the same draws with the debiased estimator leaves a bias in the fourth decimal at every sample size and returns coverage to a mean of 0.948 against the nominal 0.95, within the band 0.922 to 0.968 across all one hundred and twenty forest cells of the design — every data generating process, sample size, feature and outcome type. This is more than a bias correction, and the reason matters for which learners the framework can be used with. A depth-constrained forest whose plug-in error grows with n is not converging to f, and Proposition 2 says that with the representer known this costs nothing: the estimator is root-n normal about $\theta_h(f)$ even when $\hat{f}$ converges to some other function entirely. The simulation is that proposition observed rather than merely a demonstration that the correction is well calibrated. Because the inference theory consumes gradient consistency for no learner class, the Breiman forest is the natural choice for applied work rather than a concession, and what remains open for that class is the separate interpretive question of whether its window effect converges to a classical derivative as $h \to 0$.

The remainder of the paper is organized as follows. Section 2 defines the estimand, presents the estimator, and establishes its relationship to classical average marginal effects for the GLM family. Section 3 develops the asymptotic theory: the window estimand and its Riesz representation, the orthogonal score with cross-fitting, root-n asymptotic normality and semiparametric efficiency, the pairing identity for estimators of bounded variation, gradient consistency for three learner classes, and the nuisance rates that the conditions require. Section 4 presents the three empirical applications. Section 5 presents simulation evidence on finite-sample performance. Section 6 concludes.

2 Method

We propose a framework for interpreting any supervised machine learning model through the lens of average marginal effects. The framework is agnostic to the choice of learner — the same estimation procedure applies whether $\hat{f}$ is a neural network, a gradient boosted model, a random forest, or any other supervised learner. Crucially, the framework produces standard errors from the influence function of a Neyman-orthogonal score, evaluated by cross-fitting, enabling hypothesis tests and confidence intervals that existing model-agnostic interpretability methods do not support. A refitting bootstrap is retained, but as a diagnostic rather than as the basis for inference: Section 3.4 shows that it is blind to the learner's regularization bias, which is precisely the term the orthogonal correction removes. SHAP values and permutation-based feature importance measures provide point estimates of variable importance but no basis for inference. The average marginal effect estimator proposed here yields a coefficient table — estimate, standard error, and significance — directly analogous to the output of a parametric regression, with no restrictions on the functional form of $\hat{f}$.

2.1 Estimand

Let $\mathbf{x}_i = (x_{i1}, \ldots, x_{ip})^\top \in \mathbb{R}^p$ denote the feature vector for observation i, and let $y_i \in \mathbb{R}$ denote the outcome. We observe n independent draws $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$ from some population distribution P. Let $f : \mathbb{R}^p \to \mathbb{R}$ denote the conditional expectation function $f(\mathbf{x}) = E[Y \mid \mathbf{x}]$. For binary outcomes, $f(\mathbf{x}) = P(Y = 1 \mid \mathbf{x})$ is the conditional probability of the positive class; for continuous outcomes, $f(\mathbf{x}) = E[Y \mid \mathbf{x}]$ directly. In both cases f is the same object — the conditional expectation of Y given $\mathbf{x}$ — and the method requires no modification across outcome types.

The marginal effect of variable j at observation i is the partial derivative of f evaluated at $\mathbf{x}_i$:

$$m_{ij} = \left.\frac{\partial f}{\partial x_j}\right|_{\mathbf{x}_i} \tag{1}$$

The population average marginal effect of variable j is:

$$\mathrm{AME}_j = \mathbb{E}\left[\frac{\partial f}{\partial x_j}(\mathbf{x})\right] \tag{2}$$

where the expectation is taken over the marginal distribution of $\mathbf{x}$ under P. This is the classical target. The operational estimand of the paper is its finite-window counterpart, the *window average marginal effect* $\theta_{j,h} = \mathbb{E}[w(\mathbf{x})\,\{f(\mathbf{x} + he_j) - f(\mathbf{x} - he_j)\}/2h]$, where $h > 0$ is a fixed step scaled to the spread of x_j and w is a trimming weight that excludes points within h of the boundary of the support. The window effect is well defined for any bounded prediction function — no derivative need exist anywhere, which is what a genuinely model-agnostic estimand requires — and it converges to AME_j as $h \to 0$ whenever f has one weak derivative. Both facts, and the conditions under which the estimators below converge, are established in Section 3.

2.2 Estimation

Given an estimator $\hat{f}$ of f, the sample analog is:

$$\widehat{\mathrm{AME}}_j = \frac{1}{n}\sum_{i=1}^{n} \frac{\partial \hat{f}}{\partial x_j}\bigg|_{\mathbf{x}_i} \tag{3}$$

For most supervised learners the partial derivative $\partial \hat{f}/\partial x_j$ is not available in closed form, and for tree-based models it does not exist everywhere in the classical sense. We therefore estimate it via a centered finite difference:

$$\frac{\partial \hat{f}}{\partial x_j}\bigg|_{\mathbf{x}_i} \approx \frac{\hat{f}\Big(\mathbf{x}_i^{(j,\,x_{ij}+h_j)}\Big) - \hat{f}\Big(\mathbf{x}_i^{(j,\,x_{ij}-h_j)}\Big)}{2h_j} \tag{4}$$

where $\mathbf{x}_i^{(j,v)}$ denotes the vector $\mathbf{x}_i$ with the j-th component replaced by v. The centered finite difference is preferred over a one-sided difference because its approximation error relative to the pointwise derivative is $O(h_j^2)$ rather than $O(h_j)$, which becomes material when h_j is not negligibly small. Relative to the window estimand $\theta_{j,h}$ the centered difference involves no approximation at all: it is the defining operation of the estimand, computed exactly, and h_j is a reported feature of the target — the perturbation scale at which effects are measured — rather than a numerical tolerance.

The step size h_j is chosen adaptively as:

$$h_j = \max\big(10^{-4},\ 0.05 \cdot \hat{\sigma}_j\big) \tag{5}$$

where $\hat{\sigma}_j$ is the sample standard deviation of the j-th feature. Scaling h_j to the spread of x_j ensures the finite difference is neither so small as to introduce numerical instability nor so large as to obscure local behavior. The floor of 10^{-4} guards against degenerate cases where $\hat{\sigma}_j \approx 0$. Observations within h_j of the boundary of the observed support receive trimming weight zero, so that $\hat{f}$ is never evaluated at points where the regression function is

unidentified; the trimmed shell carries probability mass of order h_j, and with the default step this exclusion is small. Untrimmed evaluation — the default in earlier implementations — silently extrapolates the fitted model outside the support, and the trimming weight should be understood as the repair of that extrapolation rather than as a restriction of the method.

For binary or categorical variables the partial derivative does not exist and the finite difference is replaced by a contrast. The marginal effect for observation i is:

$$m_{ij} = \hat{f}\left(\mathbf{x}_i^{(j,1)}\right) - \hat{f}\left(\mathbf{x}_i^{(j,0)}\right) \tag{6}$$

which measures the change in predicted outcome when x_j is switched from 0 to 1 holding all other variables at their observed values. The sample average marginal effect is then $\widehat{\text{AME}}_j = n^{-1} \sum_{i=1}^{n} m_{ij}$, with interpretation identical to the continuous case: a change in x_j from 0 to 1 is associated with a $\widehat{\text{AME}}_j$ change in the predicted outcome, on average across the sample.

For binary classification models, $\hat{f}$ must return the estimated conditional probability $\hat{P}(Y = 1 \mid \mathbf{x})$ rather than the predicted class label. Class labels are discrete and their finite differences are identically zero; the method requires a continuous output. This is the predicted probability of the positive class, available as `predict_proba(X)[:,1]` in scikit-learn and equivalent interfaces in other frameworks. The AME is then interpreted as the change in predicted probability associated with a one-unit change in x_j, on average across the sample.

2.3 Relationship to Classical Average Marginal Effects

Average marginal effects are a standard tool in the econometrics of nonlinear models (Greene, 2012; Wooldridge, 2010). For parametric models the estimator $\widehat{\text{AME}}_j$ defined above reduces exactly to the classical result. Consider a generalized linear model with inverse link function μ and linear predictor $\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$, so that $f(\mathbf{x}_i) = \mu(\eta_i)$. The marginal effect of variable j at

observation i is:

$$\left.\frac{\partial f}{\partial x_j}\right|_{\mathbf{x}_i} = \beta_j \cdot \mu'(\eta_i) \tag{7}$$

and the population average marginal effect is $\text{AME}_j = \beta_j \cdot E[\mu'(\eta)]$. For the identity link $\mu'(\eta) = 1$ and $\text{AME}_j = \beta_j$ — the marginal effect is constant across observations and equals the OLS coefficient. For the logistic link $\mu'(\eta) = \Lambda(\eta)(1 - \Lambda(\eta))$ and the AME averages the logistic density over the sample (Greene, 2012). For the probit link $\mu'(\eta) = \phi(\eta)$, the standard normal density evaluated at the linear predictor (Wooldridge, 2010).

In each case $\widehat{\text{AME}}_j$ recovers the classical result automatically — the finite difference applied to the model output numerically approximates $\beta_j \cdot \mu'(\eta_i)$ without requiring knowledge of the link function. The framework therefore nests the entire GLM family as a special case and reduces to familiar output when the model is parametric. This backward compatibility is what the `marginaleffects` package in R and Python implements (Arel-Bundock et al., 2024); the present framework extends that implementation with formal asymptotic guarantees for machine learning models.

2.4 Inference

Inference is based on a debiased, cross-fitted estimator whose formal analysis is the subject of Section 3; here we describe the procedure. Partition the sample into K folds ($K = 5$ by default). For each fold k, using only the observations outside the fold: (1) fit the learner $\hat{f}^{(-k)}$ exactly as it would be fit in any application, tuning included if the learner is tuned, using only that fold's training portion; (2) fit an estimate $\hat{\alpha}^{(-k)}$ of the Riesz representer of the window effect — an explicit weight function characterized in Section 3 — by minimizing an empirical quadratic loss that requires only evaluating candidate functions at the shifted points $\mathbf{x}_i \pm h_j e_j$, with no density estimation of any kind. The estimate of $\theta_{j,h}$ averages, over the held-out observations of every fold, the finite difference of the out-of-fold learner plus a

correction term:

$$\hat{\theta}_{j,h} = \frac{1}{n}\sum_{k=1}^{K}\sum_{i\in I_k}\left[w(\mathbf{x}_i)\,\frac{\hat{f}^{(-k)}\left(\mathbf{x}_i^{(j,\,x_{ij}+h_j)}\right) - \hat{f}^{(-k)}\left(\mathbf{x}_i^{(j,\,x_{ij}-h_j)}\right)}{2h_j} + \hat{\alpha}^{(-k)}(\mathbf{x}_i)\left(y_i - \hat{f}^{(-k)}(\mathbf{x}_i)\right)\right] \tag{8}$$

The first term is the plug-in average of Section 2.2 computed on held-out data; the second term removes, at first order, the bias that regularization of the learner transmits to the plug-in average — each held-out residual, weighted by the representer, testifies to how the learner is locally mis-calibrated where that mis-calibration matters for the marginal effect. For binary and categorical variables the same formula applies with the contrast in place of the finite difference and inverse propensity weights in the role of the representer.

Standard errors require no resampling: the summands above, centered at $\hat{\theta}_{j,h}$, are estimates of the influence function values, and the standard error is the sample standard deviation of those values divided by $\sqrt{n}$. The coefficient table — estimate, standard error, t-statistic, and p-value for each feature — is therefore produced by K model fits in total, in place of the hundreds of refits that a bootstrap requires; Section 3 establishes that the resulting confidence intervals are asymptotically valid and that the estimator attains the semiparametric efficiency bound. When a full resampling distribution is wanted — for simultaneous confidence bands across features, for example — a multiplier bootstrap that perturbs the estimated influence function values with i.i.d. mean-zero weights is available at no additional model fits.

The package additionally retains a nonparametric model-refitting bootstrap (Efron, 1979; Efron and Tibshirani, 1993) as a *diagnostic* for how far the fitted function moves when the sample is perturbed — variability that the asymptotic theory conditions away. The learner's settings are held fixed across replicates; if the learner is tuned, the tuning is part of what each replicate refits. For $b = 1, \ldots, B$: draw $\mathcal{D}^{(b)}$ by resampling $(\mathbf{x}_i, y_i)$ pairs jointly with replacement; refit $\hat{f}^{(b)}$ on $\mathcal{D}^{(b)}$ from a fresh initialization at the same settings; and recompute the plug-in average marginal effect. The dispersion of the replicates measures

how sensitive the reported effects are to the joint variability of resampling and refitting, and large dispersion is a useful warning that the fitted model is unstable. It is not, however, a substitute for the influence function standard errors above, for reasons Section 3 makes precise: the refitting bootstrap recenters at the fitted model and is therefore blind to the regularization bias that the correction term removes — a blindness whose consequences for coverage are documented for random forests in Section 5 — and, because each replicate is evaluated at the original sample, it omits the component of sampling variance that comes from averaging a heterogeneous effect over a finite sample of covariate values.

3 Asymptotic Theory

3.1 Overview

The asymptotic theory must deliver three things. First, a population estimand that is well defined for any supervised learner, including learners such as trees and forests whose prediction functions are not differentiable anywhere. Second, valid root-n inference on that estimand under conditions that flexible machine learning estimators can actually satisfy — in particular, without requiring the learner itself to converge at the parametric rate, a condition that fails whenever the complexity of the function class grows with the sample size. Third, an account of what the gradient of a fitted model measures: when the average marginal effect of a fitted forest is computed, what population quantity is that number tracking? This section delivers the three in turn.

The organizing observation is structural: the average marginal effect is a *linear* functional of the regression function. Linearity has three consequences that together carry the entire theory. First, the functional admits an exact Riesz representation — there is an explicit, bounded weight function α_h such that averaging the finite difference of any function against the covariate distribution equals averaging the function itself against α_h (Lemma 1). The representation is obtained by a change of variables rather than integration by parts, so no

boundary terms arise and no derivative of the covariate density is ever needed. Second, the estimation bias of a debiased moment built from this representation is *exactly* a product of two estimation errors (Lemma 2) — not a product plus a linearization remainder, as for nonlinear functionals, but an identity. Root-n inference therefore requires only that the product of the L^2 errors of two nuisance estimates be $o_p(n^{-1/2})$ — for instance, both converging faster than $n^{-1/4}$ — rather than root-n convergence of the learner in any norm (Theorem 1). Third, the resulting estimator attains the semiparametric efficiency bound for the estimand (Theorem 2): among all regular estimators, none has smaller asymptotic variance.

The estimand the theory analyzes is exactly the quantity the estimator of Section 2 computes: the average of a centered finite difference at a fixed step h, which we call the *window average marginal effect*. Treating h as part of the estimand rather than as a numerical approximation to a derivative has two payoffs. The window effect is well defined for any bounded prediction function — no differentiability anywhere — which is what makes a genuinely model-agnostic theory possible. And the classical average derivative is recovered as the $h \to 0$ limit whenever the true regression function has one weak derivative (Proposition 1), so nothing classical is given up. A trimming weight confines the evaluation points to the interior of the support; the binary and categorical contrasts of Section 2 fit the same template, with the change of variables running over a two-point support and the representer reducing to inverse-probability weights (Remark 2).

The interpretation question is answered by a pairing identity (Theorem 3). For any estimator of bounded variation — which includes every tree-based model — the window average marginal effect of the *fitted* model equals the integral of a locally averaged covariate density against the distributional gradient of the fitted model: for a tree, a sum of leaf boundary jumps weighted by the boundary's share of smoothed covariate mass. This makes precise, apparently for the first time, which population quantity the marginal effect of a piecewise constant learner measures. The Sobolev and BV lemmas of Section 3.5 then

characterize the regime in which the learner's own gradient converges to the population gradient — the strongest interpretability guarantee available, and the condition under which the uncorrected plug-in estimator of Section 2 is already consistent. Those results describe when the correction term is a refinement; the inference theory of Theorem 1 does not depend on them.

The relationship between the debiased estimator and the plug-in-plus-bootstrap procedure of Section 2 is worth stating plainly. The plug-in average is the first component of the orthogonal score; the correction term $\alpha_h(x_i)\,(y_i - \hat{f}(x_i))$ removes, at first order, the bias transmitted to the average marginal effect by regularization of the learner. The simulation evidence in Section 5 displays exactly this mechanism: random forest confidence intervals built from the uncorrected plug-in collapse to near-zero coverage at large n because the intervals concentrate around a regularization-biased center — the bias term the correction is constructed to remove. The model-refitting bootstrap of Section 2 remains a useful diagnostic for specification uncertainty, but the inferential guarantees of this section attach to the debiased estimator and its influence function, not to the uncorrected plug-in.

3.2 Notation and Preliminaries

Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain and let $f : \Omega \to \mathbb{R}$ be the true underlying function relating covariates to outcomes. Let $\hat{f}_n : \Omega \to \mathbb{R}$ denote an estimator of f constructed from n observations $\{(x_i, y_i)\}_{i=1}^n$. We assume throughout that Ω is bounded and Lipschitz, which ensures the compactness and embedding results we rely on are available. These are mild regularity conditions satisfied by any compact covariate space with reasonable boundary behavior.

Throughout, $(\mathbf{x}_i, y_i)_{i=1}^n$ are independent draws from P, the joint distribution of (X, Y); with mild abuse of notation P also denotes the marginal distribution of the covariates, with $\|g\|_{P,2}^2 = \mathbb{E}[g(X)^2]$ the corresponding L^2 norm. We fix a coordinate $j \in \{1, \ldots, d\}$ for the remainder of the section and suppress it from the notation where no confusion arises. For a

step size $h > 0$, define the centered difference operator:

$$\Delta_{j,h} g(x) = \frac{g(x + he_j) - g(x - he_j)}{2h} \tag{9}$$

where e_j is the j-th coordinate vector, and let $\Omega_{j,h} = \{x \in \Omega : x + he_j \in \Omega \text{ and } x - he_j \in \Omega\}$ denote the set of points whose shifted evaluations remain in the domain. A *trimming weight* is a measurable function $w : \Omega \to [0, 1]$ with $\text{supp}(w) \subseteq \Omega_{j,h}$; it confines the finite difference to points where both shifted arguments lie inside the support, so that the prediction function is never evaluated where the regression function is unidentified. The following conditions are maintained throughout the section.

Assumption 1. *The data are i.i.d. draws from P with $\mathbb{E}[Y^2] < \infty$ and $\text{Var}(Y \mid X = x) \leq \bar{\sigma}^2 < \infty$ for P-almost every x. The covariate distribution admits a density p on Ω with $0 < p_{\min} \leq p(x) \leq p_{\max} < \infty$ for all $x \in \Omega$.*

The density bounds are the same condition previously used to define the average marginal effect functional and are standard in the average derivative literature (Powell et al., 1989; Newey and Stoker, 1993). They enter the theory only through the ratio $p_{\max}/p_{\min}$, which bounds the Riesz representer below.

The bounds are genuinely restrictive, and one design of interest violates them. Independent Gaussian covariates have a density that is not bounded below on $\mathbb{R}^d$, and the representer of Lemma 1 is then unbounded. That case is covered by an alternative to the density bounds, under which every result of this section continues to hold with the sup-norm conditions replaced by moment conditions.

Assumption 2 (Unbounded support)**.** *In place of the density bounds of Assumption 1: (a) $p(x) > 0$ for every $x \in \Omega$; (b) there exists $r > 1$ with*

$$\mathbb{E}\left[\left(\frac{p(X - he_j)}{p(X)}\right)^r\right] < \infty \quad \textit{and} \quad \mathbb{E}\left[\left(\frac{p(X + he_j)}{p(X)}\right)^r\right] < \infty \tag{10}$$

and (c) $\alpha_h \in L^q(P)$ for some $q > 2$. The moment conditions on Y in Assumption 1 are retained unchanged.

Condition (b) is precisely what the density bounds were doing in the proofs. Every appeal to $p_{\min}$ below reduces to controlling a shifted second moment, $\mathbb{E}[g(X \pm he_j)^2] = \mathbb{E}[g(X)^2\, p(X \mp he_j)/p(X)]$, which a uniform density bound handles by brute force and which (10) handles by Hölder, at the cost of a slightly higher integrability requirement on g. Condition (c) replaces the sup-norm bound that Lemma 1 supplies in the bounded case; it is an assumption here rather than a conclusion, because with an unbounded density the explicit formula for α_h need not be bounded even though it remains square integrable.

Section 5 exhibits both assumption sets. Its Gaussian designs satisfy Assumption 2 while failing Assumption 1; a bounded-support arm, with the covariates drawn uniformly on a box, satisfies Assumption 1 literally and carries a representer bounded by $1/(2h)$. Coverage agrees between them to within four thousandths, so nothing in what follows depends on which route is taken. For independent standard normal covariates the shift ratio is $p(u - he_j)/p(u) = \exp(hu_j - h^2/2)$, which has moments of every order, so (b) holds for every r; and the representer is $\alpha_h(u) = e^{-h^2/2} \sinh(hu_j)/h$, for which

$$\mathbb{E}\left[\alpha_h(X)^2\right] = \frac{e^{-h^2}}{h^2}\,\mathbb{E}\left[\sinh^2(hX_j)\right] = \frac{e^{-h^2}\left(e^{2h^2} - 1\right)}{2h^2} < \infty \tag{11}$$

with all higher moments finite as well, so (c) holds for every q. The representer is unbounded — sinh grows exponentially — which is exactly why the sup-norm route is unavailable and the moment route is needed.

We work with two function spaces depending on the estimator class. The choice of function space is not arbitrary—each space is the natural habitat of the estimator class in question, in the sense that estimators of that class belong to the space with probability one under mild conditions.

The **Sobolev space** $W^{k,p}(\Omega)$ for $k \geq 1$ and $p \geq 1$ consists of functions whose weak

derivatives up to order k are in $L^p(\Omega)$, with norm:

$$\|g\|_{W^{k,p}} = \sum_{|\alpha| \leq k} \|\partial^\alpha g\|_{L^p} \tag{12}$$

The gradient $\nabla g \in L^p(\Omega)^d$ is a vector valued L^p function for any $g \in W^{1,p}(\Omega)$. Sobolev spaces are the natural setting for smooth estimators because they quantify not just how well the function is approximated in L^p but how well its derivatives are approximated simultaneously. This is exactly the property needed to justify gradient based interpretation—L^p consistency of $\hat{f}_n$ alone is insufficient because differentiation is an unbounded operator on L^p, and convergence of a function does not in general imply convergence of its derivatives without additional regularity (Evans, 2010, Chapter 5).

The **space of functions of bounded variation** $BV(\Omega)$ consists of functions $g \in L^1(\Omega)$ whose distributional gradient Dg is a vector valued Radon measure with finite total variation $|Dg|(\Omega) < \infty$. The BV norm is:

$$\|g\|_{BV} = \|g\|_{L^1} + |Dg|(\Omega) \tag{13}$$

The BV framework extends the notion of gradient to functions that are not differentiable in the classical or even weak sense—precisely the situation with tree based estimators, which have jump discontinuities at leaf boundaries. For piecewise constant functions the distributional gradient is a sum of surface measures concentrated on the boundaries between constant regions, capturing the intuition that the "derivative" of a step function is a spike at the jump location. This makes BV the natural space for random forests and boosting with tree base learners. The total variation $|Dg|(\Omega)$ measures the total size of all jumps, weighted by the boundary surface area—a quantity that is finite and well controlled for any piecewise constant function with finitely many pieces. The standard reference for the theory of BV functions is Ambrosio et al. (2000).

The relationship between the two spaces is important. If $f \in W^{1,p}(\Omega)$ for $p \geq 1$, then

$f \in BV(\Omega)$ with $Df = \nabla f \cdot \mathcal{L}^d$—that is, the distributional gradient of f is absolutely continuous with respect to Lebesgue measure with density ∇f. This embedding $W^{1,p} \hookrightarrow BV$ means that assuming $f \in W^{1,p}(\Omega)$ is sufficient for both the Sobolev and BV results below. We make this assumption throughout.

Definition 1 (Well-Behaved Estimator, Sobolev Sense)**.** *We say that $\hat{f}_n$ is* well-behaved in the Sobolev sense *if:*

$$\sup_n \left\| \hat{f}_n \right\|_{W^{2,p}} < \infty \quad \textit{and} \quad \left\| \hat{f}_n - f \right\|_{L^p} \xrightarrow{p} 0 \tag{14}$$

Definition 2 (Well-Behaved Estimator, BV Sense)**.** *We say that $\hat{f}_n$ is* well-behaved in the BV sense *if:*

$$\sup_n \mathbb{E}\left[|D\hat{f}_n|(\Omega) \right] < \infty \quad \textit{and} \quad \left\| \hat{f}_n - f \right\|_{L^1} \xrightarrow{p} 0 \tag{15}$$

In both cases the well-behaved condition has the same structure: a uniform boundedness condition in a strong space combined with consistency in a weaker space. The uniform boundedness condition is the key regularity requirement—it says that the estimator sequence does not become increasingly irregular as n grows, even as it fits the data more closely. This condition is verifiable from properties of the estimator class and is established for each class in the theorems below. The consistency condition is the standard requirement that the estimator converges to the truth and is available from existing results in each case.

3.3 The Window Estimand and Its Riesz Representation

Definition 3 (Window Average Marginal Effect)**.** *Fix $h > 0$ and a trimming weight w with* $\text{supp}(w) \subseteq \Omega_{j,h}$*. The* window average marginal effect *of variable j at scale h is:*

$$\theta_h(f) = \mathbb{E}\left[w(X)\, \Delta_{j,h} f(X) \right] \tag{16}$$

Up to the trimming weight, $\theta_h(f)$ is exactly the population counterpart of the quantity

the estimator of Section 2 computes: the finite difference at step h is not a numerical approximation to a derivative that the theory then ignores, but the defining operation of the estimand itself. The step size is a fixed feature of the estimand, chosen and reported by the analyst exactly as a bandwidth is, and the default $h_j = 0.05\,\hat{\sigma}_j$ of Section 2 corresponds to asking how the predicted outcome responds to displacements of a twentieth of a standard deviation — an economically interpretable perturbation scale. Two features deserve emphasis. First, $\theta_h(f)$ is well defined for any bounded measurable f: no differentiability is assumed anywhere, which is what permits a single estimand to cover neural networks and regression trees simultaneously. Second, the trimming weight repairs a defect of the untrimmed implementation, which evaluates $\hat{f}$ at points $x_i \pm he_j$ that may fall outside the support of the covariates, where the regression function is not identified; w excises a boundary shell of probability mass $O(h)$ and confines every evaluation to the region where f means something.

Remark 1 (Data-dependent step sizes)**.** *Definition 3 fixes h, and every result below is a statement about the estimand at that fixed value. The default of Section 2 sets $h_j = \max(10^{-4},\, 0.05\,\hat{\sigma}_j)$, which is a function of the data. The two are reconciled by reading all results conditional on the realized $\hat{h}$: the reported quantity is the window effect at the step actually used, which is why h_j is reported alongside every estimate in Table 1 rather than being treated as a tuning parameter. Nothing in the argument requires h to be non-random — the representation of Lemma 1, the orthogonality of Lemma 2 and the expansion of Theorem 1 all hold pointwise in h — and the conditional reading is the natural one, because the estimand is genuinely indexed by the perturbation scale the analyst chose.*

It is worth being explicit about what the conditional reading buys, since the alternative is not free. If one insists instead that the target is θ_{h_0} at the deterministic limit $h_0 = 0.05\,\sigma_j$, then whenever $h \mapsto \theta_h(f)$ is differentiable at h_0 the delta method gives $\theta_{\hat{h}} - \theta_{h_0} = \theta'_{h_0}\,(\hat{h} - h_0) + o_p(n^{-1/2})$, and since $\hat{\sigma}_j$ is root-n consistent this is $O_p(n^{-1/2})$ — the same order as the sampling error itself. The substitution is therefore not *asymptotically negligible: it contributes a term to the limiting variance, and an unconditional treatment would have to add*

that term to V_h. We do not pursue this, because the conditional statement is the one that matches how the estimand is used and reported. An analyst who wants the unconditional reading can fix h from prior information or from a pilot sample, at which point the two coincide.

The estimand admits an exact dual representation, which is the engine of everything that follows. Extend $q = wp$ by zero outside Ω.

Lemma 1 (Riesz Representation)**.** *Let Assumption 1 hold and define:*

$$\alpha_h(u) = \frac{q(u - he_j) - q(u + he_j)}{2h\, p(u)}, \qquad u \in \Omega \tag{17}$$

Then for every $g \in L^1(P)$:

$$\mathbb{E}\left[w(X)\, \Delta_{j,h} g(X)\right] = \mathbb{E}\left[\alpha_h(X)\, g(X)\right] \tag{18}$$

Moreover $\mathbb{E}[\alpha_h(X)] = 0$, and under the density bounds of Assumption 1, $\|\alpha_h\|_\infty \leq p_{\max}/(h\, p_{\min})$. Under Assumption 2 the representer need not be bounded, and (18) *holds for every g with $\alpha_h g \in L^1(P)$.*

The proof is a change of variables in each half of the finite difference and nothing more. This is worth dwelling on, because it is the point at which the present theory parts ways with the classical average derivative literature. For the derivative functional $\mathbb{E}[\partial_j f(X)]$, the analogous representation is obtained by integration by parts, the representer is the negative score $-\partial_j \log p$ of the covariate density, and a boundary term appears that must be killed by assuming the density vanishes on the boundary — an assumption incompatible with a density bounded away from zero on a compact support, and the reason Powell et al. (1989) pass to density-weighted variants. For the window functional the integration by parts is replaced by exact summation: no boundary term ever arises, because the trimming weight guarantees that every shifted evaluation point stays inside Ω. The representer involves the

density only through ratios of its values at points $2h$ apart — no derivative of p, and hence no density derivative estimation, is ever required. The mean-zero property follows from applying (18) to $g \equiv 1$, and the uniform bound makes α_h a bounded weight function, in contrast to the unbounded score-based representers of the derivative functional. The cost of the exactness is the factor h^{-1} in the bound: smaller windows mean noisier representers, the familiar bias–variance logic of a bandwidth, here appearing in the variance of the efficient influence function rather than in a smoothing bias.

The classical estimand is recovered in the small-window limit.

Proposition 1 (Recovery of the Average Derivative)**.** *Let $f \in W^{1,1}(\Omega)$ and let w be a trimming weight with* $\operatorname{supp}(w) \subseteq \Omega_{j,h_0}$ *for some $h_0 > 0$. Then:*

$$\lim_{h \downarrow 0} \theta_h(f) = \mathbb{E}\left[w(X)\, \partial_j f(X)\right] \tag{19}$$

If moreover $(w_m)_{m \geq 1}$ is a sequence of trimming weights with $w_m \uparrow 1$ P-almost everywhere, then $\mathbb{E}[w_m\, \partial_j f] \to \mathbb{E}[\partial_j f] = \mathrm{AME}_j$.

One weak derivative — the minimal regularity under which the classical estimand exists at all — is thus the only condition separating the window effect from the average derivative. Nothing about the fitted model appears in Proposition 1: the double limit concerns the estimand, not the estimator, and in practice one fixes h and reports the window effect, treating the proposition as the license to interpret it as a derivative summary when the truth is smooth.

Remark 2 (Binary and Categorical Variables)**.** *For a binary $X_j \in \{0, 1\}$ the contrast estimand of Section 2 is $\theta^{\mathrm{bin}} = \mathbb{E}[w(X_{-j})\{f(X_{-j}, 1) - f(X_{-j}, 0)\}]$, and the same change-of-variables argument — with the integral over x_j replaced by a sum over its two support points — yields the representer:*

$$\alpha^{\mathrm{bin}}(u) = w(u_{-j}) \left[\frac{\mathbf{1}\{u_j = 1\}}{\pi(u_{-j})} - \frac{\mathbf{1}\{u_j = 0\}}{1 - \pi(u_{-j})} \right], \qquad \pi(x_{-j}) = \mathbb{P}(X_j = 1 \mid X_{-j} = x_{-j}) \tag{20}$$

These are inverse propensity weights, and the resulting orthogonal score is the augmented inverse probability weighting score of Robins et al. (1994) with X_j in the role of the treatment. The overlap condition $\varepsilon \leq \pi \leq 1-\varepsilon$ plays exactly the role the trimming weight plays in the continuous case, and every result of Section 3.4 applies verbatim with (α_h, θ_h) replaced by $(\alpha^{\text{bin}}, \theta^{\text{bin}})$. Continuous, binary, and categorical effects are thus instances of one construction rather than a definition and two patches.

3.4 Orthogonal Score, Cross-Fitting, and Efficient Inference

The Riesz representation converts the estimand into a moment condition with an exact second-order bias. Define, for candidate functions $\tilde{f}, \tilde{\alpha}$ and scalar θ, the score:

$$\psi(x, y; \tilde{f}, \tilde{\alpha}, \theta) = w(x)\, \Delta_{j,h}\tilde{f}(x) + \tilde{\alpha}(x)\left(y - \tilde{f}(x)\right) - \theta \tag{21}$$

At the truth, $\mathbb{E}[\psi(X, Y; f, \alpha_h, \theta_h(f))] = 0$: the first term has mean $\theta_h(f)$ by definition and the correction term has conditional mean zero. The content of the construction is what happens away from the truth.

Lemma 2 (Exact Product Bias)**.** *Let Assumption 1 hold and let $\tilde{f} \in L^2(\Omega)$, $\tilde{\alpha} \in L^2(P)$. Then:*

$$\mathbb{E}\left[\psi(X, Y; \tilde{f}, \tilde{\alpha}, \theta_h(f))\right] = \mathbb{E}\left[\left(\alpha_h(X) - \tilde{\alpha}(X)\right)\left(\tilde{f}(X) - f(X)\right)\right] \tag{22}$$

In particular the bias is bounded by $\|\alpha_h - \tilde{\alpha}\|_{P,2} \cdot \left\|\tilde{f} - f\right\|_{P,2}$.

Lemma 2 is an identity, not an expansion: because θ_h is linear in f, the moment bias equals the product of the two nuisance errors with no linearization remainder whatsoever. Neyman orthogonality — the vanishing of the Gateaux derivative of the moment in each nuisance direction at the truth (Chernozhukov et al., 2018) — is the special case obtained by setting one error to zero, but the exact form is stronger and is what makes the subsequent analysis short. The practical meaning: an error in the fitted learner harms the estimate only

to the extent that the representer is also misestimated, and vice versa. It is this multiplication of errors, not any smoothness of the learner, that buys the parametric rate.

The estimator combines the score with cross-fitting. Partition $\{1, \ldots, n\}$ into K folds $I_1, \ldots, I_K$ of approximately equal size ($K = 5$ is the package default), and for each k let $\hat{f}^{(-k)}$ and $\hat{\alpha}^{(-k)}$ be estimates of f and α_h computed from the observations outside fold k — the learner exactly as in Section 2, the representer as described in Section 3.7. The debiased estimator is:

$$\hat{\theta}_h = \frac{1}{n} \sum_{k=1}^{K} \sum_{i \in I_k} \left[w(\mathbf{x}_i)\, \Delta_{j,h} \hat{f}^{(-k)}(\mathbf{x}_i) + \hat{\alpha}^{(-k)}(\mathbf{x}_i) \left(y_i - \hat{f}^{(-k)}(\mathbf{x}_i) \right) \right] \tag{23}$$

Cross-fitting decouples the randomness of the nuisance estimates from the observations at which the score is evaluated, so that no entropy or Donsker condition on the learner's function class is needed (Chernozhukov et al., 2018) — a point of substance here, since the function classes of boosted trees and deep networks are exactly the classes for which such conditions fail.

Assumption 3. *For each fold k: (a) $\left\| \hat{f}^{(-k)} - f \right\|_{P,2} = o_p(1)$ and $\left\| \hat{\alpha}^{(-k)} - \alpha_h \right\|_{P,2} = o_p(1)$; (b) $\left\| \hat{f}^{(-k)} - f \right\|_{P,2} \cdot \left\| \hat{\alpha}^{(-k)} - \alpha_h \right\|_{P,2} = o_p(n^{-1/2})$; (c) $\left\| \hat{\alpha}^{(-k)} \right\|_{\infty} \leq \bar{A} < \infty$ almost surely.*

Condition (c) is without loss of generality: α_h obeys the known bound of Lemma 1, so any estimate may be truncated at that bound without degrading its L^2 error. Conditions (a) and (b) are rate conditions on *regression-type* L^2 errors only. The sufficient condition for (b) that we verify for specific learners in Section 3.7 is that both nuisances converge faster than $n^{-1/4}$; the product form also permits trading accuracy in one nuisance against the other.

Under Assumption 2 in place of the density bounds, the three conditions are read as follows. Condition (a) is strengthened to $\left\| \hat{f}^{(-k)} - f \right\|_{P,2+\delta} = o_p(1)$ for a $\delta > 0$ with $(2+\delta)/2$ conjugate to the r of (10); condition (c) is replaced by $\left\| \hat{\alpha}^{(-k)} \right\|_{P,q} \leq \bar{A}_q < \infty$ for the q of Assumption 2(c); and condition (b) is unchanged.

That last point is the one worth emphasizing. What changes between the two assumption

sets are a consistency requirement and a boundedness requirement; the product condition (b), which governs how fast the nuisances must converge, is identical in both. The reason is the exactness of Lemma 2: the bias is a product of two L^2 errors with no linearization remainder to control, so the step of the argument that carries the rates never touches a density bound. Everything in Section 3.7 therefore applies verbatim under either assumption set. Had the bias carried the usual remainder term, the higher integrability would have propagated into the rate conditions and the thresholds of Section 3.7 would have moved.

Where the learner's error is uniformly bounded — as it is in the compactly supported settings in which the rate results of Section 3.7 are proved, whose estimators are truncated at a known bound on f — the strengthening of (a) is free. Writing $\Delta = \hat{f}^{(-k)} - f$, interpolation gives

$$\|\Delta\|_{P,2+\delta} \leq \|\Delta\|_{\infty}^{\delta/(2+\delta)} \; \|\Delta\|_{P,2}^{2/(2+\delta)} \tag{24}$$

so L^2 consistency implies $L^{2+\delta}$ consistency with no additional assumption. On unbounded support it is a genuine additional requirement, though a mild one, and it holds for the learners used in the Gaussian design of Section 5.

Theorem 1, Theorem 2 and Proposition 2 hold under either assumption set. The appendix proofs carry the two cases side by side at the three points where they differ.

Theorem 1 (Asymptotic Linearity and Normality)**.** *Let Assumptions 1 and 3 hold, write* $\psi_i = \psi(\mathbf{x}_i, y_i; f, \alpha_h, \theta_h(f))$*, and suppose* $V_h = \mathrm{Var}(\psi_i) > 0$*. Then:*

$$\sqrt{n}\left(\hat{\theta}_h - \theta_h(f)\right) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi_i + o_p(1) \;\xrightarrow{d}\; N(0, V_h) \tag{25}$$

Moreover the plug-in variance estimator:

$$\hat{V}_h = \frac{1}{n} \sum_{k=1}^{K} \sum_{i \in I_k} \left[w(\mathbf{x}_i)\, \Delta_{j,h} \hat{f}^{(-k)}(\mathbf{x}_i) + \hat{\alpha}^{(-k)}(\mathbf{x}_i) \left(y_i - \hat{f}^{(-k)}(\mathbf{x}_i)\right) - \hat{\theta}_h \right]^2 \tag{26}$$

satisfies $\hat{V}_h \xrightarrow{p} V_h$*.*

No smoothness of the fitted learner appears anywhere in Theorem 1 or its proof. The only property of the finite difference operator that is used is that it is bounded on $L^2(P)$ after trimming:

$$\|w\,\Delta_{j,h}g\|_{P,2} \le \frac{1}{h}\left(\frac{p_{\max}}{p_{\min}}\right)^{1/2}\|g\|_{P,2} \tag{27}$$

so L^2 convergence of the learner transfers to L^2 convergence of its window effects automatically — precisely the step that fails for the derivative operator, which is unbounded on every L^p space, and the reason the classical route requires function-space convergence of the learner's gradient. The condition that the earlier version of this theory imposed at this point — root-n convergence of $\hat{f}_n$ to f in a Sobolev or BV norm — is not merely weakened but absent: it has been replaced by Assumption 3(b), a product of L^2 rates that growing-complexity learners can attain.

The exactness of Lemma 2 yields a second dividend beyond weakened rates: the score is doubly robust, and one direction of that robustness removes the consistency requirement on the learner altogether. Setting $\tilde{\alpha} = \alpha_h$ in Lemma 2 makes the moment bias vanish identically for *every* $\tilde{f}$ — not approximately, not at first order, but as an identity — and cross-fitting converts this into exact finite-sample unbiasedness of $\hat{\theta}_h$. The learner's failures are then paid entirely in variance, never in location.

Proposition 2 (Double Robustness: Inference without Learner Consistency)**.** *Let Assumption 1 hold and suppose that for each fold: (a)* $\left\|\hat{f}^{(-k)} - g\right\|_{P,2} = o_p(1)$ *for some fixed* $g \in L^2(\Omega)$*,* not necessarily equal to f*; (b)* $\left\|\hat{\alpha}^{(-k)} - \alpha_h\right\|_{P,2} = o_p(n^{-1/2})$ *— in particular* $\hat{\alpha}^{(-k)} = \alpha_h$ *when the covariate density is known; and (c)* $\left\|\hat{\alpha}^{(-k)}\right\|_\infty \le \bar{A} < \infty$ *almost surely. Define* $m_g = w\,\Delta_{j,h}g + \alpha_h\,(f - g)$*, which satisfies* $\mathbb{E}[m_g(X)] = \theta_h(f)$ *for every* g*, and:*

$$V_h(g) = \mathrm{Var}\big(m_g(X)\big) + \mathbb{E}\big[\alpha_h(X)^2\,\sigma^2(X)\big] \tag{28}$$

If $V_h(g) > 0$, then:

$$\sqrt{n}\left(\hat{\theta}_h - \theta_h(f)\right) \xrightarrow{d} N\big(0, V_h(g)\big), \qquad \hat{V}_h \xrightarrow{p} V_h(g) \tag{29}$$

so the confidence intervals of Corollary 1 retain asymptotic coverage. When $\hat{\alpha}^{(-k)} = \alpha_h$ exactly, $\mathbb{E}[\hat{\theta}_h] = \theta_h(f)$ at every sample size.

The practical reach of Proposition 2 is wider than its hypotheses may suggest, because condition (a) is the condition that off-the-shelf machine learning defaults actually satisfy. A depth-capped forest, an early-stopped boosting run, a network of fixed width — fixed-complexity learners generally — converge not to f but to a pseudo-true limit g within their approximation class, so Assumption 3(a) fails and Theorem 1 is silent about them. The proposition covers exactly this case whenever the representer side is strong: in simulation designs and designed experiments the covariate density is known and α_h is computable in closed form; and in the binary case of Remark 2, a randomized X_j means a *known* propensity score, so the marginal effect of a randomized treatment is validly covered for any black-box learner with no consistency requirement whatsoever — the model-agnostic analog of the classical robustness of randomization (Robins et al., 1994). The mirror-image direction of the double robustness is worth recording too: since the bias identity is symmetric, a learner converging at the parametric rate — a correctly specified GLM — tolerates an arbitrarily poor representer estimate, which is how the classical delta-method validity of parametric average marginal effects is nested here as the opposite corner of the same identity.

Two honest qualifications. First, when $g \neq f$ the estimator is asymptotically linear with influence function different from the efficient one; there is no conflict with Theorem 2 below, because such an estimator is not regular, and efficiency is restored exactly when $g = f$, at which point (28) reduces to the bound. The noise component of the variance is invariant in g; misspecification of the learner is paid entirely through the dispersion of m_g, typically as wider intervals. Second, in observational data the density is unknown and condition (b)

demands a parametric-rate representer, which Riesz regression delivers only under strong structure; the proposition is therefore a complement to Theorem 1, not a replacement — the theorem splits the burden multiplicatively across the two nuisances, the proposition shifts it entirely onto one of them.

Theorem 2 (Semiparametric Efficiency)**.** *Let Assumption 1 hold. In the nonparametric model in which the distribution of (X, Y) is restricted only by Assumption 1, the efficient influence function of the functional $P \mapsto \theta_h(f_P)$ is $\psi(x, y; f, \alpha_h, \theta_h(f))$, and the semiparametric efficiency bound is:*

$$V_h = \mathrm{Var}\big(w(X)\,\Delta_{j,h} f(X)\big) + \mathbb{E}\big[\alpha_h(X)^2\,\sigma^2(X)\big] \tag{30}$$

where $\sigma^2(x) = \mathrm{Var}(Y \mid X = x)$. The estimator $\hat{\theta}_h$ of Theorem 1 attains the bound.

The variance decomposition (30) is the honest accounting of what inference on a marginal effect costs. The first term is the heterogeneity of the effect itself across the covariate distribution — irreducible, present even if f were known. The second is the price of not knowing f: the conditional noise amplified by the squared representer, which scales as h^{-2} in the worst case through the bound of Lemma 1. Narrow windows purchase proximity to the derivative estimand at the cost of a larger efficiency bound — the bias–variance tradeoff of the problem, located in the estimand rather than hidden in an estimator's tuning. No estimator that is regular for θ_h can do better than (30); the question the earlier version of this theory could not answer — whether the procedure is optimal in any sense — has the sharpest available affirmative answer for the window estimand.

Corollary 1 (Confidence Intervals)**.** *Under the conditions of Theorem 1, $\hat{\theta}_h \pm z_{1-\gamma/2}\,(\hat{V}_h/n)^{1/2}$ is an asymptotically valid $1-\gamma$ confidence interval for $\theta_h(f)$, and the associated t-statistic is asymptotically standard normal under $H_0 : \theta_h(f) = 0$.*

Remark 3 (Resampling-Free and Resampling-Based Inference)**.** *Corollary 1 requires no resampling: the coefficient table of Section 2 is produced from a single pass of cross-fitting,*

with K model fits in place of the B refits of the bootstrap — a reduction of one to two orders of magnitude in computation. When a resampling distribution is nonetheless wanted (for simultaneous bands across coefficients, for example), asymptotic linearity delivers one for free: the multiplier bootstrap that perturbs the estimated influence function values $\hat{\psi}_i$ by i.i.d. mean-zero weights is consistent (van der Vaart and Wellner, 1996, Chapter 3.6) under the conditions of Theorem 1, and costs no additional model fits.

Remark 4 (The Model-Refitting Bootstrap)**.** *The refitting bootstrap of Section 2, applied to the uncorrected plug-in average, is not covered by these guarantees, for two reasons that the present framework makes visible. First, it recenters at the fitted model, so it measures variability around $\mathbb{E}[w\Delta_{j,h}\hat{f}]$ rather than around $\theta_h(f)$: the regularization bias $\mathbb{E}[\alpha_h(\hat{f} - f)]$ — first order for slowly converging learners — is invisible to it. This is the mechanism behind the collapse of random forest coverage documented in Section 5. Second, evaluating each refitted model at the original sample holds the empirical covariate distribution fixed and thereby omits the first variance component of* (30) *from the bootstrap distribution. The refitting bootstrap retains a legitimate role as a diagnostic for how far the fitted function moves under resampling, variability which the asymptotic theory conditions away; it should be read as that diagnostic, not as the inferential procedure.*

3.5 What Learner Gradients Estimate

The inference theory of Section 3.4 used no property of the fitted learner beyond an L^2 rate. The question it leaves open is interpretive: the first component of the score averages the finite differences of the fitted model, and one wants to know what that object measures about the model — and when the model's gradient information converges to the population's. This subsection answers both questions in the function spaces the learners naturally inhabit. Four lemmas establish that gradients exist and converge in the Sobolev space of smooth estimators and the BV space of piecewise constant ones; a pairing identity (Theorem 3) then expresses the window effect of any BV estimator exactly as its jump measure integrated against a

locally averaged covariate density, which is, to our knowledge, the first characterization of the population quantity that the marginal effect of a tree-based model measures. All proofs are in the Appendix.

The parallel structure of Lemmas 3 through 6 reflects the parallel structure of the Sobolev and BV frameworks and is the formal expression of the analogy between smooth and piecewise constant estimators. None of these results is needed for the validity of Theorem 1; they characterize the regime in which the *uncorrected* plug-in average is already consistent, and hence in which the correction term of (23) is a refinement rather than a repair.

Lemma 3 (Derivative Existence in Sobolev Spaces)**.** *Let $\hat{f}_n \in W^{2,p}(\Omega)$ for some $p > 1$. Then $\nabla \hat{f}_n \in W^{1,p}(\Omega)^d$ exists as a well defined L^p function and satisfies:*

$$\left\|\nabla \hat{f}_n\right\|_{L^p} \leq \left\|\hat{f}_n\right\|_{W^{1,p}} \leq \left\|\hat{f}_n\right\|_{W^{2,p}} \tag{31}$$

Lemma 3 is essentially definitional—membership in $W^{2,p}(\Omega)$ implies by definition that the first and second weak derivatives exist and are in L^p. The content of the lemma is the norm bound, which establishes that the gradient operator $\nabla : W^{2,p} \to W^{1,p}$ is bounded and linear. This boundedness is what makes the subsequent consistency argument work—a bounded linear operator preserves convergence, so once we have convergence of $\hat{f}_n$ in the right space, convergence of $\nabla \hat{f}_n$ follows automatically. The reason we require $\hat{f}_n \in W^{2,p}(\Omega)$ rather than just $W^{1,p}(\Omega)$ is that we need one additional degree of differentiability to apply the compactness argument in Lemma 4. The relevant theory of Sobolev spaces and their norms is developed in Evans (2010, Chapter 5).

Lemma 4 (Derivative Consistency in Sobolev Spaces)**.** *Let $f \in W^{1,p}(\Omega)$ and suppose $\hat{f}_n$ is well-behaved in the Sobolev sense. Then:*

$$\left\|\nabla \hat{f}_n - \nabla f\right\|_{L^p} \xrightarrow{p} 0 \tag{32}$$

Lemma 4 is the harder result and the one that does the real work in the Sobolev setting. The difficulty is that L^p consistency of $\hat{f}_n$ alone does not imply L^p consistency of $\nabla \hat{f}_n$—differentiation is an unbounded operator on L^p and limits and derivatives cannot be exchanged without additional structure. The uniform $W^{2,p}(\Omega)$ boundedness condition provides that structure. The proof applies the Rellich–Kondrachov compact embedding theorem, which states that a uniformly bounded sequence in $W^{2,p}(\Omega)$ has a subsequence converging strongly in $W^{1,p}(\Omega)$ (Evans, 2010, Theorem 5, Section 5.7). Combined with the L^p consistency condition identifying the limit as f, this gives strong convergence of $\hat{f}_n$ to f in $W^{1,p}(\Omega)$, from which L^p convergence of the gradients follows immediately from the boundedness of ∇ established in Lemma 3. The key insight is that the uniform boundedness condition ensures the estimator sequence lives in a compact subset of $W^{1,p}(\Omega)$, preventing the gradients from escaping to infinity even as the estimator fits the data more closely.

Lemma 5 (Derivative Existence in BV)**.** *Let $\hat{f}_n \in BV(\Omega)$. Then the distributional gradient $D\hat{f}_n$ exists as a well defined vector valued Radon measure in $\mathcal{M}(\Omega)^d$ with:*

$$|D\hat{f}_n|(\Omega) < \infty \tag{33}$$

For piecewise constant estimators with leaf partition $\{R_j\}$ and leaf means $\{\bar{y}_j\}$, the distributional gradient takes the explicit form:

$$D\hat{f}_n = \sum_{j \sim k} (\bar{y}_k - \bar{y}_j) \cdot \nu_{jk} \cdot \mathcal{H}^{d-1} \lfloor_{\partial R_j \cap \partial R_k} \tag{34}$$

where ν_{jk} is the unit normal between adjacent leaves and $\mathcal{H}^{d-1}$ is the $(d-1)$-dimensional Hausdorff measure on leaf boundaries.

Lemma 5 is the BV analog of Lemma 3 and is similarly structural in nature. The first part—that BV membership implies the distributional gradient exists as a Radon measure—follows from the definition of BV as developed in Ambrosio et al. (2000, Chapter 3). The

content of the lemma is the explicit formula for piecewise constant estimators, which makes concrete what the distributional gradient actually looks like for a tree based model. The formula says that the gradient is not a function at all but a measure concentrated on the boundaries between leaves, with mass proportional to the jump in response values across the boundary weighted by the boundary surface area. This is the precise sense in which a random forest or boosted tree has a gradient—not a pointwise derivative, which does not exist, but a distributional object that captures where and how much the estimator changes. This explicit representation is also what makes the total variation $|D\hat{f}_n|(\Omega)$ computable in terms of observable quantities—the leaf means and the partition geometry—which is essential for verifying the well-behaved condition in the theorems below.

Lemma 6 (Derivative Consistency in BV)**.** *Let $f \in W^{1,p}(\Omega)$ for some $p > 1$, so that $f \in BV(\Omega)$ with $Df = \nabla f \cdot \mathcal{L}^d$. Suppose $\hat{f}_n$ is well-behaved in the BV sense. Then:*

$$D\hat{f}_n \xrightarrow{w*} Df \quad \text{in } \mathcal{M}(\Omega)^d \tag{35}$$

If additionally $\mathbb{E}[|D\hat{f}_n|(\Omega)] \to |Df|(\Omega)$, then convergence is strict.

Lemma 6 is the hardest of the four function space lemmas and the BV analog of Lemma 4. The convergence is necessarily weaker than in the Sobolev case—weak-$*$ convergence of measures rather than strong L^p convergence—because the BV space is not reflexive and the stronger norm convergence is not available in general. Weak-$*$ convergence means that for any continuous test function ϕ:

$$\int_\Omega \phi \, d(D\hat{f}_n) \to \int_\Omega \phi \, d(Df) \tag{36}$$

which is the appropriate notion of convergence for the distributional gradients of piecewise constant estimators. In the context of average marginal effects, this means that integrated gradient information—which is exactly what the AME computes— converges to the correct

population quantity. The proof applies the BV compactness theorem of Ambrosio et al. (2000, Theorem 3.23), which is the analog of Rellich–Kondrachov in the BV setting: a uniformly bounded sequence in BV has a subsequence converging in L^1, and the L^1 consistency condition identifies the limit as f. The strict convergence upgrade follows from the lower semicontinuity of the total variation functional and is the stronger result one would want for applications requiring convergence of the total size of gradient information rather than just its distribution.

The parallel between Lemmas 4 and 6 is worth noting explicitly. In both cases the argument has the same structure: uniform boundedness in a strong space gives compactness, consistency in a weak space identifies the limit, and the combination gives convergence of the derivative in the appropriate topology. The difference is in the topology of the conclusion—strong L^p convergence in the Sobolev case, weak-$*$ convergence of measures in the BV case—and this difference is fundamental rather than technical. It reflects the fact that smooth estimators carry gradient information in a stronger sense than piecewise constant estimators, which is itself a reflection of the different inductive biases of the estimator classes.

The pairing identity is the capstone of the interpretation theory. Recall from Lemma 5 that the distributional gradient of a piecewise constant estimator is a measure concentrated on leaf boundaries. The window effect of such an estimator integrates that measure against a smoothed density: let $K_h \star q$ denote the one-dimensional moving average of $q = wp$ (extended by zero) along coordinate j:

$$(K_h \star q)(u) = \frac{1}{2h} \int_{-h}^{h} q(u + s\, e_j)\, ds \tag{37}$$

Theorem 3 (Pairing Identity)**.** *Let Assumption 1 hold, let w be a trimming weight with* $\mathrm{supp}(w) \subseteq \Omega_{j,h}$*, and let $g \in BV(\Omega)$ (identified with its precise representative). Then:*

$$\mathbb{E}\left[w(X)\, \Delta_{j,h} g(X)\right] = \int_{\Omega} (K_h \star q)(u)\; dD_j g(u) \tag{38}$$

where $D_j g$ is the j-th component of the distributional gradient of g. In particular, for a piecewise constant g with leaf partition $\{R_l\}$ and leaf values $\{\bar{y}_l\}$, by Lemma 5:

$$\mathbb{E}\left[w(X)\,\Delta_{j,h} g(X)\right] = \sum_{l \sim m} (\bar{y}_m - \bar{y}_l)\,(\nu_{lm})_j \int_{\partial R_l \cap \partial R_m} (K_h \star q)\; d\mathcal{H}^{d-1} \tag{39}$$

Corollary 2. *If in addition q extends to a continuous function of compact support (for instance w continuous and p continuous on Ω), then for fixed $g \in BV(\Omega)$:*

$$\lim_{h \downarrow 0} \int_\Omega (K_h \star q)\; dD_j g = \int_\Omega q\; dD_j g \tag{40}$$

and when $g \in W^{1,1}(\Omega)$ the right side equals $\mathbb{E}[w(X)\,\partial_j g(X)]$, consistent with Proposition 1.

Theorem 3 settles a question that the interpretability literature has left informal: what does the average marginal effect of a random forest *measure*? The answer given by (39) is exact and geometric: it is the total jump of the fitted surface across leaf boundaries in direction j, with each boundary segment weighted by its share of the covariate mass in an h-window — the places where the model changes its prediction, weighted by how many observations live near those places. The identity reconciles the two readings of a marginal effect that Section 2 could only juxtapose. Read with $g = \hat{f}$, (38) is a descriptive statement about the fitted model, available in finite samples with no assumptions at all. Read through Theorem 1, the same plug-in quantity, corrected by the orthogonal score, estimates the population functional $\theta_h(f)$ at the parametric rate. The descriptive and the inferential interpretations are the two sides of (23), and the pairing identity is what makes the first of them precise.

3.6 Gradient Convergence of Specific Learners

The theorems of this subsection verify the well-behaved conditions for three estimator classes and conclude, via the lemma pairs above, that the learner's own gradient information con-

verges to the population's. This is the strongest interpretability property a learner can have — the fitted model's local behavior, not merely its window averages, tracks the truth — and it implies consistency of the uncorrected plug-in average for the small-h estimand. We emphasize the division of labor: these results calibrate how much interpretive weight the fitted gradients themselves can bear, while the inference of Section 3.4 requires none of them. In each case the argument has two components: verifying that the estimator satisfies the well-behaved condition, and citing the appropriate lemma pair. The first component is the estimator-specific contribution; the second is the functional analytic machinery established above.

Theorem 4 (Gradient Consistency for Neural Networks)**.** *Let $\hat{f}_n$ be a neural network estimator with smooth activations $\sigma \in C^2$, bounded weight norms $\sup_n \prod_\ell \left\|W_\ell^{(n)}\right\| < \infty$, and $\left\|\hat{f}_n - f\right\|_{L^p} \xrightarrow{p} 0$ for $p > d$. Then $\hat{f}_n$ is well-behaved in the Sobolev sense and:*

$$\left\|\nabla \hat{f}_n - \nabla f\right\|_{L^p} \xrightarrow{p} 0 \tag{41}$$

The conditions of Theorem 4 are natural and verifiable. Smooth activations — sigmoid, tanh, GELU — are standard in practice and ensure the network function is twice differentiable. The bounded weight norm condition formalizes the effect of L^2 regularization during training, which prevents the network weights from growing without bound. The L^p consistency condition is the standard requirement that the network is learning the true function. Hornik et al. (1990) establish that multilayer feedforward networks with smooth activation functions are capable of approximating an arbitrary function and its derivatives to arbitrary accuracy, providing the foundation for the consistency condition and establishing that the function class is rich enough to contain the estimators we consider. The condition $p > d$ is required by the pointwise differentiability theorem of Evans and Gariepy (2015, Theorem 6.5) and is satisfied for all p larger than the covariate dimension — a mild regularity condition that is implicit in the classical Sobolev embedding theory.

Proof sketch. The chain rule applied through the network layers gives $\hat{f}_n \in W^{2,p}(\Omega)$ with:

$$\left\|\hat{f}_n\right\|_{W^{2,p}} \lesssim \prod_{\ell} \left\|W_\ell^{(n)}\right\|^2 \cdot \|\sigma''\|_\infty \tag{42}$$

which is uniformly bounded by the weight norm assumption. The L^p consistency condition and Lemma 4 then give the result. Full proof in Appendix L. □

Remark 5. *Theorem 4 is stated for smooth activations. ReLU networks require a modified argument because ReLU is not C^2 and the chain rule calculation does not apply directly. Extension to ReLU networks is possible through a distributional argument but complicates the presentation and is omitted here.*

Theorem 5 (Gradient Consistency for Gradient Boosted Models)**.** *Let $\hat{f}_n = \hat{F}_m = (I-(I-S)^{m+1})Y$ be an L_2Boost estimator with symmetric linear learner S satisfying the eigenvalue condition $0 < \lambda_k \leq 1$ for all k, and with base learners h_m that are either smooth ($h_m \in W^{2,p}$) or tree stumps. Suppose $\left\|\hat{f}_n - f\right\|_{L^1} \xrightarrow{p} 0$. Then:*

(i) If base learners are smooth, $\hat{f}_n$ is well-behaved in the Sobolev sense and $\left\|\nabla\hat{f}_n - \nabla f\right\|_{L^p} \xrightarrow{p} 0$.

(ii) If base learners are tree stumps, $\hat{f}_n$ is well-behaved in the BV sense and $D\hat{f}_n \xrightarrow{w*} Df$ in $\mathcal{M}(\Omega)^d$.

Theorem 5 covers both cases that arise in practice and illustrates how the function space framework adapts to the base learner. XGBoost and LightGBM — the implementations used in Section 3 — use shallow trees as base learners and therefore fall under case (ii). The two cases in the theorem are not merely technical alternatives — they reflect a genuine difference in what the gradient of the estimator is. For smooth base learners the gradient is an ordinary function and L^p convergence is available. For tree base learners the gradient is a measure and only weak-$*$ convergence can be established. The average marginal effects are well defined and consistent in both cases, but the sense in which they are consistent differs.

The eigenvalue condition $0 < \lambda_k \leq 1$ is the key regularity condition on the base learner. Bühlmann and Yu (2003) impose this condition explicitly in their Theorem 1 and establish that under it the bias decays exponentially fast while the variance increases only by exponentially diminishing amounts — the exponential bias–variance trade-off that makes L_2Boost analytically tractable. The condition holds for smoothing spline learners and, generically, for tree stump learners whose splits have positive but incomplete explanatory power, which is the typical case in practice. Learners with any eigenvalue equal to zero are degenerate in the sense of Corollary 1 of Bühlmann and Yu (2003): $\mathcal{B}_m \equiv S$ for all m and boosting does not improve on the base learner. The functional gradient descent perspective of Mason et al. (2000) provides the conceptual framework for understanding the boosting path as gradient descent in function space.

Proof sketch. For smooth base learners the $W^{2,p}(\Omega)$ bound follows from linearity of the estimator:

$$\left\| \hat{f}_n \right\|_{W^{2,p}} \leq \sum_m \eta_m \left\| h_m \right\|_{W^{2,p}} < \infty \tag{43}$$

For tree stump base learners the BV bound uses the eigenvalue structure of L_2Boost. By Proposition 1 of Bühlmann and Yu (2003), the residual at step m is $u_m = (I - S)^m Y$. By Proposition 2, under the eigenvalue condition $0 < \lambda_k \leq 1$, Theorem 1 of Bühlmann and Yu (2003) gives exponential decay of residuals:

$$\left\| u_m \right\|_{L^2} \leq C\rho^m, \quad \rho = \max_k (1 - \lambda_k) < 1 \tag{44}$$

Each stump's BV norm is controlled by its residual magnitude via Lemma 5, and the total BV norm of the ensemble satisfies:

$$\mathbb{E}[|D\hat{f}_n|(\Omega)] \leq C \sum_{m=1}^{\infty} \rho^m = \frac{C\rho}{1 - \rho} < \infty \tag{45}$$

The geometric series converges because $\rho < 1$, giving uniform BV boundedness without any

step size condition. Lemma 6 then gives $D\hat{f}_n \xrightarrow{w*} Df$. Full proof in Appendix M. □

Theorem 6 (Gradient Consistency for Mondrian Forests)**.** *Let $\hat{f}_n$ be a Mondrian random forest estimator with budget $\lambda_n \asymp n^{1/(d+2)}$, $B \gg n^{2(d-\alpha)/(d+2)}$ trees where $\alpha = 1 - d/p$, f Lipschitz on $[0,1]^d$, and $f \in W^{1,p}(\Omega)$ for $p > d$. Then $\hat{f}_n$ is well-behaved in the BV sense and:*

$$D\hat{f}_n \xrightarrow{w*} Df \quad \text{in } \mathcal{M}(\Omega)^d \tag{46}$$

Theorem 6 is the most technically demanding result in the paper. Random forests are genuinely different from neural networks and boosted models in a way that makes gradient consistency harder to establish — the partition structure is random and the response averages within leaves depend on the data in a complex way. The restriction to Mondrian forests, where splits are made independently of the data, is the key simplification that makes the proof tractable. For Mondrian forests the partition geometry and the response variation can be controlled separately, whereas for Breiman forests with data-dependent splits the two are coupled in a way that resists clean analysis. The L^2 consistency result under the budget $\lambda_n \asymp n^{1/(d+2)}$ follows from Mourtada et al. (2017), who establish the minimax rate $O(n^{-2/(d+2)})$ for Lipschitz regression functions, and L^1 consistency follows by Cauchy–Schwarz on the bounded domain Ω.

The budget condition $\lambda_n \asymp n^{1/(d+2)}$ and the tree count condition $B \gg n^{2(d-\alpha)/(d+2)}$ are natural and interpretable. The budget determines the approximation rate — balancing the bias from coarse partitions against the variance from fine partitions at the minimax optimal rate. The tree count condition ensures that the forest average has sufficient cancellation across trees to control the total variation; it is mild in practice, satisfied by standard forest sizes for moderate dimension and sample size.

Proof sketch. Each tree is piecewise constant and therefore in BV by Lemma 5. The key difficulty is that individual tree BV norms grow with n, so the triangle inequality applied to the forest average is too crude. Instead we work with the second moment of the total

variation. Writing $D\hat{f}_n = \frac{1}{B}\sum_b DT_b$:

$$\mathbb{E}[|D\hat{f}_n|(\Omega)] \leq \sqrt{\mathbb{E}[|D\hat{f}_n|(\Omega)^2]} = \sqrt{\frac{1}{B^2}\mathbb{E}\left[\left|\sum_b DT_b\right|(\Omega)^2\right]} \tag{47}$$

The cross terms between trees $b \neq b'$ vanish by independence of the Mondrian partitions and the zero-mean condition on the jumps. The diagonal terms are bounded using the cell diameter result $\mathbb{E}[D_{\lambda_n}(x)^2] \leq 4d/\lambda_n^2$ of Roy and Teh (2009) and the split count bound $\mathbb{E}[K_{\lambda_n}] \leq (e(\lambda_n+1))^d$ of Mourtada et al. (2017), giving:

$$\mathbb{E}[|D\hat{f}_n|(\Omega)^2] = \frac{1}{B}\,\mathbb{E}[|DT_1|(\Omega)^2] \leq \frac{1}{B}\,O(n^{2(d-\alpha)/(d+2)}) \tag{48}$$

Under the condition $B \gg n^{2(d-\alpha)/(d+2)}$ this is uniformly bounded. Lemma 6 then gives $D\hat{f}_n \xrightarrow{w*} Df$ in $\mathcal{M}(\Omega)^d$. Full proof in Appendix N. □

Remark 6. *Extension of Theorem 6 to Breiman forests with data-dependent splits requires additional control over the interaction between partition geometry and response variation. The data-dependent splitting rule couples these two quantities in a way that makes the cross-term cancellation argument unavailable — splits are chosen precisely to maximize response differences across boundaries, which is the worst case for BV control. This extension is an important direction for future work and is discussed further in Section 6.*

3.7 Nuisance Estimation and Rates

Assumption 3 asks for two regression-type L^2 rates whose product is $o_p(n^{-1/2})$; the benchmark sufficient condition is that each nuisance converge faster than $n^{-1/4}$. This subsection records when the learners of Section 2 meet the benchmark for $\hat{f}$, and how the representer α_h is estimated without ever estimating a density.

For the regression function, the required rates are exactly the L^2 risk bounds of the modern nonparametric literature, and a single smoothness threshold organizes all three learner

classes. For deep neural networks with ReLU activations, Schmidt-Hieber (2020) establishes the rate $n^{-s/(2s+d)}$ up to logarithmic factors for s-Hölder regression functions (with the effective d reduced under compositional structure), so the $n^{-1/4}$ benchmark is met precisely when:

$$\frac{s}{2s+d} > \frac{1}{4} \quad \Longleftrightarrow \quad s > \frac{d}{2} \tag{49}$$

For Mondrian forests, the minimax-optimal rates of Mourtada et al. (2017) take the same form $n^{-s/(2s+d)}$ for Hölder smoothness $s \leq 2$, so the benchmark is again (49): at Lipschitz smoothness $s = 1$ this requires $d < 2$, and exploiting forest smoothing up to $s = 2$ accommodates $d \leq 3$. For L_2Boost, Bühlmann and Yu (2003) establish minimax rates for smoothing-spline base learners with the boosting iteration acting as the regularization parameter, and the benchmark corresponds to smoothness exceeding one half in their univariate and additive settings — with additive structure, as usual, breaking the dependence on the ambient dimension. The reading we intend is not that these conditions are innocuous — (49) is a real restriction, and in high dimensions it binds — but that the currency in which the conditions are denominated has changed: what inference requires of the learner is an L^2 error rate of the kind the learning-theory literature actually proves, rather than root-n convergence in a Sobolev norm, which no growing-complexity learner satisfies. For Breiman forests with data-dependent splits, L^2 consistency is available under structural assumptions (Scornet et al., 2015), and polynomial L^2 rates for the original algorithm — sample CART splits and column subsampling included — have been established under sufficient-impurity-decrease conditions by Chi et al. (2022); the resulting exponents can fall short of the $n^{-1/4}$ benchmark, but the product form of Assumption 3(b) permits pairing a slow forest rate with a fast representer estimate, and Proposition 2 covers the known-representer case with no forest rate at all. The open question relevant to this paper is therefore an L^2 rate question, a substantially more tractable target than the gradient-consistency question that Section 3.6 leaves open for that class.

The representer is estimated by Riesz regression (Chernozhukov et al., 2022), which

exploits the same linearity that produced Lemma 2. By Lemma 1, for any candidate $\alpha \in L^2(P)$:

$$\mathbb{E}\left[\alpha(X)^2\right] - 2\,\mathbb{E}\left[w(X)\,\Delta_{j,h}\alpha(X)\right] = \|\alpha - \alpha_h\|_{P,2}^2 - \|\alpha_h\|_{P,2}^2 \tag{50}$$

so the population loss on the left is minimized uniquely at α_h, and its sample analog:

$$\hat{L}(\alpha) = \frac{1}{n}\sum_{i=1}^{n}\left[\alpha(\mathbf{x}_i)^2 - 2\,w(\mathbf{x}_i)\,\frac{\alpha(\mathbf{x}_i + he_j) - \alpha(\mathbf{x}_i - he_j)}{2h}\right] \tag{51}$$

is computable directly: minimizing $\hat{L}$ over a class of functions — penalized linear sieves, neural networks, or forests — requires only the ability to evaluate candidates at shifted points, and at no stage is the density p, let alone a derivative of it, estimated. Chernozhukov et al. (2022) provide L^2 convergence guarantees for such estimators over exactly these classes, with rates governed by the complexity of the class and the approximability of α_h. Since α_h obeys the explicit bound of Lemma 1, the minimizer is truncated at that bound, which enforces Assumption 3(c) by construction without affecting the L^2 error. Two further remarks. First, the product form of Assumption 3(b) permits asymmetry: a learner with a fast rate for f tolerates a slow Riesz estimate, and conversely. Second, in designed or simulated data where the covariate density is known — as in the Monte Carlo study of Section 5 — α_h is available in closed form and the representer error is identically zero; Proposition 2 then applies, and valid inference requires no rate — indeed no consistency — from the learner at all.

4 Empirical Examples

Across the three empirical applications we fit four model classes: logistic or linear regression as the parametric baseline, a random forest, XGBoost, and a feedforward neural network. The random forest uses 200 trees with a maximum depth of 8. XGBoost uses 200 boosting rounds with a maximum depth of 4 and a learning rate of 0.05. The neural network architecture and training schedule depend on the outcome type. For binary classification, the

network is a feedforward architecture with two hidden layers of 64 and 32 units trained with the Adam optimizer at a learning rate of 10^{-3}, with early stopping on a 10% validation split with patience of 20 epochs. For continuous regression, the network is deeper — four hidden layers of 256, 128, 64, and 32 units — trained at a learning rate of 3×10^{-4} with patience of 50 epochs, reflecting the larger capacity required to fit the continuous outcome surface and the longer training horizon needed for the optimization to converge. For regression outcomes, the network includes the input normalization layer and target standardization wrapper described in Section 2 to ensure scale-invariant gradient updates and predictions on the natural outcome scale. Estimation follows the debiased cross-fitted procedure of Section 2.4 with $K = 5$ folds: within each fold the learner and the Riesz representer are fit on the other four folds and evaluated on the held-out fold, and standard errors come from the sample variance of the orthogonal score. No learner is ever differenced at an observation it was trained on. Every model class is refit from scratch on each fold, including the neural networks, whose early stopping schedule is applied within the fold. The fit statistics reported at the foot of each table are computed from a single full-sample fit and are descriptive of the learner rather than of the estimator; they are not used in the estimation. Where applicable, predicted probabilities for classification models are extracted from `predict_proba(X)[:,1]` as discussed in Section 2.

The window estimand requires two settings to be fixed before estimation, and because $\theta_{j,h}$ is defined by them they are reported rather than treated as tuning. The step size is the default $h_j = \max(10^{-4}, 0.05\hat{\sigma}_j)$, floored at 0.5 for integer-valued features. The floor matters for count variables such as bedrooms, bathrooms, years of education and months delayed: a count has no mass strictly between its levels, so a window narrower than one unit has an empty interior, and for a piecewise constant learner the difference across it vanishes except where the window happens to straddle a split. At $h_j = 0.5$ the window is exactly the one-unit contrast, which is also the quantity of substantive interest for a count. The second setting is the trimming weight, which gives weight zero to observations lying within h_j of the boundary

of the observed support of x_j, where $x \pm h_j e_j$ would require the learner to extrapolate. For most features this removes a negligible share of the sample, but for variables with mass sitting exactly on a boundary it does not: capital gains and capital losses are zero for over ninety percent of the Adult sample, and zero is the minimum of their observed support, so those AMEs are effects on the subpopulation with nonzero gains or losses rather than on the full sample. Table 1 reports h_j and the trimmed fraction for every feature in the three applications, and the discussion below flags the cases where the trimming is large enough to change the interpretation.

4.1 UCI Adult Income

The first dataset is the UCI Adult Income dataset (Dua and Graff, 2019; Kohavi, 1996), a widely used benchmark for binary classification in the social sciences. We combine the training and test sets for a total of 48,842 observations. The outcome is whether an individual earns more than $50,000 per year. Features are organized into three panels: demographics (age, sex, years of education), work characteristics (hours worked per week, occupation group, work class), and financial variables (capital gains, capital losses, marital status). The specification search considers logistic regression and XGBoost across four specifications, allowing the method to be evaluated both where the parametric model is approximately correctly specified and where the black-box learner is substantially better fitted. Years of education takes the floored step size $h_j = 0.5$, so its effect is the one-year contrast; age and hours per week have enough spread that the adaptive rule already exceeds the floor, at $h_j = 0.686$ and 0.620. Trimming is negligible for every feature except capital gains and capital losses, which lose 92.3% and 95.3% of the sample for the reason discussed below.

Table 2 presents the specification search results for the UCI Adult Income dataset. This is the largest of the three applications at 48,842 observations, and the estimates are correspondingly the most precisely determined in the paper. The demographic panel tells a consistent story across all specifications and both model classes. Age and education have

Table 1: Window settings for the empirical applications. Step size h_j and the fraction of observations receiving trimming weight zero.

Dataset	Feature	h_j	Trimmed
UCI Adult Income	age	0.6855	1.3%
	female	—	0.0%
	education_num	0.5	1.4%
	hours_per_week	0.6196	0.3%
	government	—	0.0%
	self_employed	—	0.0%
	white_collar	—	0.0%
	blue_collar	—	0.0%
	service	—	0.0%
	capital_gain	373	92.3%
	capital_loss	20.15	95.3%
	married	—	0.0%
UCI Credit Card Default	age	0.5	0.2%
	female	—	0.0%
	education	0.5	36.8%
	married	—	0.0%
	avg_pay_status	0.04911	7.0%
	months_delayed	0.5	70.9%
	avg_bill_amt	3,163	0.0%
	avg_pay_amt	507	11.7%
	pay_ratio	0.01839	18.4%
Ames Housing	GrLivArea	26.27	0.1%
	BedroomAbvGr	0.5	0.5%
	FullBath	0.5	2.9%
	HalfBath	0.5	63.4%
	OverallQual	0.5	1.4%
	OverallCond	0.5	1.6%
	YearBuilt	1.51	1.4%
	YearRemodAdd	1.032	14.5%
	LotArea	499	1.2%
	GarageArea	10.69	5.6%
	nbhd_high	—	0.0%
	nbhd_mid	—	0.0%

Notes: h_j is the default adaptive step size, $\max(10^{-4}, 0.05\hat{\sigma}_j)$, floored at 0.5 for integer-valued features so that a count is contrasted over a whole unit. Features declared binary use the level contrast $f(x \mid x_j = 1) - f(x \mid x_j = 0)$, for which h_j is undefined and no trimming applies. The trimmed fraction is the share of observations lying within h_j of the boundary of the observed support of x_j, which receive weight zero; for those features the estimand is the window effect on the untrimmed subpopulation.

Table 2: UCI Adult Income: Specification Search. Average marginal effects on P(Income > $50k).

	Logistic				XGBoost			
	Spec A	A+B	A+C	A+B+C	Spec A	A+B	A+C	A+B+C
Panel A: Demographics								
age	0.0084***	0.0081***	0.0041***	0.0043***	0.0059***	0.0059***	0.0031***	0.0033***
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
female	-0.1749***	-0.1670***	-0.0122***	-0.0153**	-0.1604***	-0.1507***	-0.0089**	-0.0128**
	(0.0034)	(0.0045)	(0.0046)	(0.0061)	(0.0033)	(0.0041)	(0.0042)	(0.0056)
education_num	0.0464***	0.0351***	0.0371***	0.0284***	0.0606***	0.0458***	0.0417***	0.0325***
	(0.0007)	(0.0008)	(0.0006)	(0.0007)	(0.0007)	(0.0008)	(0.0006)	(0.0007)
Panel B: Work Characteristics								
hours_per_week	–	0.0035***	–	0.0027***	–	0.0084***	–	0.0038***
		(0.0002)		(0.0001)		(0.0002)		(0.0001)
government	–	0.0059	–	-0.0004	–	-0.0072	–	-0.0060
		(0.0067)		(0.0061)		(0.0066)		(0.0055)
self_employed	–	0.0218***	–	-0.0085**	–	0.0192***	–	-0.0080**
		(0.0058)		(0.0043)		(0.0056)		(0.0040)
white_collar	–	0.0649***	–	0.0561***	–	0.0480***	–	0.0478***
		(0.0053)		(0.0049)		(0.0051)		(0.0045)
blue_collar	–	-0.0246***	–	-0.0248***	–	-0.0482***	–	-0.0300***
		(0.0036)		(0.0031)		(0.0034)		(0.0028)
service	–	-0.0744***	–	-0.0456***	–	-0.0795***	–	-0.0342***
		(0.0024)		(0.0023)		(0.0024)		(0.0020)
Panel C: Financial								
capital_gain	–	–	0.0000***	0.0000***	–	–	0.0000***	0.0000***
			(0.0000)	(0.0000)			(0.0000)	(0.0000)
capital_loss	–	–	0.0000***	0.0000***	–	–	-0.0001***	-0.0000***
			(0.0000)	(0.0000)			(0.0000)	(0.0000)
married	–	–	0.2842***	0.2785***	–	–	0.2442***	0.2423***
			(0.0048)	(0.0054)			(0.0045)	(0.0047)
McFadden R^2	0.199	0.233	0.376	0.400	0.257	0.296	0.467	0.492
Accuracy	0.797	0.806	0.841	0.847	0.805	0.818	0.864	0.872

Notes: ***p<0.01, **p<0.05, *p<0.10. Estimates are the debiased cross-fitted window AME with $K = 5$ folds; standard errors from the orthogonal score in parentheses. Step sizes h_j and trimmed fractions are reported in Table 1.

positive effects on the probability of earning above $50,000, and the female indicator is large and negative in the specifications that omit marital status, at −0.175 for logistic regression and −0.160 for XGBoost, collapsing to −0.012 and −0.009 once marital status enters in specification A+C. That collapse is the single most dramatic omitted variable pattern in the paper, and it is legible only because the AME puts every specification on the same probability scale. Education runs from 0.046 to 0.028 for logistic regression and 0.061 to 0.033 for XGBoost as controls accumulate. Marital status itself carries much the largest effect in the table, 0.278 and 0.242 in the full specification.

The two model classes track each other closely throughout, and their standard errors are now of the same order everywhere — 0.0061 and 0.0056 on the female indicator in the full specification, 0.0007 and 0.0007 on education. Under the refitting bootstrap the black-box standard errors were sometimes an order of magnitude smaller than the parametric ones, which the earlier draft read as an efficiency gain from the better-fitting model. That reading was mistaken. The two models estimate the same $\theta_{j,h}$ at the same sample size, so their standard errors ought to be similar, and the earlier gap measured how stable each fitting procedure was under resampling rather than how precisely either pinned down the target. The influence-function standard errors do behave as sampling variability should: they shrink with n, they are largest for the features carrying the least independent variation, and they do not reward a model for being reproducibly wrong.

The financial panel illustrates the limits of what this design can say about capital gains and losses. Both variables are measured in raw dollars, so the reported effect is the response to a one-dollar change, and it displays as zero to four decimals in every cell but one — as it did under the plug-in. The estimates are nonetheless significant at the one percent level, because the standard errors are smaller still; with nearly fifty thousand observations a substantively negligible effect can be sharply distinguished from zero, and this panel is a reminder that the stars answer a different question from the one the reader usually cares about. They also carry by far the heaviest trimming in the paper. Both are zero for the overwhelming

majority of the sample and zero is the minimum of their observed support, so 92.3% and 95.3% of observations receive weight zero and the reported effect is identified only among filers who have nonzero gains or losses. The sign reversal between logistic regression and XGBoost that the earlier draft flagged in specification A+C does not survive the correction; both are zero to the displayed precision. Nothing here should be read as evidence about capital income, and a study interested in that question would want the variable re-expressed on a scale where a unit change is meaningful.

Table 3: UCI Adult Income: Model Comparison. Average marginal effects on P(Income > \$50k), full specification (A+B+C).

Feature	Logistic	Random Forest	XGBoost	Neural Net
age	0.0043***	0.0056***	0.0033***	0.0041***
	(0.0001)	(0.0001)	(0.0001)	(0.0001)
female	-0.0153**	-0.0210***	-0.0128**	-0.0166***
	(0.0061)	(0.0058)	(0.0056)	(0.0059)
education_num	0.0284***	0.0336***	0.0325***	0.0284***
	(0.0007)	(0.0007)	(0.0007)	(0.0007)
hours_per_week	0.0027***	0.0077***	0.0038***	0.0049***
	(0.0001)	(0.0001)	(0.0001)	(0.0001)
government	-0.0004	-0.0025	-0.0060	-0.0060
	(0.0061)	(0.0056)	(0.0055)	(0.0060)
self_employed	-0.0085**	-0.0103**	-0.0080**	-0.0071*
	(0.0043)	(0.0041)	(0.0040)	(0.0042)
white_collar	0.0561***	0.0606***	0.0478***	0.0383***
	(0.0049)	(0.0047)	(0.0045)	(0.0047)
blue_collar	-0.0248***	-0.0316***	-0.0300***	-0.0533***
	(0.0031)	(0.0030)	(0.0028)	(0.0031)
service	-0.0456***	-0.0262***	-0.0342***	-0.0758***
	(0.0023)	(0.0021)	(0.0020)	(0.0023)
capital_gain	0.0000***	0.0000***	0.0000***	0.0000***
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
capital_loss	0.0000***	-0.0000***	-0.0000***	0.0000***
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
married	0.2785***	0.2504***	0.2423***	0.2438***
	(0.0054)	(0.0048)	(0.0047)	(0.0048)
McFadden R^2	0.400	0.450	0.492	0.449
Accuracy	0.847	0.863	0.872	0.858

Notes: ***p<0.01, **p<0.05, *p<0.10. Estimates are the debiased cross-fitted window AME with $K = 5$ folds; standard errors from the orthogonal score in parentheses. Step sizes h_j and trimmed fractions are reported in Table 1.

Table 3 compares all four model classes on the full specification, and with nearly fifty thousand observations it is the cleanest test in the paper of whether the correction delivers a common estimand. It does. The education AME is 0.028, 0.034, 0.033 and 0.028 for logistic regression, random forest, XGBoost and the neural network, and marital status is 0.279, 0.250, 0.242 and 0.244 — a spread of 0.005 and 0.036 respectively across four architectures with McFadden R^2 ranging from 0.400 to 0.492. The female indicator, which the plug-in spread from -0.027 down to -0.009 and left insignificant for the neural network, is -0.015, -0.021, -0.013 and -0.017 after correction, all four significant and all four within a standard error or two of one another. The variable the earlier draft singled out as sensitive to model specification turns out not to have been.

The standard errors tell the same story from the other side. For a common target at a common sample size, four estimators using different nuisances should report standard errors of similar magnitude, and here they do: 0.0056 to 0.0061 on the female indicator, 0.0047 to 0.0054 on marital status, 0.0020 to 0.0023 on the service indicator. Under the plug-in they did not. The random forest reported 0.0001 on the female indicator and 0.0006 on marital status, one to two orders of magnitude below every other model, which the earlier draft attributed to ensemble averaging producing a smoother prediction surface and therefore more stable finite differences across bootstrap replicates. That description was accurate as a description of the bootstrap and wrong as a statement about precision: the replicates agreed with each other because every one of them was regularized the same way, not because the quantity was well determined. The convergence of the four standard-error columns to a common magnitude is what the orthogonal score is supposed to produce, and it is the more convincing for happening at the sample size where the asymptotics have the best claim to apply.

Table 4 compares AMEs against SHAP values and partial dependence plot slopes across all four models. The SHAP column measures something else and is reported here to make the difference concrete rather than to adjudicate between the two. Mean signed SHAP values

Table 4: UCI Adult Income: Method Comparison. AME, SHAP, and PDP estimates for P(Income > $50k), full specification (A+B+C).

	Logistic			Random Forest			XGBoost			Neural Net		
	AME	SHAP	PDP	AME	SHAP	PDP	AME	SHAP	PDP	AME	SHAP	PDP
age	0.0043	-0.0270	0.0028	0.0056	-0.0068	0.0024	0.0033	-0.0097	0.0036	0.0041	0.0044	0.0033
female	-0.0153	0.0022	-0.0270	-0.0210	0.0011	-0.0218	-0.0128	0.0015	-0.0122	-0.0166	-0.0001	-0.0222
education_num	0.0284	-0.1186	0.0269	0.0336	-0.0077	0.0137	0.0325	-0.0093	0.0170	0.0284	-0.0051	0.0207
hours_per_week	0.0027	-0.0691	0.0033	0.0077	-0.0054	0.0020	0.0038	-0.0080	0.0028	0.0049	-0.0032	0.0028
government	-0.0004	-0.0009	0.0026	-0.0025	-0.0003	0.0051	-0.0060	-0.0008	-0.0003	-0.0060	0.0002	0.0082
self_employed	-0.0085	0.0016	-0.0265	-0.0103	-0.0003	0.0032	-0.0080	-0.0002	-0.0055	-0.0071	-0.0008	-0.0045
white_collar	0.0561	0.0127	0.0538	0.0606	0.0031	0.0600	0.0478	0.0025	0.0474	0.0383	0.0007	0.0312
blue_collar	-0.0248	0.0136	-0.0273	-0.0316	0.0010	-0.0240	-0.0300	0.0004	-0.0251	-0.0533	-0.0008	-0.0531
service	-0.0456	-0.0027	-0.0427	-0.0262	0.0003	-0.0176	-0.0342	0.0005	-0.0343	-0.0758	0.0011	-0.0558
capital_gain	0.0000	-0.2627	0.0000	0.0000	-0.0224	0.0000	0.0000	-0.0264	-0.0001	0.0000	0.0042	0.0000
capital_loss	0.0000	0.0027	–	-0.0000	-0.0034	–	-0.0000	-0.0062	–	0.0000	-0.0115	–
married	0.2785	-0.0768	0.2749	0.2504	-0.0064	0.2446	0.2423	-0.0049	0.2357	0.2438	0.0149	0.1958

Notes: AME = Average Marginal Effect (marginfx). SHAP = mean signed SHAP value (GradientExplainer). PDP = slope of partial dependence plot. Full specification (A+B+C) only. No standard errors reported for SHAP and PDP.

are on a different scale entirely and in several cases carry the opposite sign: for logistic regression the SHAP value for education is -0.119 against an AME of 0.028, and for marital status it is -0.077 against an AME of 0.279 — opposite in sign on the two largest effects in the application. The SHAP value for age is negative for three of the four models while every AME is positive. The most striking entry is capital gains, whose SHAP value of -0.263 for logistic regression is the largest SHAP magnitude anywhere in the table, attached to a variable whose AME is zero to four decimals and 92% of whose observations are trimmed. None of this indicates an error in SHAP or in its implementation. A Shapley value is an additive attribution of a prediction relative to a reference distribution, and its average over the sample is not an estimate of the average rate of change of the prediction in a feature. The two quantities answer different questions and there is no reason to expect them to agree in magnitude or in sign. The point of the comparison is interpretive rather than evaluative: SHAP summaries are frequently read as though larger values meant larger effects on the outcome, and that reading is not supported by what the quantity measures.

The partial dependence column behaves as it did in the other two applications: it agrees with the debiased AME where the underlying plug-in is nearly unbiased and falls short of it elsewhere. For logistic regression, the best-specified model here, agreement is close on the large effects — marital status 0.279 against 0.275, the white collar indicator 0.056 against 0.054, service -0.046 against -0.043, education 0.028 against 0.027. For the two tree ensembles the same features separate substantially: the random forest education AME of 0.034 sits against a PDP slope of 0.014, and its hours-per-week AME of 0.008 against a slope of 0.002. A piecewise constant fit has little local slope to report between splits, and the partial dependence average inherits that flatness while the orthogonal correction does not. Partial dependence slopes for capital losses are unavailable for every model, a limitation of the grid construction when almost all mass sits at a single value. As with SHAP, no standard errors accompany either column, so a researcher working from them has point estimates of unknown precision and no way to test whether any variable has a statistically distinguishable

effect.

4.2 UCI Credit Default

The second dataset is the UCI Default of Credit Card Clients dataset (Yeh, 2009; Dua and Graff, 2019), comprising 30,000 observations of Taiwanese credit card holders. The outcome is whether a cardholder defaulted on their payment in the following month. Features are organized into demographics (age, sex, education, marital status), payment history (average payment status, months delayed), and bill and payment amounts (average bill amount, average payment amount, payment ratio). The specification search considers logistic regression and a TensorFlow neural network, providing a direct comparison between a classical parametric model and a deep learning approach on the same inferential task. Two features here carry substantial trimming. Months delayed takes seven integer values and 70.9% of the sample sits on one of the two outer ones, so its effect is identified among cardholders strictly inside that range; education takes four levels and is trimmed at 36.8%. Both take the floored step size $h_j = 0.5$, so their reported effects are one-level contrasts. The payment ratio, which turns out to carry the largest effect in the application, is continuous and trimmed at 18.4%.

Table 5 presents the specification search. The dominant finding is the importance of payment history and of the payment ratio. Months delayed enters at 0.035 for logistic regression and 0.030 for the neural network in specification A+B and remains at 0.030 and 0.025 in the full specification, while the payment ratio, once it is available in specifications A+C and A+B+C, is much the largest effect in the table: -0.265 for logistic regression and -0.194 for the neural network in the full specification. The demographic panel tells a different story from the Adult Income dataset: age is statistically indistinguishable from zero in nearly every specification for both models, and the female indicator, while consistently negative and significant, shrinks substantially once payment history is controlled for, falling from -0.036 to -0.020 for logistic regression and from -0.036 to -0.018 for the neural network. This pattern of attenuation is cleanly legible in the AME framework because the

Table 5: UCI Credit Card Default: Specification Search. Average marginal effects on P(Default).

	Logistic				Neural Net			
	Spec A	A+B	A+C	A+B+C	Spec A	A+B	A+C	A+B+C
Panel A: Demographics								
age	-0.0015***	-0.0002	-0.0005	0.0001	-0.0026***	-0.0006*	-0.0010***	-0.0000
	(0.0004)	(0.0003)	(0.0004)	(0.0003)	(0.0004)	(0.0003)	(0.0004)	(0.0003)
female	-0.0358***	-0.0206***	-0.0291***	-0.0198***	-0.0364***	-0.0205***	-0.0265***	-0.0176***
	(0.0050)	(0.0046)	(0.0049)	(0.0046)	(0.0050)	(0.0046)	(0.0049)	(0.0045)
education	-0.0001	-0.0027	-0.0054**	-0.0040*	-0.0008	-0.0035	-0.0048**	-0.0040*
	(0.0022)	(0.0022)	(0.0023)	(0.0022)	(0.0021)	(0.0021)	(0.0022)	(0.0022)
married	0.0253***	0.0189***	0.0303***	0.0206***	0.0260***	0.0152**	0.0269***	0.0186***
	(0.0068)	(0.0064)	(0.0066)	(0.0063)	(0.0068)	(0.0064)	(0.0067)	(0.0063)
Panel B: Payment History								
avg_pay_status	–	-0.0140***	–	-0.0218***	–	0.0073*	–	0.0040
		(0.0041)		(0.0075)		(0.0041)		(0.0072)
months_delayed	–	0.0347***	–	0.0301***	–	0.0298***	–	0.0254***
		(0.0011)		(0.0012)		(0.0011)		(0.0012)
Panel C: Bill and Payment Amounts								
avg_bill_amt	–	–	-0.0000***	-0.0000**	–	–	-0.0000	-0.0000*
			(0.0000)	(0.0000)			(0.0000)	(0.0000)
avg_pay_amt	–	–	-0.0000***	0.0000	–	–	-0.0000***	-0.0000***
			(0.0000)	(0.0000)			(0.0000)	(0.0000)
pay_ratio	–	–	-0.3521***	-0.2651***	–	–	-0.2572***	-0.1943**
			(0.0731)	(0.0730)			(0.0792)	(0.0773)
McFadden R^2	0.003	0.130	0.031	0.137	0.010	0.155	0.055	0.166
Accuracy	0.779	0.803	0.779	0.803	0.779	0.808	0.779	0.808

Notes: ***p<0.01, **p<0.05, *p<0.10. Estimates are the debiased cross-fitted window AME with $K = 5$ folds; standard errors from the orthogonal score in parentheses. Step sizes h_j and trimmed fractions are reported in Table 1.

estimates are on a common probability scale across specifications.

What is most striking in this table is how little separates the two columns. In specification A the female AME is -0.0358 for logistic regression and -0.0364 for the neural network; married is 0.0253 and 0.0260; in the full specification the pair is -0.0198 and -0.0176, and 0.0206 and 0.0186. A parametric model and a four-layer network, differing in McFadden R^2 by three percentage points, produce estimates that agree to the third decimal place across sixteen model-specification cells. This is what the orthogonal score is supposed to deliver and what the plug-in did not: since each column estimates the same $\theta_{j,h}$ and the correction absorbs the first-order effect of the fitted regression, the remaining gap between the two columns reflects only the second-order term and sampling noise. The standard errors are also now nearly identical between the two models — 0.0046 and 0.0045 on the female indicator, 0.0012 and 0.0012 on months delayed — rather than differing by the order of magnitude the refitting bootstrap reported, where the network's standard error for months delayed was 0.032 against the logistic model's 0.002. Both figures were describing the stability of a particular fitting procedure under resampling rather than the sampling variability of an estimate of a fixed target.

Table 6 compares all four models on the full specification, and it is the sharpest illustration in the paper of what the orthogonal correction changes. Under the plug-in the four models flatly contradicted one another on the two variables that matter. Months delayed was 0.083 for logistic regression, 0.018 for the random forest, 0.021 for XGBoost and 0.112 for the neural network, a sixfold spread; the payment ratio was -0.431 and -0.440 for the two tree ensembles, -0.045 for logistic regression, and 0.0009 — indistinguishable from no effect at all — for the neural network. A researcher reading that table had to conclude either that the models were detecting genuinely different structure, or that at least three of the four were wrong, with no way to tell which.

The debiased estimates resolve the contradiction rather than splitting the difference. Months delayed becomes 0.030, 0.013, 0.019 and 0.025, and the payment ratio becomes

Table 6: UCI Credit Card Default: Model Comparison. Average marginal effects on P(Default), full specification (A+B+C).

Feature	Logistic	Random Forest	XGBoost	Neural Net
age	0.0001	-0.0002	-0.0005	-0.0000
	(0.0003)	(0.0003)	(0.0003)	(0.0003)
female	-0.0198***	-0.0161***	-0.0175***	-0.0176***
	(0.0046)	(0.0045)	(0.0045)	(0.0045)
education	-0.0040*	-0.0018	0.0034	-0.0040*
	(0.0022)	(0.0022)	(0.0022)	(0.0022)
married	0.0206***	0.0222***	0.0205***	0.0186***
	(0.0063)	(0.0062)	(0.0062)	(0.0063)
avg_pay_status	-0.0218***	0.0061	0.0907***	0.0040
	(0.0075)	(0.0074)	(0.0075)	(0.0072)
months_delayed	0.0301***	0.0134***	0.0191***	0.0254***
	(0.0012)	(0.0011)	(0.0011)	(0.0012)
avg_bill_amt	-0.0000**	-0.0000***	-0.0000***	-0.0000*
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
avg_pay_amt	0.0000	-0.0000***	-0.0000***	-0.0000***
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
pay_ratio	-0.2651***	-0.4890***	-0.5836***	-0.1943**
	(0.0730)	(0.0670)	(0.0685)	(0.0773)
McFadden R^2	0.137	0.210	0.194	0.166
Accuracy	0.803	0.821	0.815	0.808

Notes: ***p<0.01, **p<0.05, *p<0.10. Estimates are the debiased cross-fitted window AME with $K = 5$ folds; standard errors from the orthogonal score in parentheses. Step sizes h_j and trimmed fractions are reported in Table 1.

-0.265, -0.489, -0.584 and -0.194: every model now agrees that a higher payment ratio is associated with a substantially lower probability of default, including the neural network, whose plug-in estimate had put the effect at zero. The demographic panel converges the same way. The female AME was -0.020, -0.006, -0.015 and -0.014 under the plug-in and is -0.0198, -0.0161, -0.0175 and -0.0176 under the correction; married was 0.022, 0.005, 0.014 and 0.019 and is now 0.0206, 0.0222, 0.0205 and 0.0186. The variation that the earlier draft read as four models weighing the covariates differently was in the main the regularization bias $\mathbb{E}[\alpha_h(\hat{f} - f)]$, which differs across learners because their regularization differs, and which the correction removes. What survives is a common estimate of a common target.

Convergence is not complete, and the payment ratio shows where it stops. The four debiased estimates still span -0.194 to -0.584, a range far wider than their standard errors of roughly 0.07, so the models have not been brought into agreement to within sampling error on that variable. Payment ratio is the most skewed feature in the design and is trimmed at 18.4%, and the second-order product term that Theorem 1 requires to be $o_p(n^{-1/2})$ is plausibly not negligible here for the weaker nuisance fits. The correct reading is that the correction removes the first-order bias and leaves a residual that the theory bounds but does not eliminate, which is visible precisely where the nuisance problem is hardest. Finally, the random forest standard errors, which the plug-in bootstrap reported as 0.0000 to four decimal places on the female indicator, are 0.0045 under the influence-function formula — essentially identical to the other three models, as they should be for a common estimand at a common sample size.

Table 7 again shows the two quantities parting company. For the payment history variables the separation is large: the AME for months delayed is 0.030 for logistic regression while the mean signed SHAP value is -0.015, opposite in sign. The payment ratio illustrates the most consequential divergence: the random forest and XGBoost AMEs are -0.489 and -0.584, indicating that a higher payment ratio is associated with a large reduction in default

Table 7: UCI Credit Card Default: Method Comparison. AME, SHAP, and PDP estimates for P(Default), full specification (A+B+C).

	Logistic			Random Forest			XGBoost			Neural Net		
	AME	SHAP	PDP	AME	SHAP	PDP	AME	SHAP	PDP	AME	SHAP	PDP
age	0.0001	-0.0003	0.0002	-0.0002	0.0011	-0.0000	-0.0005	0.0004	-0.0002	-0.0000	-0.0010	0.0012
female	-0.0198	-0.0005	-0.0198	-0.0161	0.0001	-0.0062	-0.0175	0.0002	-0.0146	-0.0176	0.0009	-0.0200
education	-0.0040	0.0002	0.0002	-0.0018	0.0006	-0.0057	0.0034	-0.0005	-0.0356	-0.0040	-0.0016	0.0082
married	0.0206	-0.0038	0.0220	0.0222	0.0002	0.0054	0.0205	0.0002	0.0142	0.0186	0.0022	0.0217
avg_pay_status	-0.0218	-0.0144	-0.0246	0.0061	0.0029	0.0610	0.0907	0.0027	0.0404	0.0040	-0.0116	0.0097
months_delayed	0.0301	-0.0148	0.1107	0.0134	-0.0008	0.0456	0.0191	0.0001	0.0404	0.0254	-0.0154	0.0653
avg_bill_amt	-0.0000	-0.0001	0.0000	-0.0000	-0.0018	-0.0000	-0.0000	-0.0046	-0.0000	-0.0000	0.0036	0.0000
avg_pay_amt	0.0000	0.0026	-0.0000	-0.0000	-0.0006	-0.0000	-0.0000	-0.0005	-0.0000	-0.0000	-0.0025	-0.0000
pay_ratio	-0.2651	0.0100	-0.0444	-0.4890	0.0012	-0.0998	-0.5836	0.0005	-0.0953	-0.1943	0.0015	-0.0170

Notes: AME = Average Marginal Effect (marginfx). SHAP = mean signed SHAP value (GradientExplainer). PDP = slope of partial dependence plot. Full specification (A+B+C) only. No standard errors reported for SHAP and PDP.

probability, while the corresponding SHAP values are 0.0012 and 0.0005, near zero and of the opposite sign. This is again a difference of estimand rather than a defect: a feature can carry a large average derivative while contributing little on average to the displacement of predictions from their baseline. The practical concern is that a researcher reading the SHAP column as a marginal effect would conclude that the payment ratio is essentially irrelevant to default risk, and, with no standard errors attached to either the SHAP or the PDP column, would have no statistical basis on which to check that conclusion.

The partial dependence column separates from the AME column here in the same pattern as in the Ames application, and for the same reason: a partial dependence slope is a plug-in average over a single full-sample fit, so it carries the regularization bias that the correction removes. The separation tracks the quality of the underlying fit. On the female indicator, where the logistic model is close to correctly specified, the AME of -0.0198 and the PDP slope of -0.0198 agree exactly; for the random forest on the same variable they are -0.0161 and -0.0062. On the payment ratio, the hardest nuisance problem in the design, the gap is largest and always toward zero: -0.265 against -0.044 for logistic regression, -0.489 against -0.100 for the random forest, -0.584 against -0.095 for XGBoost, -0.194 against -0.017 for the neural network. Months delayed runs the other way, with PDP slopes of 0.111, 0.046, 0.040 and 0.065 against debiased AMEs of 0.030, 0.013, 0.019 and 0.025, which is what one expects when an uncorrected estimator distributes a shared effect between two correlated features differently than the corrected one does. Taken across all three applications, the attenuation is the dominant pattern rather than a universal one: restricting attention to the effects the debiased estimator resolves away from zero, the partial dependence slope is smaller in absolute value than the AME in 62 of 84 cases, with a median ratio of 0.47 on Ames Housing, 0.60 here and 0.95 on Adult Income. It is most pronounced for the random forest, the learner whose regularization bias the simulations isolate.

4.3 Ames Housing

The third dataset is the Ames Housing dataset (De Cock, 2011), comprising approximately 1,460 observations of residential home sales in Ames, Iowa between 2006 and 2010. The outcome is sale price in dollars, so average marginal effects are interpreted in dollar units — a one-unit change in a continuous feature is associated with a $\widehat{\theta}_{j,h}$ change in predicted sale price, on average across the untrimmed sample. Features are organized into physical characteristics (above-ground living area, number of bedrooms, number of bathrooms), quality and condition (overall quality rating, condition rating, year built, year remodeled), and lot and location (lot area, garage area, neighborhood tier indicators). The specification search considers linear regression and a random forest, covering the regression case and completing the sweep across outcome types. Six of the ten continuous features here are integer counts or ratings and so take the floored step size $h_j = 0.5$, which makes their reported effects one-unit contrasts: one more bedroom, one more point of overall quality. One feature requires a narrower reading than the rest. Half bathrooms take only the values zero, one and two, and both outer levels lie on the boundary of the observed support, so 63.4% of the sample is trimmed and the half-bath effect is identified only among homes that have exactly one. The remaining features in this application are trimmed at 15% or below.

Table 8 presents the specification search. The regression setting allows AMEs to be interpreted directly in dollar units, which makes the substantive findings immediately legible. Above-ground living area carries an AME of \$139 per square foot in the physical-characteristics specification for linear regression, falling to \$106, \$85 and finally \$68 as quality, condition, and location controls are added — a pattern of positive omitted variable bias that makes intuitive sense, since larger homes tend to be in better neighborhoods and of higher quality. The random forest traces the same path from \$129 to \$81. That the two model classes track each other this closely across the sequence is the first thing to note about the debiased estimates: because the orthogonal correction removes the first-order dependence on the fitted regression, the two columns are estimating the same window effect rather than

Table 8: Ames Housing: Specification Search. Average marginal effects on Sale Price (USD).

	Linear				Random Forest			
	Spec A	A+B	A+C	A+B+C	Spec A	A+B	A+C	A+B+C
Panel A: Physical Characteristics								
GrLivArea	138.96***	106.16***	84.82***	68.05***	128.77***	103.42***	90.19***	81.50***
	(20.51)	(15.58)	(14.25)	(12.14)	(13.06)	(8.98)	(8.97)	(8.77)
BedroomAbvGr	-33209.66***	-14046.40*	-24282.55***	-26454.60*	-28632.01***	-11415.18***	-11292.79**	-9212.02
	(5594.51)	(7435.20)	(9027.62)	(14881.54)	(3382.73)	(3920.43)	(5576.82)	(8925.07)
FullBath	18589.00***	-10533.55**	7261.14	-10578.42	16236.01***	-6616.40*	5652.46	7858.42
	(5990.54)	(4912.27)	(4987.91)	(9969.12)	(3779.18)	(3670.56)	(3754.82)	(8988.79)
HalfBath	-11228.94**	-8180.81**	-9092.88**	1028.31	-7189.23*	-6638.05***	-11578.88**	-10317.96**
	(4624.11)	(3242.49)	(4451.09)	(5697.41)	(4264.04)	(2525.05)	(4601.22)	(4713.92)
Panel B: Quality and Condition								
OverallQual	–	15819.18***	–	5436.28	–	11911.79***	–	16255.82***
		(1589.02)		(4964.23)		(1342.87)		(4829.24)
OverallCond	–	6477.48*	–	2890.60	–	8281.97***	–	9190.11***
		(3687.03)		(5496.46)		(2434.71)		(3492.01)
YearBuilt	–	443.39	–	-141.08	–	1033.46***	–	1323.39***
		(277.71)		(404.21)		(190.55)		(419.89)
YearRemodAdd	–	192.69	–	1743.45**	–	556.20***	–	-311.44
		(161.82)		(740.09)		(173.98)		(753.53)
Panel C: Lot and Location								
LotArea	–	–	5.51	23.02**	–	–	-7.53	-12.51
			(3.93)	(11.37)			(8.06)	(11.14)
GarageArea	–	–	40.08*	-102.78	–	–	103.31***	90.33
			(20.94)	(71.57)			(34.64)	(59.74)
nbhd_high	–	–	128801.76***	89812.31**	–	–	66978.55***	43512.42*
			(34103.92)	(38393.43)			(16110.93)	(24635.87)
nbhd_mid	–	–	77339.38**	74336.31*	–	–	42305.81***	33767.43
			(33663.73)	(39182.81)			(15387.40)	(26676.67)
R^2	0.584	0.758	0.739	0.802	0.829	0.937	0.926	0.953
RMSE	51,243.9	39,080.1	40,596.4	35,355.5	32,830.9	19,881.6	21,534.2	17,244.4

Notes: ***p<0.01, **p<0.05, *p<0.10. Estimates are the debiased cross-fitted window AME with $K = 5$ folds; standard errors from the orthogonal score in parentheses. Step sizes h_j and trimmed fractions are reported in Table 1.

reporting the slope each model happens to believe, and the linear column is no longer simply the ordinary least squares coefficient. The standard errors are now of comparable size across the two model classes — \$12 for linear regression and \$9 for the random forest on GrLivArea in the full specification — rather than differing by the order of magnitude that the refitting bootstrap produced, where the ensemble's smooth prediction surface made every replicate agree with every other and drove the reported standard error to \$1.93. Those intervals were narrow around a biased center; the influence-function intervals are wider and honest.

The cost of that honesty is visible in the quality panel. The overall quality rating is estimated at \$15,819 per point for linear regression and \$11,912 for the random forest in specification A+B, both precisely determined, but in the full specification the linear estimate falls to \$5,436 with a standard error of \$4,964 and is no longer distinguishable from zero, while the random forest gives \$16,256 with a standard error of \$4,829. With $n = 1{,}460$ observations, twelve features and a second-order sieve for the representer, the full specification is close to the limit of what this sample can support: the estimator reports that limit rather than concealing it. The neighborhood tier indicators show the same attenuation as before — the high neighborhood effect falls from \$128,802 to \$89,812 for linear regression and from \$66,979 to \$43,512 for the random forest as quality controls are added — but with standard errors near \$25,000 to \$38,000, the attenuation is a pattern in the point estimates rather than a difference the data can resolve.

Table 9 compares all four models on the full specification. All four are well fitted, with R^2 ranging from 0.802 for linear regression to 0.954 for XGBoost, the random forest at 0.953 and the neural network at 0.932 with normalization. The GrLivArea AME is \$68, \$81, \$70, and \$66 for linear regression, random forest, XGBoost, and neural network respectively — a range of \$15 across four models that differ fundamentally in their functional form, and all four intervals overlap comfortably. Under the debiased estimator this agreement means something different from what it meant under the plug-in. Every column is now an estimate of the same $\theta_{j,h}$, so agreement is evidence that the four nuisance estimates are all good

Table 9: Ames Housing: Model Comparison. Average marginal effects on Sale Price (USD), full specification (A+B+C).

Feature	Linear	Random Forest	XGBoost	Neural Net
GrLivArea	68.05***	81.50***	70.27***	66.20***
	(12.14)	(8.77)	(6.60)	(6.24)
BedroomAbvGr	-26454.60*	-9212.02	-3464.31	-12638.60***
	(14881.54)	(8925.07)	(4508.19)	(3830.14)
FullBath	-10578.42	7858.42	4761.27	-5055.91
	(9969.12)	(8988.79)	(7661.72)	(5080.88)
HalfBath	1028.31	-10317.96**	-10906.00***	-2175.44
	(5697.41)	(4713.92)	(3940.81)	(3286.03)
OverallQual	5436.28	16255.82***	14913.45***	8524.39***
	(4964.23)	(4829.24)	(4271.98)	(2394.41)
OverallCond	2890.60	9190.11***	12102.59***	7234.42**
	(5496.46)	(3492.01)	(2519.57)	(2920.49)
YearBuilt	-141.08	1323.39***	1352.73***	230.04
	(404.21)	(419.89)	(366.04)	(267.61)
YearRemodAdd	1743.45**	-311.44	-289.89	942.55***
	(740.09)	(753.53)	(673.39)	(351.83)
LotArea	23.02**	-12.51	-10.83	10.04**
	(11.37)	(11.14)	(9.91)	(4.81)
GarageArea	-102.78	90.33	92.29*	-6.11
	(71.57)	(59.74)	(49.36)	(27.32)
nbhd_high	89812.31**	43512.42*	33476.93**	35458.39**
	(38393.43)	(24635.87)	(16103.49)	(17705.63)
nbhd_mid	74336.31*	33767.43	17468.12	36951.86**
	(39182.81)	(26676.67)	(18286.73)	(17998.56)
R^2	0.802	0.953	0.954	0.932
RMSE	35,355.5	17,244.4	17,036.6	20,699.9

Notes: ***p<0.01, **p<0.05, *p<0.10. Estimates are the debiased cross-fitted window AME with $K = 5$ folds; standard errors from the orthogonal score in parentheses. Step sizes h_j and trimmed fractions are reported in Table 1.

enough for the correction to do its work, not evidence that four different models happen to have similar slopes. Disagreement, correspondingly, is now diagnostic of nuisance quality rather than of functional form.

The most consequential change from the plug-in results is in the standard errors, and it runs in the opposite direction to the one the refitting bootstrap suggested. Under that bootstrap the neural network's standard errors were implausibly tight — on the order of single dollars for several variables, \$7 for bedrooms and \$8 for overall quality — because resampling propagated uncertainty through the model fitting but the smooth differentiable surface produced nearly identical finite differences across replicates. The influence-function standard errors for the same quantities are \$3,830 and \$2,394. The random forest shows the same correction: the high-neighborhood standard error rises from \$256 to \$24,636. Those earlier figures were not precision but the bootstrap's inability to see a bias that every replicate shared, which is the failure mode of Remark 4, and their disappearance is the clearest single indication in the empirical results that the orthogonal score is doing what it was constructed to do.

Read with those intervals attached, the bedroom variable is more equivocal than the plug-in made it look. The AME is −\$26,455 for linear regression, −\$9,212 for the random forest, −\$3,464 for XGBoost, and −\$12,639 for the neural network. All four are negative, consistent with the well-known hedonic finding that holding square footage fixed, more bedrooms means smaller rooms, but only the neural network estimate is significant at conventional levels and the linear estimate only marginally so. The plug-in reported all four as significant with standard errors between \$7 and \$3,042. The honest reading is that the sign is stable across architectures while this sample cannot pin down the magnitude, which is a weaker conclusion than the paper could previously state and a better supported one.

Table 10 makes the difference between the two quantities easiest to see, since the units are dollars and the magnitudes are large enough to be concrete. SHAP values are unavailable for the neural network in this table, reflecting a limitation of the GradientExplainer

Table 10: Ames Housing: Method Comparison. AME, SHAP, and PDP estimates for Sale Price (USD), full specification (A+B+C).

	Linear			Random Forest			XGBoost			Neural Net		
	AME	SHAP	PDP	AME	SHAP	PDP	AME	SHAP	PDP	AME	SHAP	PDP
GrLivArea	68.05	3423.49	65.57	81.50	61.69	38.64	70.27	-971.88	39.06	66.20	–	58.62
BedroomAbvGr	-26454.60	145.82	-6189.02	-9212.02	90.87	-1272.27	-3464.31	158.44	-4456.61	-12638.60	–	-9294.43
FullBath	-10578.42	31.01	-6287.42	7858.42	34.06	543.63	4761.27	54.86	193.04	-5055.91	–	2427.62
HalfBath	1028.31	-298.76	-6967.90	-10317.96	50.46	-125.75	-10906.00	30.53	-1495.20	-2175.44	–	-2723.69
OverallQual	5436.28	1635.24	16465.22	16255.82	596.70	20776.14	14913.45	713.13	20130.92	8524.39	–	10802.73
OverallCond	2890.60	980.44	6311.48	9190.11	161.26	2030.44	12102.59	-481.60	5071.47	7234.42	–	6365.50
YearBuilt	-141.08	-415.01	440.47	1323.39	288.29	209.05	1352.73	569.00	397.33	230.04	–	421.27
YearRemodAdd	1743.45	86.86	101.51	-311.44	373.04	118.06	-289.89	499.84	152.00	942.55	–	561.16
LotArea	23.02	267.44	0.61	-12.51	-183.50	1.33	-10.83	216.28	2.38	10.04	–	2.35
GarageArea	-102.78	93.22	38.68	90.33	-884.31	23.77	92.29	-704.59	28.98	-6.11	–	41.48
nbhd_high	89812.31	77.50	51433.02	43512.42	-506.36	5058.64	33476.93	50.70	20307.34	35458.39	–	12399.98
nbhd_mid	74336.31	266.38	7508.07	33767.43	40.68	5949.97	17468.12	-134.71	9744.97	36951.86	–	11392.44

Notes: AME = Average Marginal Effect (marginfx). SHAP = mean signed SHAP value (GradientExplainer). PDP = slope of partial dependence plot. Full specification (A+B+C) only. No standard errors reported for SHAP and PDP.

implementation for this architecture. For the other three models the two columns are far apart. The SHAP value for GrLivArea is \$3,423 for the linear model against an AME of \$68, roughly fifty times larger — which is what one should expect, since the SHAP figure is the average absolute displacement a typical house's living area contributes to its predicted price, not the price change associated with one additional square foot. The SHAP values for garage area (−\$884), lot area (−\$183), and the high neighborhood indicator (−\$506) are negative for the random forest while the AME estimates are positive for all three. And as before, no standard errors attach to the SHAP column, so a reader who did want to treat those figures as effects would have no way to gauge their precision.

The partial dependence column requires a different reading than it did under the plug-in, and the change is itself informative. A partial dependence slope is computed by perturbing a single full-sample fit and averaging, which makes it a plug-in average of exactly the kind the orthogonal correction is designed to repair; it carries no correction term and no standard error. Where the plug-in is nearly unbiased the two therefore continue to agree closely: on above-ground living area, the best supported and smoothest feature in the data, the AME is \$68 and the PDP slope \$66 for linear regression, \$66 and \$59 for the neural network. Where the regularization bias bites, they separate, and they separate in the direction the theory predicts — toward zero for the uncorrected quantity. The bedroom AME of −\$26,455 for linear regression sits against a PDP slope of −\$6,189; the high-neighborhood AME of \$43,512 for the random forest sits against a PDP slope of \$5,059, and the mid-neighborhood AME of \$33,767 against \$5,950. These are the count and binary contrasts, where the fitted surface is flattest between levels and where the plug-in has the most to lose. The pattern that once read as mutual corroboration between two estimators now reads as a measurement of the bias separating them, which is the same quantity Simulation 2 isolates in the coverage of the refitting bootstrap. The Ames Housing results complete the sweep across outcome types: the framework produces dollar-denominated marginal effects with valid standard errors for a continuous regression outcome, using exactly the same estimation procedure as the binary

classification examples, with no modification required.

5 Simulation

We evaluate the estimator through three simulation studies, which together separate what the window average recovers from what can be said about it. Simulation 1 examines the bias and root mean squared error of the uncorrected plug-in window average as a function of sample size, across three data generating processes and two outcome types. Simulation 2 asks whether the refitting bootstrap of Section 3.4 delivers valid intervals for that plug-in, comparing empirical coverage of nominal 95% intervals against the target. Simulation 3 repeats both exercises for the debiased cross-fitted estimator of Section 3.4, holding the designs, learners and windows fixed so that the two estimators differ only in the orthogonal correction. The three are best read in order: the first locates a bias, the second shows that resampling cannot see it, and the third removes it. All features x_1, x_2, x_3, x_4 are drawn independently from $N(0,1)$. Features x_1 and x_2 have nonzero true effects; x_3 and x_4 are pure noise with true AME of zero. Three data generating processes are considered. The linear DGP sets $\eta = 2x_1 + 3x_2$. The nonlinear DGP sets $\eta = 2x_1^2 + 3x_2$, so that the true AME of x_1 is zero in the regression case by symmetry of $N(0,1)$ — $\mathbb{E}[4x_1] = 0$ — but nonzero in the classification case where the sigmoid breaks that symmetry. The interaction DGP sets $\eta = 2x_1 + 3x_2 + 2x_1x_2$, which preserves the marginal AMEs of x_1 and x_2 at 2.0 and 3.0 respectively in the regression case since $\mathbb{E}[2x_2] = \mathbb{E}[2x_1] = 0$. For regression, the outcome is $y = \eta + \varepsilon$ where $\varepsilon \sim N(0,1)$. For classification, $P(y = 1) = \sigma(\eta)$ where σ denotes the logistic function. True AMEs for all DGPs and outcome types are computed via Monte Carlo integration using $n = 1{,}000{,}000$ observations drawn from the true prediction function directly, treating the result as ground truth. Sample sizes of $n \in \{250, 500, 1\,000, 2\,500, 5\,000\}$ are evaluated over 500 Monte Carlo iterations for classification and 1,000 iterations for regression.

The simulation study fits four model classes with hyperparameters chosen to balance

computational tractability against representativeness of typical applied use. The random forest uses 100 trees with a maximum depth of 6, and XGBoost uses 100 boosting rounds with a maximum depth of 4 and a learning rate of 0.05 — conservative settings that prevent overfitting on the small sample sizes considered while remaining flexible enough to capture the nonlinear and interaction structures in the data generating processes. The neural network is a feedforward architecture with two hidden layers of 32 and 16 units, trained for 50 epochs with batch size 64 and learning rate 10^{-3} using the Adam optimizer. For regression outcomes, the network includes the input normalization layer and target standardization wrapper described in Section 2 to ensure scale-invariant gradient updates. Logistic and linear regression are included as baselines and use no hyperparameter tuning. All model fits are performed independently on each Monte Carlo iteration with no warm starting between iterations, so the variance estimates reflect the full sampling distribution of the AME estimator including model fitting variability. The Monte Carlo ground truth AMEs are computed once at the start of the study using $n = 1{,}000{,}000$ observations drawn from the true prediction function directly, with the resulting AMEs treated as known quantities for the bias and coverage calculations. Simulation 2 uses 1,000 bootstrap resamples per Monte Carlo iteration and 1,000 Monte Carlo iterations per cell for the linear, logistic, random forest and XGBoost learners. The neural network uses 200 resamples and 500 iterations: every resample is a fresh training run, which makes the network roughly nine tenths of the total cost of the study, and the calibration result the argument rests on is the forest's. Percentile intervals are used throughout, which is why the resample count is set well above the couple of hundred that would suffice for a standard error alone. Simulation 3 uses 1,000 Monte Carlo iterations per cell for every learner; it needs no resampling, since its standard errors come from the influence function.

Table 11: Simulation 1: AME Recovery — Linear DGP (Classification). Mean bias and RMSE (in parentheses) across 500 Monte Carlo iterations.

		Bias (RMSE)				
Variable	True AME	$n = 250$	$n = 500$	$n = 1,000$	$n = 2,500$	$n = 5,000$
Panel: Logistic						
x1	0.199	-0.0058	-0.0034	-0.0014	-0.0008	-0.0006
		(0.0201)	(0.0146)	(0.0106)	(0.0064)	(0.0048)
x2	0.298	-0.0109	-0.0057	-0.0026	-0.0010	-0.0005
		(0.0229)	(0.0153)	(0.0105)	(0.0066)	(0.0046)
x3	0.000	0.0012	0.0006	0.0006	0.0003	0.0002
		(0.0198)	(0.0136)	(0.0099)	(0.0063)	(0.0045)
x4	0.000	0.0008	-0.0001	-0.0008	-0.0001	0.0003
		(0.0197)	(0.0136)	(0.0100)	(0.0062)	(0.0045)
Panel: Random Forest						
x1	0.199	0.0126	0.0044	-0.0018	-0.0111	-0.0157
		(0.0412)	(0.0239)	(0.0160)	(0.0142)	(0.0168)
x2	0.298	0.0199	0.0113	-0.0008	-0.0102	-0.0158
		(0.0500)	(0.0309)	(0.0174)	(0.0143)	(0.0172)
x3	0.000	0.0007	0.0010	0.0005	0.0002	0.0002
		(0.0284)	(0.0171)	(0.0104)	(0.0056)	(0.0034)
x4	0.000	-0.0008	-0.0013	-0.0012	-0.0001	0.0002
		(0.0267)	(0.0177)	(0.0106)	(0.0055)	(0.0034)
Panel: XGBoost						
x1	0.199	0.0590	0.0265	0.0107	-0.0016	-0.0049
		(0.0782)	(0.0401)	(0.0223)	(0.0108)	(0.0083)
x2	0.298	0.0643	0.0315	0.0085	-0.0015	-0.0046
		(0.0893)	(0.0490)	(0.0218)	(0.0116)	(0.0083)
x3	0.000	0.0008	-0.0002	0.0006	0.0002	0.0002
		(0.0332)	(0.0193)	(0.0110)	(0.0056)	(0.0033)
x4	0.000	0.0008	-0.0009	-0.0006	0.0000	0.0002
		(0.0344)	(0.0198)	(0.0116)	(0.0054)	(0.0032)
Panel: Neural Net						
x1	0.199	-0.0078	-0.0007	0.0005	0.0003	-0.0002
		(0.0222)	(0.0156)	(0.0115)	(0.0074)	(0.0058)
x2	0.298	-0.0162	-0.0017	0.0004	0.0004	0.0002
		(0.0276)	(0.0174)	(0.0116)	(0.0073)	(0.0052)
x3	0.000	0.0014	0.0010	0.0007	0.0003	0.0004
		(0.0204)	(0.0144)	(0.0103)	(0.0069)	(0.0057)
x4	0.000	0.0009	-0.0003	-0.0007	-0.0001	0.0004
		(0.0203)	(0.0143)	(0.0105)	(0.0070)	(0.0055)

Notes: Mean bias and RMSE (in parentheses) computed over 500 Monte Carlo iterations. True AMEs computed via Monte Carlo integration with n = 1,000,000 observations. Noise features x3 and x4 have true AME of zero.

5.1 Simulation 1: Recovering the Window Effect

Table 11 reports bias and RMSE for the linear classification DGP, where the true AMEs of x_1 and x_2 are 0.199 and 0.298 — these are not 2.0 and 3.0 because the sigmoid compresses the scale. Logistic regression recovers the true AMEs with negligible bias at all sample sizes, and RMSE declines from around 0.020 at $n = 250$ to 0.005 at $n = 5{,}000$ at the expected $O(n^{-1/2})$ rate. The neural network behaves similarly, with small bias at $n = 250$ that disappears by $n = 500$. XGBoost shows larger bias at small samples — around 0.06 for both x_1 and x_2 at $n = 250$ — reflecting the well-known tendency of gradient boosted models to underfit at small sample sizes, but bias falls sharply and RMSE converges by $n = 2{,}500$. Random forests show a different pattern: bias is small at $n = 250$ and $n = 500$ but turns negative and persistent at $n = 2{,}500$ and $n = 5{,}000$, reaching around -0.016 for both signal features at the largest sample size. This reflects the regularization bias inherent in random forests — the tree depth and subsampling constraints impose shrinkage that does not vanish with sample size at the default tuning settings. The noise features x_3 and x_4 have bias and RMSE near zero throughout for all model classes, confirming that the estimator does not spuriously attribute effects to irrelevant features.

Table 12 reports results for the nonlinear classification DGP, where $P(y = 1) = \sigma(2x_1^2 + 3x_2)$. The true AME of x_1 is zero despite the quadratic term, because the sigmoid and quadratic effects cancel in expectation at these parameter values as confirmed by the Monte Carlo ground truth. All four model classes recover this near-zero AME accurately, with bias well below 0.005 across all sample sizes. The more important result is x_2, which has true AME of 0.298. Logistic regression misspecifies the DGP — the true relationship is nonlinear in x_1 — but still recovers the x_2 AME accurately because x_2 enters linearly and the misspecification in x_1 does not propagate to the x_2 marginal effect. This is an important practical result: AMEs from a misspecified parametric model can still be reliable for variables that enter the true DGP in a manner consistent with the model's functional form. The black-box models show the same pattern as the linear DGP: XGBoost has elevated RMSE at small

Table 12: Simulation 1: AME Recovery — Nonlinear DGP (Classification). Mean bias and RMSE (in parentheses) across 500 Monte Carlo iterations.

		Bias (RMSE)				
Variable	True AME	$n = 250$	$n = 500$	$n = 1,000$	$n = 2,500$	$n = 5,000$
Panel: Logistic						
x1	0.000	0.0002	0.0015	0.0009	0.0002	-0.0001
		(0.0230)	(0.0166)	(0.0123)	(0.0078)	(0.0055)
x2	0.298	-0.0077	-0.0042	-0.0017	-0.0007	-0.0005
		(0.0215)	(0.0152)	(0.0109)	(0.0065)	(0.0048)
x3	0.000	0.0003	0.0001	-0.0002	-0.0001	0.0003
		(0.0229)	(0.0158)	(0.0112)	(0.0074)	(0.0052)
x4	0.000	-0.0012	0.0005	0.0006	-0.0001	-0.0000
		(0.0252)	(0.0159)	(0.0108)	(0.0075)	(0.0051)
Panel: Random Forest						
x1	0.000	0.0014	0.0001	0.0000	-0.0003	-0.0000
		(0.0407)	(0.0256)	(0.0173)	(0.0114)	(0.0082)
x2	0.298	0.0177	0.0060	-0.0040	-0.0120	-0.0155
		(0.0521)	(0.0278)	(0.0182)	(0.0157)	(0.0171)
x3	0.000	0.0019	-0.0000	0.0003	0.0001	0.0003
		(0.0285)	(0.0166)	(0.0100)	(0.0052)	(0.0031)
x4	0.000	-0.0026	-0.0007	0.0005	0.0000	-0.0001
		(0.0295)	(0.0165)	(0.0099)	(0.0054)	(0.0031)
Panel: XGBoost						
x1	0.000	0.0039	0.0003	0.0004	-0.0000	-0.0003
		(0.0549)	(0.0324)	(0.0209)	(0.0123)	(0.0079)
x2	0.298	0.0471	0.0221	0.0066	-0.0028	-0.0054
		(0.0787)	(0.0396)	(0.0208)	(0.0114)	(0.0087)
x3	0.000	0.0038	0.0007	0.0007	0.0002	0.0003
		(0.0307)	(0.0172)	(0.0108)	(0.0051)	(0.0028)
x4	0.000	-0.0016	0.0010	0.0006	0.0001	0.0001
		(0.0329)	(0.0184)	(0.0105)	(0.0052)	(0.0027)
Panel: Neural Net						
x1	0.000	0.0006	0.0015	0.0002	0.0001	0.0000
		(0.0234)	(0.0189)	(0.0140)	(0.0094)	(0.0069)
x2	0.298	-0.0132	-0.0026	0.0002	0.0005	0.0001
		(0.0249)	(0.0155)	(0.0112)	(0.0077)	(0.0064)
x3	0.000	0.0001	0.0002	0.0007	0.0003	0.0003
		(0.0215)	(0.0138)	(0.0101)	(0.0068)	(0.0053)
x4	0.000	-0.0007	0.0004	0.0005	-0.0001	0.0002
		(0.0230)	(0.0144)	(0.0100)	(0.0070)	(0.0056)

Notes: Mean bias and RMSE (in parentheses) computed over 500 Monte Carlo iterations. True AMEs computed via Monte Carlo integration with n = 1,000,000 observations. Noise features x3 and x4 have true AME of zero.

samples, random forests develop negative bias at large samples, and neural networks perform comparably to logistic regression throughout.

Table 13 reports results for the interaction DGP, where $P(y = 1) = \sigma(2x_1 + 3x_2 + 2x_1x_2)$. The true AMEs of x_1 and x_2 remain 0.199 and 0.298 — the same as the linear DGP — because the interaction term $2x_1x_2$ has zero expectation under $N(0, 1)$ and the Monte Carlo ground truth confirms near-identical values. The key finding is that all four model classes recover these AMEs with bias and RMSE patterns essentially identical to the linear DGP, despite the true DGP including a multiplicative interaction. This is by design of the AME estimand: the average marginal effect averages over the joint distribution of the features, so the interaction term is absorbed into the average derivative rather than requiring explicit modeling. A logistic regression that omits the interaction term recovers the correct AME because the interaction's contribution to the marginal effect of x_1 is $\mathbb{E}[2x_2] = 0$ and similarly for x_2. This is an important robustness result — AMEs are not sensitive to interaction misspecification when the interactions involve zero-mean features, which is the common case after centering.

Table 14 turns to the regression setting with the linear DGP, $y = 2x_1 + 3x_2 + \varepsilon$. Linear regression recovers the true AMEs of 2.0 and 3.0 with negligible bias across all sample sizes, and RMSE declines from around 0.066 at $n = 250$ to 0.014 at $n = 5{,}000$, consistent with $O(n^{-1/2})$ convergence. The regression setting reveals the regularization bias of random forests more starkly than the classification setting: bias reaches -0.091 for x_1 and -0.095 for x_2 at $n = 5{,}000$, and RMSE is not declining at the largest sample sizes, indicating that the bias is not vanishing with n at default tuning. This is a known property of random forests in the regression case — the ensemble averaging and subsampling impose shrinkage toward zero that does not vanish at these sample sizes. XGBoost shows large positive bias at small samples — 0.41 for x_1 at $n = 250$ — but converges quickly, with bias falling below 0.05 by $n = 2{,}500$. The neural network shows moderate negative bias at small samples that falls toward zero by $n = 2{,}500$, though RMSE does not decline monotonically at the

Table 13: Simulation 1: AME Recovery — Interaction DGP (Classification). Mean bias and RMSE (in parentheses) across 500 Monte Carlo iterations.

		Bias (RMSE)				
Variable	True AME	$n = 250$	$n = 500$	$n = 1,000$	$n = 2,500$	$n = 5,000$
Panel: Logistic						
x1	0.199	-0.0065	-0.0042	-0.0021	-0.0008	-0.0007
		(0.0219)	(0.0153)	(0.0112)	(0.0071)	(0.0051)
x2	0.298	-0.0083	-0.0035	-0.0018	-0.0008	-0.0007
		(0.0222)	(0.0154)	(0.0102)	(0.0071)	(0.0050)
x3	0.000	0.0009	-0.0002	0.0005	0.0003	-0.0001
		(0.0206)	(0.0155)	(0.0107)	(0.0070)	(0.0046)
x4	0.000	0.0007	-0.0000	-0.0001	0.0001	0.0002
		(0.0212)	(0.0159)	(0.0106)	(0.0066)	(0.0047)
Panel: Random Forest						
x1	0.199	0.0114	0.0051	-0.0031	-0.0110	-0.0154
		(0.0423)	(0.0264)	(0.0169)	(0.0144)	(0.0167)
x2	0.298	0.0178	0.0104	-0.0023	-0.0101	-0.0140
		(0.0498)	(0.0320)	(0.0200)	(0.0148)	(0.0158)
x3	0.000	0.0019	0.0004	0.0002	0.0002	-0.0000
		(0.0276)	(0.0170)	(0.0104)	(0.0059)	(0.0034)
x4	0.000	-0.0005	-0.0011	-0.0001	0.0001	-0.0000
		(0.0267)	(0.0177)	(0.0111)	(0.0057)	(0.0033)
Panel: XGBoost						
x1	0.199	0.0511	0.0225	0.0079	-0.0014	-0.0046
		(0.0737)	(0.0392)	(0.0219)	(0.0105)	(0.0082)
x2	0.298	0.0496	0.0228	0.0042	-0.0021	-0.0047
		(0.0796)	(0.0439)	(0.0227)	(0.0125)	(0.0091)
x3	0.000	0.0004	0.0005	0.0004	0.0000	0.0000
		(0.0310)	(0.0205)	(0.0111)	(0.0064)	(0.0035)
x4	0.000	-0.0012	-0.0015	-0.0004	0.0001	0.0001
		(0.0338)	(0.0205)	(0.0118)	(0.0060)	(0.0037)
Panel: Neural Net						
x1	0.199	-0.0089	-0.0013	0.0001	0.0000	-0.0004
		(0.0233)	(0.0158)	(0.0117)	(0.0079)	(0.0061)
x2	0.298	-0.0114	-0.0002	0.0007	0.0005	-0.0002
		(0.0257)	(0.0170)	(0.0131)	(0.0085)	(0.0067)
x3	0.000	0.0008	0.0004	0.0008	0.0001	0.0000
		(0.0193)	(0.0148)	(0.0099)	(0.0070)	(0.0053)
x4	0.000	0.0005	-0.0002	0.0000	-0.0001	0.0006
		(0.0206)	(0.0151)	(0.0103)	(0.0069)	(0.0057)

Notes: Mean bias and RMSE (in parentheses) computed over 500 Monte Carlo iterations. True AMEs computed via Monte Carlo integration with n = 1,000,000 observations. Noise features x3 and x4 have true AME of zero.

Table 14: Simulation 1: AME Recovery — Linear DGP (Regression). Mean bias and RMSE (in parentheses) across 1,000 Monte Carlo iterations.

		Bias (RMSE)				
Variable	True AME	$n = 250$	$n = 500$	$n = 1,000$	$n = 2,500$	$n = 5,000$
Panel: Linear						
x1	2.000	-0.0017	-0.0001	-0.0000	-0.0006	-0.0003
		(0.0659)	(0.0445)	(0.0322)	(0.0207)	(0.0137)
x2	3.000	-0.0043	-0.0017	0.0007	-0.0004	-0.0003
		(0.0636)	(0.0453)	(0.0316)	(0.0197)	(0.0140)
x3	0.000	-0.0001	-0.0006	-0.0005	0.0012	-0.0003
		(0.0650)	(0.0457)	(0.0312)	(0.0199)	(0.0140)
x4	0.000	-0.0034	0.0006	-0.0002	-0.0003	-0.0001
		(0.0642)	(0.0452)	(0.0315)	(0.0196)	(0.0139)
Panel: Random Forest						
x1	2.000	-0.0973	-0.0332	-0.0387	-0.0703	-0.0911
		(0.2064)	(0.1189)	(0.0850)	(0.0824)	(0.0956)
x2	3.000	-0.0215	-0.0105	-0.0433	-0.0797	-0.0953
		(0.2338)	(0.1456)	(0.0997)	(0.0958)	(0.1025)
x3	0.000	-0.0015	-0.0005	-0.0005	0.0001	0.0000
		(0.0562)	(0.0290)	(0.0137)	(0.0036)	(0.0011)
x4	0.000	-0.0021	-0.0010	-0.0005	-0.0002	0.0000
		(0.0574)	(0.0297)	(0.0138)	(0.0037)	(0.0011)
Panel: XGBoost						
x1	2.000	0.4114	0.1780	0.0366	-0.0337	-0.0467
		(0.4644)	(0.2227)	(0.0917)	(0.0554)	(0.0547)
x2	3.000	0.3122	0.0948	-0.0172	-0.0558	-0.0601
		(0.4103)	(0.1884)	(0.0947)	(0.0754)	(0.0677)
x3	0.000	0.0007	0.0016	-0.0003	0.0011	-0.0001
		(0.1090)	(0.0550)	(0.0313)	(0.0153)	(0.0083)
x4	0.000	-0.0080	0.0002	0.0011	-0.0006	0.0001
		(0.1077)	(0.0597)	(0.0314)	(0.0149)	(0.0081)
Panel: Neural Net						
x1	2.000	-0.0659	-0.0208	-0.0063	-0.0021	-0.0008
		(0.1161)	(0.0544)	(0.0344)	(0.0293)	(0.0329)
x2	3.000	-0.1231	-0.0349	-0.0102	-0.0031	-0.0006
		(0.1765)	(0.0638)	(0.0361)	(0.0324)	(0.0372)
x3	0.000	0.0014	-0.0006	-0.0003	0.0009	-0.0006
		(0.0788)	(0.0484)	(0.0330)	(0.0250)	(0.0280)
x4	0.000	-0.0002	0.0008	-0.0003	-0.0004	-0.0017
		(0.0752)	(0.0496)	(0.0336)	(0.0247)	(0.0287)

Notes: Mean bias and RMSE (in parentheses) computed over 1,000 Monte Carlo iterations. True AMEs computed via Monte Carlo integration with $n = 1{,}000{,}000$ observations. Noise features x3 and x4 have true AME of zero.

largest sample sizes, suggesting residual variance from the optimization landscape. The noise features have RMSE near zero for linear regression and declining rapidly for the black-box models, confirming that the estimator correctly attributes near-zero effects to irrelevant features in the regression setting.

Table 15 reports results for the nonlinear regression DGP, $y = 2x_1^2 + 3x_2 + \varepsilon$, where the true AME of x_1 is zero by symmetry of $N(0,1)$ and the true AME of x_2 is 3.0. Linear regression recovers the x_2 AME accurately with RMSE declining at the standard rate, and correctly attributes a near-zero AME to x_1 despite misspecifying its functional form — the same robustness result observed in the classification case. However, the RMSE for x_1 under linear regression is substantially larger than in the linear DGP — 0.387 at $n = 250$ versus 0.066 — reflecting increased estimation variance from the misspecified model. The most striking result is random forests: the bias for x_2 reaches -0.185 at $n = 5{,}000$ and is not declining, indicating that the regularization bias is amplified in the nonlinear setting where the forest must capture the quadratic relationship in x_1 while simultaneously estimating the linear effect of x_2. XGBoost and the neural network both show elevated RMSE relative to the linear DGP but declining bias, suggesting that these models are better able to adapt to the nonlinear DGP with sufficient data. The x_1 AME of zero is recovered accurately by all model classes at moderate sample sizes, which is a reassuring result for practitioners who want to test whether nonlinear features have nonzero average effects.

Table 16 reports results for the interaction regression DGP, $y = 2x_1 + 3x_2 + 2x_1x_2 + \varepsilon$, with true AMEs of 2.0 and 3.0 for x_1 and x_2. Linear regression recovers these AMEs accurately despite omitting the interaction term, confirming the robustness result from the classification case: the interaction contribution to the marginal effect of x_1 is $\mathbb{E}[2x_2] = 0$ and similarly for x_2, so the misspecified linear model nevertheless recovers the correct AME. The RMSE for the noise features under linear regression is elevated relative to the linear DGP — around 0.14 at $n = 250$ — because the interaction term inflates the residual variance and increases estimation uncertainty for all coefficients. Random forests again show persistent negative

Table 15: Simulation 1: AME Recovery — Nonlinear DGP (Regression). Mean bias and RMSE (in parentheses) across 1,000 Monte Carlo iterations.

		Bias (RMSE)				
Variable	True AME	$n = 250$	$n = 500$	$n = 1,000$	$n = 2,500$	$n = 5,000$
Panel: Linear						
x1	0.000	0.0241	0.0261	0.0148	0.0048	0.0015
		(0.3868)	(0.2820)	(0.1966)	(0.1272)	(0.0904)
x2	3.000	-0.0145	-0.0041	-0.0012	-0.0002	0.0001
		(0.1921)	(0.1276)	(0.0930)	(0.0572)	(0.0417)
x3	0.000	-0.0025	0.0001	-0.0018	-0.0020	-0.0009
		(0.1889)	(0.1343)	(0.0941)	(0.0576)	(0.0414)
x4	0.000	-0.0114	-0.0003	0.0023	0.0006	-0.0018
		(0.1857)	(0.1328)	(0.0927)	(0.0594)	(0.0430)
Panel: Random Forest						
x1	0.000	0.0071	0.0097	0.0053	0.0006	0.0007
		(0.3433)	(0.2547)	(0.1836)	(0.1185)	(0.0903)
x2	3.000	0.0142	-0.1152	-0.1701	-0.1867	-0.1849
		(0.2704)	(0.2008)	(0.2004)	(0.1974)	(0.1916)
x3	0.000	0.0001	0.0008	0.0003	-0.0002	0.0000
		(0.0549)	(0.0274)	(0.0138)	(0.0055)	(0.0027)
x4	0.000	-0.0009	-0.0007	-0.0001	-0.0000	-0.0000
		(0.0573)	(0.0278)	(0.0138)	(0.0054)	(0.0025)
Panel: XGBoost						
x1	0.000	0.0552	0.0038	-0.0117	-0.0137	-0.0057
		(0.4662)	(0.2744)	(0.1829)	(0.1040)	(0.0735)
x2	3.000	0.2725	0.0816	-0.0183	-0.0702	-0.0794
		(0.3937)	(0.1981)	(0.1058)	(0.0885)	(0.0862)
x3	0.000	-0.0040	-0.0022	-0.0020	-0.0011	-0.0006
		(0.1124)	(0.0704)	(0.0446)	(0.0191)	(0.0097)
x4	0.000	-0.0052	-0.0028	-0.0021	0.0003	0.0006
		(0.1121)	(0.0636)	(0.0381)	(0.0189)	(0.0096)
Panel: Neural Net						
x1	0.000	0.0201	0.0178	0.0065	-0.0003	-0.0013
		(0.3086)	(0.1925)	(0.1390)	(0.0911)	(0.0760)
x2	3.000	-0.1545	-0.0202	-0.0150	-0.0098	-0.0022
		(0.2601)	(0.0710)	(0.0461)	(0.0343)	(0.0395)
x3	0.000	0.0017	0.0004	0.0007	0.0009	-0.0004
		(0.1419)	(0.0634)	(0.0400)	(0.0293)	(0.0321)
x4	0.000	-0.0028	-0.0001	-0.0013	-0.0004	-0.0016
		(0.1410)	(0.0626)	(0.0406)	(0.0281)	(0.0323)

Notes: Mean bias and RMSE (in parentheses) computed over 1,000 Monte Carlo iterations. True AMEs computed via Monte Carlo integration with $n = 1{,}000{,}000$ observations. Noise features x3 and x4 have true AME of zero.

Table 16: Simulation 1: AME Recovery — Interaction DGP (Regression). Mean bias and RMSE (in parentheses) across 1,000 Monte Carlo iterations.

		Bias (RMSE)				
Variable	True AME	$n = 250$	$n = 500$	$n = 1,000$	$n = 2,500$	$n = 5,000$
Panel: Linear						
x1	2.000	-0.0041	0.0005	0.0000	0.0006	0.0011
		(0.2284)	(0.1531)	(0.1129)	(0.0715)	(0.0495)
x2	3.000	0.0040	0.0087	0.0075	0.0012	0.0009
		(0.2194)	(0.1627)	(0.1135)	(0.0727)	(0.0514)
x3	0.000	-0.0114	-0.0088	-0.0054	0.0010	-0.0009
		(0.1388)	(0.1002)	(0.0725)	(0.0450)	(0.0329)
x4	0.000	-0.0073	0.0004	-0.0006	0.0013	0.0005
		(0.1443)	(0.1018)	(0.0690)	(0.0453)	(0.0321)
Panel: Random Forest						
x1	2.000	-0.0755	-0.0310	-0.0433	-0.0724	-0.0804
		(0.2566)	(0.1663)	(0.1169)	(0.0979)	(0.0925)
x2	3.000	-0.0634	-0.0690	-0.0922	-0.1112	-0.1143
		(0.2806)	(0.1913)	(0.1497)	(0.1306)	(0.1244)
x3	0.000	-0.0031	-0.0004	-0.0009	0.0003	-0.0001
		(0.0612)	(0.0352)	(0.0182)	(0.0067)	(0.0029)
x4	0.000	-0.0023	0.0004	-0.0001	-0.0002	0.0000
		(0.0665)	(0.0349)	(0.0184)	(0.0069)	(0.0030)
Panel: XGBoost						
x1	2.000	0.3184	0.1016	-0.0009	-0.0406	-0.0429
		(0.4539)	(0.2184)	(0.1163)	(0.0758)	(0.0602)
x2	3.000	0.3672	0.1135	-0.0083	-0.0572	-0.0589
		(0.4920)	(0.2373)	(0.1227)	(0.0898)	(0.0746)
x3	0.000	-0.0031	0.0011	-0.0017	0.0005	-0.0001
		(0.1040)	(0.0551)	(0.0319)	(0.0151)	(0.0078)
x4	0.000	-0.0082	0.0003	-0.0003	-0.0004	-0.0001
		(0.1028)	(0.0564)	(0.0314)	(0.0149)	(0.0080)
Panel: Neural Net						
x1	2.000	-0.0842	-0.0083	-0.0056	0.0003	0.0022
		(0.2380)	(0.1114)	(0.0779)	(0.0573)	(0.0536)
x2	3.000	-0.1231	-0.0068	-0.0052	0.0004	0.0026
		(0.2467)	(0.1120)	(0.0783)	(0.0579)	(0.0533)
x3	0.000	-0.0026	-0.0016	-0.0024	0.0004	-0.0004
		(0.0986)	(0.0551)	(0.0383)	(0.0293)	(0.0316)
x4	0.000	-0.0079	0.0015	0.0002	-0.0005	-0.0019
		(0.0981)	(0.0565)	(0.0387)	(0.0284)	(0.0331)

Notes: Mean bias and RMSE (in parentheses) computed over 1,000 Monte Carlo iterations. True AMEs computed via Monte Carlo integration with n = 1,000,000 observations. Noise features x3 and x4 have true AME of zero.

bias that grows rather than shrinks with sample size, reaching -0.080 and -0.114 for x_1 and x_2 respectively at $n = 5{,}000$. XGBoost shows the same large positive bias at small samples followed by convergence, but the bias does not fully vanish at $n = 5{,}000$ — around -0.043 for x_1 and -0.059 for x_2 — suggesting that the interaction structure requires larger samples for the gradient boosting to fully adapt. The neural network performs well, with bias below 0.003 for both signal features at $n = 2{,}500$ and $n = 5{,}000$, and RMSE declining steadily. Taken together the six bias and RMSE tables establish that the AME estimator is consistent for logistic regression, neural networks, and XGBoost across all three DGPs and both outcome types, with random forests showing persistent regularization bias in the regression setting that practitioners should be aware of when reporting AMEs from default-tuned forest models.

5.2 Simulation 2: The Refitting Bootstrap

Table 17 reports coverage of nominal 95% intervals for the linear classification design. Two distinct failures are visible, and Section 3.4 predicts both.

The first is the one the paper is built around. Under the random forest, coverage at $n = 5{,}000$ falls to 0.186 for x_1 and 0.268 for x_2 while remaining near nominal for the noise features, whose true effect is zero and whose regularization bias is therefore also zero. This is the mechanism of Remark 4: the refitting bootstrap is consistent for the dispersion of the plug-in average around its own expectation, but that expectation is displaced from $\theta_{j,h}$ by $\mathbb{E}[\alpha_h(\hat{f} - f)]$, and resampling cannot see the displacement because every replicate recentres at a similarly regularized fit. The interval shrinks at the usual rate; the bias does not shrink at all; coverage follows the ratio down. XGBoost, whose bias does vanish with n, keeps coverage between 0.877 and 0.976 throughout, so the failure is specific to learners whose regularization bias persists rather than general to flexible learners.

The second failure is milder, and it affects even a correctly specified model. Logistic regression covers at 0.881, 0.878 and 0.886 for x_1 and at 0.726, 0.766 and 0.781 for x_2, with

Table 17: Simulation 2: Bootstrap SE Calibration — Linear DGP (Classification). Nominal coverage target: 95%. Coverage rate and mean 95% CI width (in parentheses) across 500 Monte Carlo iterations.

		Coverage (CI Width)		
Variable	True AME	$n = 250$	$n = 1,000$	$n = 5,000$
Panel: Logistic				
x1	0.199	0.881	0.878	0.886
		(0.0652)	(0.0334)	(0.0150)
x2	0.298	0.726	0.766	0.781
		(0.0505)	(0.0257)	(0.0115)
x3	0.000	0.944	0.939	0.946
		(0.0752)	(0.0386)	(0.0174)
x4	0.000	0.933	0.954	0.944
		(0.0754)	(0.0386)	(0.0174)
Panel: Random Forest				
x1	0.199	0.968	0.933	0.186
		(0.1343)	(0.0553)	(0.0205)
x2	0.298	0.961	0.958	0.268
		(0.1500)	(0.0614)	(0.0219)
x3	0.000	0.962	0.942	0.949
		(0.0976)	(0.0386)	(0.0132)
x4	0.000	0.969	0.964	0.953
		(0.0978)	(0.0386)	(0.0132)
Panel: XGBoost				
x1	0.199	0.926	0.947	0.877
		(0.1864)	(0.0702)	(0.0230)
x2	0.298	0.908	0.954	0.910
		(0.2136)	(0.0794)	(0.0239)
x3	0.000	0.976	0.961	0.960
		(0.1139)	(0.0436)	(0.0141)
x4	0.000	0.973	0.969	0.959
		(0.1149)	(0.0436)	(0.0142)
Panel: Neural Net				
x1	0.199	0.870	0.870	0.958
		(0.0679)	(0.0351)	(0.0203)
x2	0.298	0.658	0.774	0.894
		(0.0586)	(0.0287)	(0.0154)
x3	0.000	0.936	0.936	0.986
		(0.0764)	(0.0391)	(0.0225)
x4	0.000	0.938	0.936	0.982
		(0.0772)	(0.0392)	(0.0225)

Notes: Coverage rate and mean 95% CI width (in parentheses). Intervals are bootstrap percentile intervals. Logistic, random forest and XGBoost use 1,000 Monte Carlo iterations and 1,000 resamples; the neural network uses 500 and 200, since every one of its resamples is a fresh training run. True AMEs computed via Monte Carlo integration with $n = 1{,}000{,}000$ observations. Noise features x3 and x4 have true AME of zero.

no tendency to improve as the sample grows, while covering at roughly the nominal rate for the two noise features. The pattern orders the features by coefficient size — $\beta_2 = 3$, $\beta_1 = 2$, and zero for x_3 and x_4 — and that ordering identifies the cause. As Section 3.4 notes, each replicate is evaluated at the observed sample rather than at the resample, so the bootstrap distribution omits the sampling variability of the average of the derivative over the covariates. That omitted component is identically zero when the fitted derivative does not vary with x and grows with how much it does, and under a logistic link the derivative varies most for the feature with the largest coefficient. The estimated standard errors are correspondingly too small: averaged over cells the ratio of the bootstrap standard error to the Monte Carlo standard deviation is 0.86 for logistic regression, 0.89 for the network and 0.90 for the forest. Neither failure is detectable from the bootstrap output alone, which is the practical point: both produce intervals that look unremarkable.

Table 18 reports the corresponding regression results, where the two mechanisms separate cleanly. Linear regression covers between 0.928 and 0.957 at every sample size and every feature, and its bootstrap standard error matches the Monte Carlo standard deviation to within one percent. That is the second mechanism switching off exactly where the theory says it must: the fitted derivative of a linear model is constant in x, so the variance component that the evaluation point omits is identically zero, and the understatement that afflicted every classification panel disappears. Comparing the two tables therefore isolates the cause more sharply than either does alone.

The first mechanism is correspondingly starker. The forest covers at 0.092 for x_1 and 0.323 for x_2 at $n = 5{,}000$, and XGBoost — which was well behaved in classification — falls to 0.521 and 0.503. Here the forest's bootstrap standard error is if anything conservative, averaging 1.24 times the Monte Carlo standard deviation, so the coverage failure cannot be attributed to an understated standard error. It is bias alone. The neural network sits at the opposite extreme, covering at 1.000 for every feature at $n = 5{,}000$: its estimate is dominated by optimization noise rather than sampling noise at that sample size, and because each

Table 18: Simulation 2: Bootstrap SE Calibration — Linear DGP (Regression). Nominal coverage target: 95%. Coverage rate and mean 95% CI width (in parentheses) across 500 Monte Carlo iterations.

		Coverage (CI Width)		
Variable	True AME	$n = 250$	$n = 1,000$	$n = 5,000$
Panel: Linear				
x1	2.000	0.928	0.945	0.954
		(0.2478)	(0.1235)	(0.0553)
x2	3.000	0.955	0.948	0.956
		(0.2489)	(0.1239)	(0.0552)
x3	0.000	0.951	0.947	0.948
		(0.2495)	(0.1234)	(0.0552)
x4	0.000	0.949	0.948	0.957
		(0.2489)	(0.1234)	(0.0552)
Panel: Random Forest				
x1	2.000	0.870	0.886	0.092
		(0.7447)	(0.2990)	(0.1170)
x2	3.000	0.967	0.962	0.323
		(0.9327)	(0.3632)	(0.1441)
x3	0.000	0.996	0.996	0.999
		(0.2555)	(0.0707)	(0.0082)
x4	0.000	0.992	0.995	1.000
		(0.2566)	(0.0708)	(0.0083)
Panel: XGBoost				
x1	2.000	0.849	0.982	0.521
		(0.8174)	(0.3009)	(0.1041)
x2	3.000	0.917	0.990	0.503
		(0.9709)	(0.3552)	(0.1184)
x3	0.000	0.988	0.969	0.971
		(0.3863)	(0.1217)	(0.0382)
x4	0.000	0.978	0.975	0.969
		(0.3778)	(0.1210)	(0.0382)
Panel: Neural Net				
x1	2.000	0.950	0.978	1.000
		(0.3442)	(0.1497)	(0.1938)
x2	3.000	0.936	0.974	1.000
		(0.4000)	(0.1537)	(0.2004)
x3	0.000	0.972	0.970	1.000
		(0.3075)	(0.1471)	(0.1887)
x4	0.000	0.968	0.972	1.000
		(0.3116)	(0.1474)	(0.1869)

Notes: Coverage rate and mean 95% CI width (in parentheses). Intervals are bootstrap percentile intervals. Logistic, random forest and XGBoost use 1,000 Monte Carlo iterations and 1,000 resamples; the neural network uses 500 and 200, since every one of its resamples is a fresh training run. True AMEs computed via Monte Carlo integration with $n = 1{,}000{,}000$ observations. Noise features x3 and x4 have true AME of zero.

replicate re-draws the initialization independently of the point fit, the replicate distribution centres near the truth rather than near the estimate. Its intervals are therefore wide enough to be useless rather than wrong, which the bootstrap output again does not reveal.

5.3 Simulation 3: The Debiased Estimator

Simulation 3 repeats the design of Simulation 2 for the debiased cross-fitted estimator of Section 3.4. The covariates are independent standard normals, so the Riesz representer of Lemma 1 is available in closed form, $\alpha_h(u) = e^{-h^2/2}\sinh(hu_j)/h$, and the estimator requires no representer estimation at all; a second arm reported at the end of this subsection replaces it with a sieve estimate, to show what is lost when the covariate density is unknown. Cross-fitting uses $K = 5$ folds. The data generating processes, learners, hyperparameters, sample sizes, window h and Monte Carlo ground truth are all held at the values used in the two preceding studies, so any difference between Simulation 3 and its predecessors is attributable to the orthogonal correction and to nothing else. Trimming is inactive throughout because the covariate support is unbounded and the trimming weight of Section 2.1 is therefore identically one. For the same reason the design satisfies Assumption efass:unbounded rather than the density bounds of Assumption efass:sampling: a Gaussian density is not bounded below on $\mathbb{R}^d$, and the representer $\alpha_h(u) = e^{-h^2/2}\sinh(hu_j)/h$ is unbounded. It is square integrable, with second moment (11), and the shift ratio $\exp(hu_j - h^2/2)$ has moments of every order, so both conditions of Assumption efass:unbounded hold for every r and q. The theorems demonstrated here are therefore the ones proved under that assumption set, and the closed-form representer used below is the exact one for this design rather than a bounded approximation to it.

Table 19 reports bias and RMSE for the linear regression DGP. The result to compare against is the random forest panel of Table 14, where the plug-in's bias was -0.097 at $n = 250$ and, rather than shrinking, grew to -0.091 for x_1 and -0.095 for x_2 at $n = 5{,}000$. Under the correction that bias is -0.002 at $n = 250$ and 0.000 at $n = 5{,}000$, and the RMSE now

Table 19: Simulation 3: debiased estimator, linear regression DGP. Bias and RMSE (in parentheses) by sample size.

		Bias (RMSE)				
Variable	True AME	$n = 250$	$n = 500$	$n = 1,000$	$n = 2,500$	$n = 5,000$
Panel: Linear						
x1	2.000	-0.002	-0.000	-0.000	-0.001	-0.000
		(0.067)	(0.044)	(0.032)	(0.021)	(0.014)
x2	3.000	-0.004	-0.002	0.001	-0.000	-0.000
		(0.064)	(0.045)	(0.032)	(0.020)	(0.014)
x3	0.000	-0.000	-0.001	-0.000	0.001	-0.000
		(0.065)	(0.046)	(0.031)	(0.020)	(0.014)
x4	0.000	-0.003	0.001	-0.000	-0.000	-0.000
		(0.065)	(0.045)	(0.031)	(0.020)	(0.014)
Panel: Random Forest						
x1	2.000	-0.002	0.002	-0.002	0.000	0.000
		(0.171)	(0.106)	(0.074)	(0.044)	(0.031)
x2	3.000	-0.001	0.010	0.003	0.001	-0.000
		(0.228)	(0.147)	(0.094)	(0.056)	(0.038)
x3	0.000	0.002	0.001	-0.001	0.001	-0.000
		(0.087)	(0.054)	(0.035)	(0.022)	(0.015)
x4	0.000	-0.002	0.000	-0.001	-0.001	-0.000
		(0.089)	(0.056)	(0.036)	(0.021)	(0.015)
Panel: XGBoost						
x1	2.000	-0.002	0.001	-0.001	-0.002	0.000
		(0.230)	(0.142)	(0.088)	(0.048)	(0.030)
x2	3.000	-0.002	0.007	0.001	0.001	0.001
		(0.280)	(0.173)	(0.101)	(0.054)	(0.034)
x3	0.000	-0.001	0.000	-0.000	0.001	-0.000
		(0.095)	(0.058)	(0.036)	(0.022)	(0.015)
x4	0.000	-0.004	0.001	0.000	-0.000	-0.000
		(0.099)	(0.061)	(0.036)	(0.021)	(0.014)
Panel: Neural Net						
x1	2.000	-0.002	-0.001	0.000	-0.001	-0.000
		(0.084)	(0.051)	(0.034)	(0.022)	(0.014)
x2	3.000	-0.004	-0.002	0.001	-0.001	-0.000
		(0.090)	(0.055)	(0.036)	(0.021)	(0.015)
x3	0.000	0.000	-0.001	-0.000	0.001	-0.000
		(0.079)	(0.049)	(0.033)	(0.021)	(0.014)
x4	0.000	-0.003	0.001	-0.001	0.000	-0.000
		(0.080)	(0.051)	(0.033)	(0.020)	(0.014)

Notes: Debiased cross-fitted estimator with $K = 5$ folds and the closed-form Riesz representer $\alpha_h(u) = e^{-h^2/2}\sinh(hu_j)/h$, exact for independent standard normal covariates. Trimming is inactive because the support is unbounded. Bias is the mean of $\hat{\theta}_{j,h} - \theta_{j,h}$ and RMSE its root mean square, over the Monte Carlo iterations. True values computed by Monte Carlo integration at n = 1,000,000. Noise features x_3 and x_4 have a true effect of zero.

falls monotonically from 0.171 to 0.031, at the $O(n^{-1/2})$ rate the plug-in never achieved. The correction does not merely shrink the bias at a given sample size; it restores the rate, which is what inference requires. The other learners were already close to unbiased on this design and are essentially unchanged, as they should be: the orthogonal score corrects a first-order term that is near zero for them.

Table 20: Coverage of nominal 95% confidence intervals, linear regression DGP: refitting bootstrap against the influence function.

		Refitting bootstrap			Influence function		
Variable	True AME	$n = 250$	$n = 1,000$	$n = 5,000$	$n = 250$	$n = 1,000$	$n = 5,000$
Panel: Linear							
x1	2.000	0.928	0.945	0.954	0.930	0.951	0.957
x2	3.000	0.955	0.948	0.956	0.955	0.946	0.960
x3	0.000	0.951	0.947	0.948	0.944	0.949	0.950
x4	0.000	0.949	0.948	0.957	0.941	0.949	0.956
Panel: Random Forest							
x1	2.000	0.870	0.886	0.092	0.936	0.939	0.946
x2	3.000	0.967	0.962	0.323	0.936	0.952	0.952
x3	0.000	0.996	0.996	0.999	0.953	0.963	0.950
x4	0.000	0.992	0.995	1.000	0.940	0.950	0.964
Panel: XGBoost							
x1	2.000	0.849	0.982	0.521	0.932	0.942	0.938
x2	3.000	0.917	0.990	0.503	0.948	0.946	0.941
x3	0.000	0.988	0.969	0.971	0.957	0.952	0.952
x4	0.000	0.978	0.975	0.969	0.940	0.957	0.962
Panel: Neural Net							
x1	2.000	0.950	0.978	1.000	0.940	0.948	0.956
x2	3.000	0.936	0.974	1.000	0.950	0.939	0.959
x3	0.000	0.972	0.970	1.000	0.947	0.951	0.948
x4	0.000	0.968	0.972	1.000	0.949	0.952	0.946

Notes: Empirical coverage of nominal 95% intervals. The bootstrap columns resample and refit the learner, centring on the plug-in average; the influence-function columns use the debiased cross-fitted estimator with $K = 5$ folds and the closed-form representer. Both are evaluated against the same window estimand $\theta_{j,h}$ and the same Monte Carlo ground truth. Noise features x_3 and x_4 have a true effect of zero.

Table 20 places the two inference procedures side by side on the same designs and the same ground truth. The bootstrap columns range from 0.092 to 1.000; the influence-function columns range from 0.930 to 0.964. Across all 300 cells of the closed-form arm — three data

generating processes, two outcome types, five learners, five sample sizes and four features — empirical coverage has mean 0.948 and lies between 0.920 and 0.970, with 98.7% of cells inside $[0.93, 0.97]$. No learner is an exception: mean coverage by model runs from 0.947 to 0.949. The forest cell that motivates the paper moves from 0.092 to 0.946.

The forest panel carries a second reading. These forests are capped at depth six, and a depth-capped forest does not converge to the regression function; it converges to the best depth-limited approximation of it. The estimator is therefore consistent for neither nuisance in the usual sense, and yet its intervals cover at the nominal rate. This is Proposition 2 operating exactly as stated: with the representer known, $\sqrt{n}$-normality survives an outcome model that converges to the wrong function, and the learner's approximation error is paid in the width of the interval rather than in its location. The forest's RMSE at $n = 5{,}000$ is 0.031 against the linear model's 0.014 — wider, and correctly centred. That is the practical content of double robustness here, and it is why the choice of learner becomes a question of efficiency rather than of validity.

Tables 21 and 22 report the corresponding classification results, where the same pattern holds on a compressed scale: the forest's coverage at $n = 5{,}000$ moves from 0.186 to 0.944 for x_1 and from 0.268 to 0.946 for x_2, and the debiased intervals are centred for every learner at every sample size. The nonlinear and interaction designs behave identically and are reported in the supplement.

The closed-form representer used above is a luxury of the simulation design, and the natural question is what the estimator costs when α_h must itself be estimated. The second arm answers it by replacing the closed form with a sieve estimate, holding everything else fixed. Coverage survives, with a mild conservatism: over the same cells the sieve arm has mean coverage 0.958 against the closed form's 0.948. The cost appears where theory says it should, in the product bias of Assumption 3(b) and in efficiency, and it vanishes with sample size. Mean absolute bias is 3.3 times the closed form's at $n = 250$, then 1.6, 1.2, 1.1, and 0.9 at $n = 5{,}000$ — indistinguishable by the largest sample size. Standard errors are 15% wider

Table 21: Simulation 3: debiased estimator, linear classification DGP. Bias and RMSE (in parentheses) by sample size.

		Bias (RMSE)				
Variable	True AME	$n = 250$	$n = 500$	$n = 1,000$	$n = 2,500$	$n = 5,000$
Panel: Logistic						
x1	0.1986	0.0019	0.0005	0.0006	0.0000	-0.0002
		(0.0199)	(0.0144)	(0.0105)	(0.0064)	(0.0048)
x2	0.2978	0.0004	-0.0001	0.0003	0.0002	0.0002
		(0.0206)	(0.0142)	(0.0102)	(0.0065)	(0.0046)
x3	0.0000	0.0013	0.0007	0.0007	0.0003	0.0001
		(0.0208)	(0.0139)	(0.0100)	(0.0063)	(0.0045)
x4	0.0000	0.0008	-0.0002	-0.0008	-0.0001	0.0003
		(0.0206)	(0.0139)	(0.0101)	(0.0062)	(0.0045)
Panel: Random Forest						
x1	0.1986	0.0034	-0.0009	0.0008	0.0003	-0.0003
		(0.0339)	(0.0223)	(0.0152)	(0.0091)	(0.0060)
x2	0.2978	-0.0003	0.0024	0.0004	0.0006	0.0004
		(0.0437)	(0.0269)	(0.0176)	(0.0101)	(0.0063)
x3	0.0000	0.0003	0.0009	0.0003	0.0003	0.0003
		(0.0246)	(0.0162)	(0.0111)	(0.0068)	(0.0047)
x4	0.0000	-0.0001	-0.0010	-0.0010	-0.0002	0.0002
		(0.0249)	(0.0168)	(0.0113)	(0.0065)	(0.0046)
Panel: XGBoost						
x1	0.1986	0.0050	-0.0025	0.0013	0.0001	-0.0003
		(0.0495)	(0.0304)	(0.0193)	(0.0107)	(0.0069)
x2	0.2978	-0.0003	0.0035	0.0005	0.0010	0.0003
		(0.0630)	(0.0395)	(0.0220)	(0.0119)	(0.0069)
x3	0.0000	0.0011	0.0007	0.0004	0.0002	0.0002
		(0.0260)	(0.0171)	(0.0112)	(0.0068)	(0.0047)
x4	0.0000	0.0004	-0.0003	-0.0004	-0.0001	0.0003
		(0.0268)	(0.0174)	(0.0117)	(0.0065)	(0.0046)
Panel: Neural Net						
x1	0.1986	0.0019	0.0005	0.0004	0.0001	-0.0001
		(0.0217)	(0.0158)	(0.0115)	(0.0068)	(0.0049)
x2	0.2978	0.0004	-0.0003	0.0000	0.0004	0.0002
		(0.0229)	(0.0171)	(0.0117)	(0.0072)	(0.0048)
x3	0.0000	0.0012	0.0010	0.0005	0.0004	0.0002
		(0.0216)	(0.0144)	(0.0101)	(0.0063)	(0.0045)
x4	0.0000	0.0008	-0.0005	-0.0008	-0.0001	0.0003
		(0.0216)	(0.0141)	(0.0105)	(0.0063)	(0.0045)

Notes: Debiased cross-fitted estimator with $K = 5$ folds and the closed-form Riesz representer $\alpha_h(u) = e^{-h^2/2} \sinh(hu_j)/h$, exact for independent standard normal covariates. Trimming is inactive because the support is unbounded. Bias is the mean of $\hat{\theta}_{j,h} - \theta_{j,h}$ and RMSE its root mean square, over the Monte Carlo iterations. True values computed by Monte Carlo integration at n = 1,000,000. Noise features x_3 and x_4 have a true effect of zero.

Table 22: Coverage of nominal 95% confidence intervals, linear classification DGP: refitting bootstrap against the influence function.

		Refitting bootstrap			Influence function		
Variable	True AME	$n = 250$	$n = 1,000$	$n = 5,000$	$n = 250$	$n = 1,000$	$n = 5,000$
Panel: Logistic							
x1	0.199	0.881	0.878	0.886	0.944	0.934	0.940
x2	0.298	0.726	0.766	0.781	0.924	0.956	0.954
x3	0.000	0.944	0.939	0.946	0.934	0.950	0.952
x4	0.000	0.933	0.954	0.944	0.948	0.950	0.944
Panel: Random Forest							
x1	0.199	0.968	0.933	0.186	0.942	0.956	0.944
x2	0.298	0.961	0.958	0.268	0.932	0.952	0.946
x3	0.000	0.962	0.942	0.949	0.944	0.960	0.948
x4	0.000	0.969	0.964	0.953	0.944	0.944	0.944
Panel: XGBoost							
x1	0.199	0.926	0.947	0.877	0.932	0.952	0.950
x2	0.298	0.908	0.954	0.910	0.934	0.968	0.972
x3	0.000	0.976	0.961	0.960	0.964	0.948	0.946
x4	0.000	0.973	0.969	0.959	0.954	0.938	0.946
Panel: Neural Net							
x1	0.199	0.870	0.870	0.958	0.940	0.930	0.942
x2	0.298	0.658	0.774	0.894	0.934	0.950	0.958
x3	0.000	0.936	0.936	0.986	0.932	0.946	0.954
x4	0.000	0.938	0.936	0.982	0.946	0.932	0.944

Notes: Empirical coverage of nominal 95% intervals. The bootstrap columns resample and refit the learner, centring on the plug-in average; the influence-function columns use the debiased cross-fitted estimator with $K = 5$ folds and the closed-form representer. Both are evaluated against the same window estimand $\theta_{j,h}$ and the same Monte Carlo ground truth. Noise features x_3 and x_4 have a true effect of zero.

at $n = 250$ and 1% wider at $n = 5{,}000$. Estimating the representer costs something real at small samples and asymptotically nothing, which is the behaviour the product structure of the bias predicts: neither nuisance need converge quickly on its own, so long as the product of their errors is $o(n^{-1/2})$.

One design remains to be accounted for. The arms reported so far draw independent standard normal covariates, and a Gaussian density is not bounded below on $\mathbb{R}^d$, so those designs satisfy Assumption 2 rather than the density bounds of Assumption 1. That is a real gap between what the theorems assume and what the simulation exhibits, and the natural way to close it is a design that meets the density-bounded hypotheses literally. The final arm does so: the same linear predictor, with the covariates drawn independently and uniformly on $[-\sqrt{3}, \sqrt{3}]$ so that each coordinate again has unit variance and the two designs are directly comparable. There the density is constant, and the representer follows from Lemma 1 with the density cancelling entirely,

$$\alpha_h(u) = \frac{w(u - he_j) - w(u + he_j)}{2h} = \frac{\mathbb{1}\{u_j \geq a + 2h\} - \mathbb{1}\{u_j \leq b - 2h\}}{2h} \tag{52}$$

on $[a, b] = [-\sqrt{3}, \sqrt{3}]$: the step function taking $-1/(2h)$ on the lower trimming shell, $+1/(2h)$ on the upper one, and zero between them. It is bounded by $1/(2h)$ and integrates to zero, so Assumption 1 and Assumption 3(c) hold as originally stated. The trimming weight is also active here rather than vacuous, which the Gaussian arms never exercise: the true effect of x_1 is 1.943 rather than 2, being exactly $2\,(1 - 2h/(b - a))$, so the boundary shell that the weight excises is visible in the target itself.

Table 23 reports bias and RMSE. The estimator behaves as it does on the unbounded designs: bias is at most 0.002 on the signal features at any sample size and for any learner, and RMSE declines at the $O(n^{-1/2})$ rate. Coverage is the quantity of interest, and across the 160 combinations of learner, sample size and feature in this arm it has mean 0.944 and lies between 0.890 and 0.970, against a mean of 0.948 over the matched Gaussian cells. The two assumption sets therefore deliver the same behaviour to within four thousandths. Since the

Table 23: Simulation 3: debiased estimator, bounded-support linear regression DGP. Bias and RMSE (in parentheses) by sample size.

		Bias (RMSE)				
Variable	True AME	$n = 250$	$n = 500$	$n = 1,000$	$n = 2,500$	$n = 5,000$
Panel: Linear						
x1	1.943	0.005	0.002	-0.004	-0.001	-0.001
		(0.151)	(0.111)	(0.075)	(0.049)	(0.034)
x2	2.913	-0.001	-0.002	-0.002	-0.001	-0.001
		(0.161)	(0.113)	(0.077)	(0.049)	(0.034)
x3	0.000	0.004	0.001	0.003	0.002	0.001
		(0.159)	(0.107)	(0.076)	(0.048)	(0.033)
x4	0.000	-0.002	0.001	-0.002	0.000	0.001
		(0.156)	(0.105)	(0.079)	(0.049)	(0.034)
Panel: Random Forest						
x1	1.943	0.003	0.002	-0.005	0.001	-0.000
		(0.252)	(0.162)	(0.107)	(0.064)	(0.044)
x2	2.913	-0.008	-0.002	0.001	-0.001	-0.000
		(0.288)	(0.188)	(0.124)	(0.070)	(0.048)
x3	0.000	0.004	-0.001	0.002	0.002	0.001
		(0.184)	(0.123)	(0.082)	(0.051)	(0.034)
x4	0.000	-0.006	0.002	-0.003	-0.000	0.001
		(0.180)	(0.120)	(0.086)	(0.051)	(0.036)
Panel: XGBoost						
x1	1.943	-0.002	-0.002	-0.003	-0.001	-0.000
		(0.302)	(0.175)	(0.111)	(0.065)	(0.042)
x2	2.913	-0.004	0.002	0.001	-0.002	0.000
		(0.334)	(0.200)	(0.127)	(0.068)	(0.043)
x3	0.000	0.006	-0.002	0.000	0.001	0.001
		(0.196)	(0.127)	(0.085)	(0.051)	(0.035)
x4	0.000	-0.004	0.002	-0.003	0.000	0.001
		(0.193)	(0.125)	(0.085)	(0.051)	(0.036)
Panel: Neural Net						
x1	1.943	0.008	0.003	-0.004	-0.001	-0.002
		(0.192)	(0.120)	(0.079)	(0.050)	(0.035)
x2	2.913	-0.002	-0.003	-0.001	-0.000	-0.001
		(0.202)	(0.119)	(0.080)	(0.050)	(0.034)
x3	0.000	0.003	0.000	0.004	0.002	0.000
		(0.184)	(0.112)	(0.078)	(0.050)	(0.034)
x4	0.000	-0.002	0.003	-0.003	0.000	0.001
		(0.182)	(0.114)	(0.081)	(0.050)	(0.035)

Notes: Debiased cross-fitted estimator with $K = 5$ folds and the closed-form Riesz representer $\alpha_h(u) = e^{-h^2/2}\sinh(hu_j)/h$, exact for independent standard normal covariates. Trimming is inactive because the support is unbounded. Bias is the mean of $\hat{\theta}_{j,h} - \theta_{j,h}$ and RMSE its root mean square, over the Monte Carlo iterations. True values computed by Monte Carlo integration at n = 1,000,000. Noise features x_3 and x_4 have a true effect of zero.

two designs share a predictor and a coordinate variance and differ only in the covariate law, that agreement isolates boundedness as making no material difference: what the moment conditions buy is coverage of the Gaussian case, not a weaker guarantee within it. The influence-function standard errors are honest on both, with a mean ratio to the Monte Carlo standard deviation of 0.99 here.

The bounded design is nonetheless the slightly harder of the two at small samples, and the pattern is worth recording rather than averaging away. Fourteen of the 160 cells fall below 0.93; thirteen are classification and eleven have $n \leq 500$. The weakest is the logistic coefficient on x_2, whose coverage runs 0.912, 0.890, 0.908, 0.948 and 0.958 across the five sample sizes while the ratio of the estimated standard error to the Monte Carlo standard deviation runs 0.87, 0.91, 0.96, 1.00 and 1.07. This is finite-sample understatement of the standard error converging to correctness, not a failure of the assumption set: the mean ratio over all cells at $n = 250$ is 0.958 and at $n = 5{,}000$ is 1.015. The trimming weight is the plausible source, since it discards a shell of probability $O(h)$ and so costs effective sample size exactly where there is least of it to spare.

6 Conclusion

This paper has proposed a framework for interpreting any supervised machine learning model through average marginal effects. The central claim is that the traditional accuracy-interpretability tradeoff is not fundamental. The move that carries the argument is to make the window the estimand rather than a numerical approximation to something else: the window average marginal effect is the average centered difference of the regression function at a fixed, analyst-chosen step, and it is well defined for a learner that is differentiable nowhere. A single estimand therefore covers trees, ensembles and networks without differentiability assumptions of any kind, and the classical average derivative is recovered in the small-window limit whenever one weak derivative exists. The estimator applies uniformly across model

classes and outcome types and produces a coefficient table — estimate, standard error, and significance level — directly analogous to the output of a parametric regression. Standard errors come from the estimated influence function of a debiased, cross-fitted estimator in a single pass, with a model-refitting bootstrap retained as a diagnostic for specification uncertainty. These are capabilities that existing interpretability methods, including SHAP (Lundberg and Lee, 2017), LIME (Ribeiro et al., 2016), and partial dependence plots (Friedman, 2001), do not provide.

The theoretical contribution proceeds in three parts. The first is the estimand and its dual: the window average marginal effect is linear in the regression function and admits an exact Riesz representation by a change of variables, with no boundary terms and no density derivatives, and a representer that is bounded whenever the covariate density is bounded away from zero — and the classical average derivative is recovered in the small-window limit under one weak derivative. The second is inference and optimality: the orthogonal score built from the representer has estimation bias exactly equal to a product of two L^2 nuisance errors, so the cross-fitted estimator is root-n consistent and asymptotically normal under $n^{-1/4}$-type L^2 rates that flexible learners can satisfy, and it attains the semiparametric efficiency bound — replacing the parametric-rate function-space condition that a functional delta method treatment would require and that growing-complexity learners cannot meet; a double robustness proposition extends validity to inconsistent, fixed-complexity learners whenever the representer is known or parametrically estimable. The third is interpretation: a pairing identity shows that the window effect of any estimator of bounded variation equals its jump measure integrated against a locally averaged covariate density, characterizing the population quantity that a tree-based model's marginal effect measures; and gradient consistency is established for neural networks in the Sobolev space $W^{1,p}$ via Rellich–Kondrachov compactness (Evans, 2010), and for gradient boosted models with tree base learners and Mondrian forests (Mourtada et al., 2017) in the space of functions of bounded variation (Ambrosio et al., 2000) via BV compactness, characterizing the regime in which the uncor-

rected plug-in is already consistent. For Breiman random forests (Breiman, 2001a; Scornet et al., 2015), gradient consistency is conjectured rather than proved — the obstruction lies in the data-dependent partition boundaries — but the inference theory does not require it, so what remains open for that class is an L^2 rate question rather than a function-space one.

The empirical results across three datasets — UCI Adult Income (Kohavi, 1996; Dua and Graff, 2019), UCI Credit Card Default (Yeh, 2009; Dua and Graff, 2019), and Ames Housing (De Cock, 2011) — validate the framework on binary classification and continuous regression outcomes. Under the debiased estimator the four model classes agree far more closely than they did under the plug-in, and they do so because they are now estimating the same quantity rather than each reporting the slope of its own fit. The credit default application shows this most sharply: the plug-in put the effect of the payment ratio at -0.431 and -0.440 for the two tree ensembles, -0.045 for logistic regression and 0.0009 — effectively nothing — for the neural network, a set of estimates that cannot all be describing the same world; the corrected estimates are -0.489, -0.584, -0.265 and -0.194, all negative, all significant, and all telling one story. The variation that the plug-in displayed was in the main the regularization bias, which differs across learners because their regularization differs.

The standard errors move in the direction the theory predicts as well, and in several places that means upward. The refitting bootstrap reported standard errors of a few dollars for the neural network on Ames Housing and of 0.0000 to four decimal places for the random forest on credit default; the influence-function standard errors for the same quantities are three to four orders of magnitude larger. Those earlier intervals were narrow around a displaced center, which is the failure mode isolated in Simulation 2. The cost of the correction is visible in the smallest application: with $n = 1{,}460$ observations, several Ames Housing effects that the plug-in reported as precisely determined are no longer distinguishable from zero, and the estimator reports that rather than concealing it.

The comparison with SHAP values shows how far a different estimand can sit from this one: SHAP values are on a different scale, sometimes of opposite sign, and carry no

inferential apparatus. The Ames Housing application is particularly informative because the dollar-denominated units make the separation concrete — the SHAP figure for above-ground living area is roughly fifty times the AME for the linear model, which is what a baseline-relative attribution should look like next to a per-square-foot rate of change, and is not a competing estimate of the same quantity. Partial dependence slopes require a subtler reading than the earlier draft gave them. A partial dependence slope is itself a plug-in average over a single full-sample fit, so it agrees with the debiased AME where the plug-in is nearly unbiased — on smooth, well-supported, continuous features — and falls short of it, in about three quarters of the resolved cases toward zero, wherever regularization flattens the fitted surface. What once read as two estimators corroborating each other is better read as a direct measurement of the bias between them. An additional finding is that Breiman random forests produce less stable plug-in AME estimates than the other estimator classes, and the simulation study traces the resulting coverage failure to the same regularization bias — the first-order term that the orthogonal correction removes, and the clearest empirical argument for reporting the debiased estimator rather than the uncorrected average.

Several important problems remain open. The most pressing concerns Breiman forests, and the present framework splits it into two questions of very different difficulty. The inferential question — valid confidence intervals for the window effect when the learner is a Breiman forest — now requires only an L^2 convergence rate for the forest, since Theorem 1 consumes nothing else; consistency is available under structural assumptions (Scornet et al., 2015), and converting it into a rate meeting the $n^{-1/4}$ benchmark, or exploiting the product form of Assumption 3(b) with a fast representer estimate, is a concrete and plausibly tractable target. Simulation 3 indicates how much of that question is already settled in the case the theory handles cleanly. With the representer known, Proposition 2 asks nothing of the forest beyond convergence to some fixed limit, and the simulation exhibits exactly that situation: a depth-constrained forest whose plug-in error grows with the sample size, and therefore is not approaching f, nevertheless yields intervals whose coverage averages 0.948 across the

whole forest design and never leaves the band 0.922 to 0.968. What the open L^2 question governs is the case where the representer must itself be estimated, and it is there rather than in the known-representer case that a rate for Breiman forests would buy something. The interpretive question — convergence of the forest's own gradient information, in the sense of Section 3.6 — remains genuinely hard: the obstruction identified in the proof of Theorem 6, that data-dependent splits couple partition geometry to response variation in a way that prevents the key decomposition, suggests that new analytical tools will be required, possibly a coupling argument relating Breiman partitions to Mondrian partitions. Separating the two questions is itself progress: the most widely used forest implementation can be brought under the inferential guarantees without waiting for the harder interpretation theory.

A second open problem is the extension to additional model classes, and the framework now presents two templates with distinct proof obligations. For inference, the obligation is minimal and uniform across classes: an $L^2(P)$ convergence rate for the learner — support vector machines with smooth kernels, Bayesian additive regression trees, generalized additive models, and kernel ridge regression all have such rates in the literature or within reach — together with a Riesz-regression class rich enough to approximate the bounded representer, for which Chernozhukov et al. (2022) provide general guarantees. For interpretation, the Sobolev and BV architecture developed here remains the template: identify the natural function space of the estimator class, establish derivative existence and consistency there, and conclude gradient convergence via the appropriate compactness theorem. Splines and ridge regression are particularly natural candidates for the second template since they live in Sobolev spaces by construction, and the pairing identity of Theorem 3 extends the first template to any learner of bounded variation without further work.

A third open problem is the extension to causal inference. The average marginal effect as defined here is a predictive quantity — the average derivative of the conditional expectation function with respect to x_j, under the observational distribution of the covariates. It is not in general a causal effect. Extending the framework to settings where causal identification

is possible, for example through instrumental variables or regression discontinuity designs, would substantially broaden its applicability to policy-relevant questions. The honest forest framework of Wager and Athey (2018) and the broader literature on causal machine learning suggest a natural path forward, but the connection to the AME estimand studied here has not been formalized.

A fourth direction concerns the discrete and mixed-type feature setting. The current framework handles binary variables through a finite difference contrast and continuous variables through an adaptive centered finite difference, but ordered categorical variables, count variables, and interactions among features of mixed type are not treated systematically. The marginal effects at the mean versus average marginal effects distinction (Greene, 2012; Wooldridge, 2010) also warrants further study in the nonparametric setting, where the mean of the covariates may lie in a sparse region of the feature space and the marginal effect at the mean may therefore be a poor summary of the average effect.

Finally, several practical questions about the debiased procedure itself deserve systematic study. The influence-function standard errors of Section 2.4 already remove the computational objection to inference — K model fits in place of B refits — but the finite-sample behavior of the procedure depends on choices the asymptotic theory does not pin down: the number of folds, the truncation level for the estimated representer, the trimming weight near the boundary, and above all the window h, which is part of the estimand and whose sensitivity should be reported the way bandwidth sensitivity is. The multiplier bootstrap of Remark 3 makes simultaneous confidence bands across features essentially free, and characterizing its finite-sample accuracy relative to the plug-in intervals, along with honest procedures when the density bounds of Assumption 1 are weak near the boundary, is a practical program we leave to future work.

A Proof of Lemma 1 (Riesz Representation)

Proof. Recall $q = wp$ extended by zero outside Ω, so that q is defined on all of $\mathbb{R}^d$, $\|q\|_\infty \leq p_{\max}$, and $q(x) \neq 0$ implies $x \in \operatorname{supp}(w) \subseteq \Omega_{j,h}$. Fix $g \in L^1(P)$ and consider the positive-shift half of the finite difference:

$$\mathbb{E}\left[w(X)\, g(X + he_j)\right] = \int_{\mathbb{R}^d} q(x)\, g(x + he_j)\, dx = \int_{\mathbb{R}^d} q(u - he_j)\, g(u)\, du \tag{53}$$

by the substitution $u = x + he_j$. The manipulation is legitimate because on the set where the integrand is nonzero we have $x \in \Omega_{j,h}$, hence $u = x + he_j \in \Omega$, so g is evaluated only on Ω; and the integral is absolutely convergent since:

$$\int q(u - he_j)\ |g(u)|\ du \leq p_{\max} \int_\Omega |g(u)|\ du \leq \frac{p_{\max}}{p_{\min}}\, \mathbb{E}\,|g(X)| < \infty \tag{54}$$

The negative-shift half is identical with $u = x - he_j$. Subtracting and dividing by $2h$:

$$\mathbb{E}\left[w(X)\, \Delta_{j,h} g(X)\right] = \int_\Omega \frac{q(u - he_j) - q(u + he_j)}{2h}\, g(u)\, du = \mathbb{E}\left[\alpha_h(X)\, g(X)\right] \tag{55}$$

where the restriction of the domain of integration to Ω in the second step uses that g is supported there, and the final step multiplies and divides by $p(u) \geq p_{\min} > 0$. The uniform bound follows from $|q(u - he_j) - q(u + he_j)| \leq 2p_{\max}$ and $p(u) \geq p_{\min}$. Applying the representation to $g \equiv 1$ gives $\mathbb{E}[\alpha_h(X)] = \mathbb{E}[w(X)\Delta_{j,h}1] = 0$ since the finite difference of a constant vanishes. □

B Proof of Proposition 1 (Recovery of the Average Derivative)

Proof. Let $f \in W^{1,1}(\Omega)$ and fix $h \le h_0$, so that $\mathrm{supp}(w) \subseteq \Omega_{j,h_0} \subseteq \Omega_{j,h}$ and every segment $\{x + te_j : |t| \le h\}$ with $x \in \mathrm{supp}(w)$ lies in Ω. Sobolev functions are absolutely continuous on almost every line parallel to a coordinate axis, with the classical derivative along the line agreeing almost everywhere with the weak derivative (Evans and Gariepy, 2015, Section 4.9). Hence for almost every $x \in \mathrm{supp}(w)$:

$$f(x + he_j) - f(x - he_j) = \int_{-h}^{h} \partial_j f(x + te_j)\, dt \tag{56}$$

and therefore:

$$\theta_h(f) - \mathbb{E}\left[w(X)\, \partial_j f(X)\right] = \int_{\mathbb{R}^d} q(x)\, \frac{1}{2h} \int_{-h}^{h} \left[\partial_j f(x + te_j) - \partial_j f(x)\right] dt\, dx \tag{57}$$

Bounding q by $p_{\max}$, applying Tonelli, and extending $\partial_j f$ by zero outside Ω:

$$|\theta_h(f) - \mathbb{E}\left[w\, \partial_j f\right]| \le p_{\max} \sup_{|t| \le h} \int_{\mathbb{R}^d} |\partial_j f(x + te_j) - \partial_j f(x)|\, \mathbf{1}\{x \in \mathrm{supp}(w)\}\, dx \tag{58}$$

where the extension introduces no error because $x + te_j \in \Omega$ for $x \in \mathrm{supp}(w)$ and $|t| \le h \le h_0$. Continuity of translation in $L^1(\mathbb{R}^d)$ applied to the extension of $\partial_j f$ sends the right side to zero as $h \downarrow 0$. For the second claim, $|w_m \partial_j f| \le |\partial_j f|$, which is P-integrable because $\partial_j f \in L^1(\Omega)$ and $p \le p_{\max}$; dominated convergence gives $\mathbb{E}[w_m \partial_j f] \to \mathbb{E}[\partial_j f] = \mathrm{AME}_j$. □

C Proof of Lemma 2 (Exact Product Bias)

Proof. Conditioning on X and using $\mathbb{E}[Y \mid X] = f(X)$:

$$\mathbb{E}\left[\tilde{\alpha}(X)\,(Y - \tilde{f}(X))\right] = \mathbb{E}\left[\tilde{\alpha}(X)\,(f(X) - \tilde{f}(X))\right] \tag{59}$$

By linearity of $\Delta_{j,h}$ and Lemma 1 applied to $g = \tilde{f} - f \in L^1(P)$:

$$\mathbb{E}\left[w(X)\,\Delta_{j,h}\tilde{f}(X)\right] - \theta_h(f) = \mathbb{E}\left[w(X)\,\Delta_{j,h}(\tilde{f} - f)(X)\right] = \mathbb{E}\left[\alpha_h(X)\,(\tilde{f}(X) - f(X))\right] \tag{60}$$

Summing the two displays gives:

$$\mathbb{E}\left[\psi(X, Y; \tilde{f}, \tilde{\alpha}, \theta_h(f))\right] = \mathbb{E}\left[(\alpha_h(X) - \tilde{\alpha}(X))(\tilde{f}(X) - f(X))\right] \tag{61}$$

and the bound is Cauchy–Schwarz. □

D Proof of Theorem 1 (Asymptotic Linearity and Normality)

D.1 Step 0: The Trimmed Difference Operator is Bounded on $L^2(P)$

For any $g \in L^2(\Omega)$, using $(a-b)^2 \le 2a^2 + 2b^2$ and the change of variables from the proof of Lemma 1:

$$\begin{aligned}\mathbb{E}\left[\left(w(X)\,\Delta_{j,h}g(X)\right)^2\right] &\le \frac{1}{2h^2}\left(\mathbb{E}\left[w(X)^2 g(X + he_j)^2\right] + \mathbb{E}\left[w(X)^2 g(X - he_j)^2\right]\right) \\ &\le \frac{1}{h^2} \cdot \frac{p_{\max}}{p_{\min}}\,\mathbb{E}\left[g(X)^2\right]\end{aligned} \tag{62}$$

which establishes (27) with constant $\kappa = h^{-1}(p_{\max}/p_{\min})^{1/2}$. This is the only property of the finite difference used below; no derivative of g appears anywhere.

D.2 Step 1: Decomposition by Folds

Write $\theta = \theta_h(f)$, $\psi_i = \psi(\mathbf{x}_i, y_i; f, \alpha_h, \theta)$, and for $i \in I_k$ let $\hat{\psi}_i = \psi(\mathbf{x}_i, y_i; \hat{f}^{(-k)}, \hat{\alpha}^{(-k)}, \theta)$. Then:

$$\sqrt{n}\,(\hat{\theta}_h - \theta) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi_i + \sum_{k=1}^{K} \sqrt{\frac{n_k}{n}} \left(\sqrt{n_k}\, A_k + \sqrt{n_k}\, B_k\right) \tag{63}$$

where, with $\mathbb{E}_k$ denoting expectation over a fresh draw (X, Y) holding the fold-k nuisances fixed:

$$A_k = \mathbb{E}_k\left[\hat{\psi} - \psi\right], \qquad B_k = \frac{1}{n_k} \sum_{i \in I_k} \left(\hat{\psi}_i - \psi_i - A_k\right) \tag{64}$$

Since K is fixed and $n_k/n \to 1/K$, it suffices that $\sqrt{n_k} A_k = o_p(1)$ and $\sqrt{n_k} B_k = o_p(1)$ for each k.

D.3 Step 2: The Bias Term

By Lemma 2 with $(\tilde{f}, \tilde{\alpha}) = (\hat{f}^{(-k)}, \hat{\alpha}^{(-k)})$:

$$|\sqrt{n_k}\, A_k| \le \sqrt{n_k}\, \left\|\hat{\alpha}^{(-k)} - \alpha_h\right\|_{P,2} \cdot \left\|\hat{f}^{(-k)} - f\right\|_{P,2} = o_p(1) \tag{65}$$

by Assumption 3(b). This is the only place the product rate is used, and it is used without any expansion: the bound is exact.

D.4 Step 3: The Empirical Process Term

Conditional on the out-of-fold data, the summands of B_k are i.i.d. and mean zero, so:

$$\mathbb{E}\left[(\sqrt{n_k}\, B_k)^2 \mid \text{out-of-fold}\right] \le \mathbb{E}_k\left[(\hat{\psi} - \psi)^2\right] \tag{66}$$

Decompose, adding and subtracting $\hat{\alpha}^{(-k)}(x)(y - f(x))$:

$$\hat{\psi} - \psi = w\,\Delta_{j,h}(\hat{f}^{(-k)} - f) + (\hat{\alpha}^{(-k)} - \alpha_h)(y - f) - \hat{\alpha}^{(-k)}\,(\hat{f}^{(-k)} - f) \tag{67}$$

and bound each piece in $L^2(P)$. The middle piece is bounded by $\bar{\sigma}\,\left\|\hat{\alpha}^{(-k)} - \alpha_h\right\|_{P,2}$ after conditioning on X under either assumption set, since it uses only the conditional variance bound of Assumption 1. The first and third pieces are where the two assumption sets diverge.

Under the density bounds of Assumption 1, the first is bounded by $\kappa\left\|\hat{f}^{(-k)} - f\right\|_{P,2}$ via (62) and the third by $\bar{A}\left\|\hat{f}^{(-k)} - f\right\|_{P,2}$ using Assumption 3(c). All three are then $o_p(1)$ by Assumption 3(a).

Under Assumption 2 neither bound is available: the representer is unbounded, and the finite-difference operator is genuinely unbounded on $L^2(P)$ rather than merely unbounded by this argument. For independent standard normal covariates, taking $g_a(x) = e^{ax_j}$ gives $\|g_a\|_{P,2}^2 = e^{2a^2}$ and $\|g_a(\cdot + he_j)\|_{P,2}^2 = e^{2ah}e^{2a^2}$, so the operator norm is $\sup_a e^{ah} = \infty$. Both pieces are instead bounded by Hölder. Writing $\Delta = \hat{f}^{(-k)} - f$ and letting r be as in (10) with conjugate r', a change of variables gives

$$\mathbb{E}\left[\Delta(X \pm he_j)^2\right] = \mathbb{E}\left[\Delta(X)^2\,\frac{p(X \mp he_j)}{p(X)}\right] \le \|\Delta\|_{P,2r'}^2\left(\mathbb{E}\left[\left(\frac{p(X \mp he_j)}{p(X)}\right)^r\right]\right)^{1/r} \tag{68}$$

which is $o_p(1)$ by Assumption 3(a) in its $L^{2+\delta}$ form, with $2r' = 2 + \delta$; the first piece follows from (68) and $|w| \le 1$. For the third, $\mathbb{E}[(\hat{\alpha}^{(-k)})^2\Delta^2] \le \left\|\hat{\alpha}^{(-k)}\right\|_{P,q}^2\|\Delta\|_{P,2s'}^2$ by Hölder with $s = q/2$, which is $o_p(1)$ by Assumption 3(c) in its moment form together with (a).

Either way $\mathbb{E}_k[(\hat{\psi} - \psi)^2] = o_p(1)$, and conditional Chebyshev gives $\sqrt{n_k}\,B_k = o_p(1)$. Note that only consistency of the nuisances is used here, never a rate; the rate enters at Step 2 alone.

D.5 Step 4: Central Limit Theorem and Variance Estimation

The ψ_i are i.i.d., mean zero by Lemma 2 at $(\tilde{f}, \tilde{\alpha}) = (f, \alpha_h)$, with variance:

$$V_h = \mathbb{E}[\psi_i^2] \leq 2\,\mathbb{E}\left[(w\Delta_{j,h}f)^2\right] + 2\,\|\alpha_h\|_\infty^2\,\mathbb{E}[\sigma^2(X)] < \infty \tag{69}$$

by (62) and Lemma 1 under the density bounds of Assumption 1. Under Assumption 2 the same finiteness follows from (68) applied to f, which is bounded, together with $\alpha_h \in L^2(P)$ from Assumption 2(c) and $\mathbb{E}[Y^2] < \infty$; for the Gaussian design the representer's second moment is (11). Either way $\mathrm{Var}(\psi) < \infty$ and the classical CLT applies. For the variance estimator, write $\hat{\psi}_i(\hat{\theta}_h) = \hat{\psi}_i - (\hat{\theta}_h - \theta)$; then:

$$\hat{V}_h = \frac{1}{n}\sum_i \hat{\psi}_i^2 - 2(\hat{\theta}_h - \theta)\,\frac{1}{n}\sum_i \hat{\psi}_i + (\hat{\theta}_h - \theta)^2 \tag{70}$$

The last two terms vanish in probability since $\hat{\theta}_h - \theta = o_p(1)$ and $n^{-1}\sum_i \hat{\psi}_i = O_p(n^{-1/2})$ by Steps 1–3. For the first term:

$$\left|\frac{1}{n}\sum_i \hat{\psi}_i^2 - \frac{1}{n}\sum_i \psi_i^2\right| \leq \frac{1}{n}\sum_i (\hat{\psi}_i - \psi_i)^2 + 2\left(\frac{1}{n}\sum_i (\hat{\psi}_i - \psi_i)^2\right)^{1/2}\left(\frac{1}{n}\sum_i \psi_i^2\right)^{1/2} \tag{71}$$

by Cauchy–Schwarz; $n^{-1}\sum_i(\hat{\psi}_i - \psi_i)^2 = o_p(1)$ by the fold-wise conditional Markov inequality and the Step 3 bound, and $n^{-1}\sum_i \psi_i^2 \xrightarrow{p} V_h$ by the law of large numbers. Hence $\hat{V}_h \xrightarrow{p} V_h$. □

E Proof of Proposition 2 (Double Robustness)

The argument retraces the proof of Theorem 1 with the pseudo-true influence function in place of the efficient one; we record the steps that change.

E.1 Step 1: The Pseudo-True Influence Function

Define $\psi_{g,i} = w(\mathbf{x}_i)\,\Delta_{j,h}g(\mathbf{x}_i) + \alpha_h(\mathbf{x}_i)\,(y_i - g(\mathbf{x}_i)) - \theta_h(f)$. By Lemma 2 with $(\tilde{f}, \tilde{\alpha}) = (g, \alpha_h)$:

$$\mathbb{E}[\psi_{g,i}] = \mathbb{E}\big[(\alpha_h - \alpha_h)(g - f)\big] = 0 \tag{72}$$

for every $g \in L^2(\Omega)$, which also establishes $\mathbb{E}[m_g(X)] = \theta_h(f)$ after conditioning on X. Conditioning likewise gives $\mathbb{E}[\psi_{g,i} \mid X] = m_g(X) - \theta_h(f)$ and $\mathrm{Var}(\psi_{g,i} \mid X) = \alpha_h(X)^2\sigma^2(X)$, so $\mathrm{Var}(\psi_{g,i}) = V_h(g)$ by the law of total variance; finiteness follows from (62), $g \in L^2(\Omega)$, the bound on α_h from Lemma 1, and $\mathbb{E}[Y^2] < \infty$ under the density bounds; under Assumption 2 it follows from (68), $\alpha_h \in L^2(P)$, and $\mathbb{E}[Y^2] < \infty$.

E.2 Step 2: The Bias Term

With $A_k = \mathbb{E}_k[\hat{\psi} - \psi_g] = \mathbb{E}_k[\hat{\psi}]$, Lemma 2 gives $A_k = \mathbb{E}[(\alpha_h - \hat{\alpha}^{(-k)})(\hat{f}^{(-k)} - f)]$, so:

$$|\sqrt{n_k}\,A_k| \le \sqrt{n_k}\,\left\|\hat{\alpha}^{(-k)} - \alpha_h\right\|_{P,2}\left(\left\|\hat{f}^{(-k)} - g\right\|_{P,2} + \|g - f\|_{P,2}\right) = \sqrt{n_k}\,o_p(n^{-1/2})\cdot O_p(1) = o_p(1) \tag{73}$$

by conditions (a) and (b); the second factor is $O_p(1)$ rather than $o_p(1)$ because g need not equal f, and it is exactly here that condition (b) must be strengthened to $o_p(n^{-1/2})$ relative to Assumption 3(b). When $\hat{\alpha}^{(-k)} = \alpha_h$, $A_k = 0$ identically at every n; averaging the fold-wise identity $\mathbb{E}[\hat{\psi}_i + \theta_h(f) \mid \text{out-of-fold data}] = \theta_h(f)$, which holds because the observations in fold k are independent of $\hat{f}^{(-k)}$, gives the exact unbiasedness claim.

E.3 Step 3: The Empirical Process Term

Decompose, for $i \in I_k$:

$$\hat{\psi}_i - \psi_{g,i} = w\,\Delta_{j,h}(\hat{f}^{(-k)} - g) + (\hat{\alpha}^{(-k)} - \alpha_h)(y_i - \hat{f}^{(-k)}) + \alpha_h\,(g - \hat{f}^{(-k)}) \tag{74}$$

The first and third terms have conditional $L^2(P)$ norms bounded by $\kappa \left\| \hat{f}^{(-k)} - g \right\|_{P,2}$ and $\|\alpha_h\|_\infty \left\| \hat{f}^{(-k)} - g \right\|_{P,2}$ respectively, both $o_p(1)$ by condition (a). For the middle term, write $y - \hat{f}^{(-k)} = (y - f) + (f - g) + (g - \hat{f}^{(-k)})$ and bound the three resulting pieces: the first by $\bar{\sigma}^2 \left\| \hat{\alpha}^{(-k)} - \alpha_h \right\|_{P,2}^2 = o_p(1)$ after conditioning on X; the third by $(\bar{A} + \|\alpha_h\|_\infty)^2 \left\| \hat{f}^{(-k)} - g \right\|_{P,2}^2 = o_p(1)$; and the second piece, $\mathbb{E}[(\hat{\alpha}^{(-k)} - \alpha_h)^2 (f-g)^2]$, tends to zero. Under the density bounds this is dominated convergence along subsequences: the factor $(\hat{\alpha}^{(-k)} - \alpha_h)^2$ is bounded by $(\bar{A} + \|\alpha_h\|_\infty)^2$ and converges to zero in P-measure by condition (b), while $(f-g)^2 \in L^1(P)$ dominates. Under Assumption 2 the domination is unavailable and Hölder is used instead: $\mathbb{E}[(\hat{\alpha}^{(-k)} - \alpha_h)^2 (f-g)^2] \leq \left\| \hat{\alpha}^{(-k)} - \alpha_h \right\|_{P,q}^2 \|f - g\|_{P,2t'}^2$ with $t = q/2$, which requires the fixed approximation error $f - g$ to lie in $L^{2t'}(P)$ — a mild condition, and one that holds whenever g and f are bounded. Conditional Chebyshev then gives $\sqrt{n_k}\, B_k = o_p(1)$ exactly as in Step 3 of the proof of Theorem 1.

E.4 Step 4: Conclusion

The $\psi_{g,i}$ are i.i.d., mean zero, with finite positive variance $V_h(g)$, so the classical CLT applies to their normalized sum, and Steps 2–3 identify $\sqrt{n}(\hat{\theta}_h - \theta_h(f))$ with that sum up to $o_p(1)$. The consistency of $\hat{V}_h$ for $V_h(g)$ repeats Step 4 of the proof of Theorem 1 verbatim with ψ_g in place of ψ, using the Step 3 bound just established. Finally, when $g \neq f$ the influence function ψ_g differs from the efficient influence function ψ of Theorem 2; since in the nonparametric model every regular asymptotically linear estimator has influence function equal to the efficient one, the estimator is not regular in this case, and no conflict with the convolution theorem arises. □

F Proof of Theorem 2 (Semiparametric Efficiency)

Proof. Consider one-dimensional submodels $dP_t = (1 + ts)\, dP$ indexed by bounded functions $s \in L_0^2(P) = \{s : \mathbb{E}[s(X,Y)] = 0\}$, which are well defined for $|t| < 1/\|s\|_\infty$ and differentiable

in quadratic mean at $t = 0$ with score s; bounded mean-zero functions are dense in $L_0^2(P)$, so the tangent space of the nonparametric model is all of $L_0^2(P)$ (van der Vaart, 1998, Chapter 25). Write $\bar{s}_X(x) = \mathbb{E}[s(X,Y) \mid X = x]$ for the induced score of the covariate marginal and $f_t(x) = \mathbb{E}_t[Y \mid X = x]$ for the regression function along the path. Differentiating the Bayes formula for f_t at $t = 0$ gives, for P-almost every x:

$$\dot{f}(x) := \frac{\partial}{\partial t} f_t(x)\Big|_{t=0} = \mathbb{E}\left[(Y - f(X))\, s(X,Y) \mid X = x\right] \tag{75}$$

where differentiation under the integral is justified by $\mathbb{E}[Y^2] < \infty$ and the boundedness of s. The functional along the path is $\theta(t) = \mathbb{E}_t[w(X)\, \Delta_{j,h} f_t(X)]$, and the product rule gives:

$$\frac{d\theta}{dt}\Big|_{t=0} = \mathbb{E}\left[w\, \Delta_{j,h} f(X) \cdot \bar{s}_X(X)\right] + \mathbb{E}\left[w\, \Delta_{j,h} \dot{f}(X)\right] \tag{76}$$

For the first term, note $\mathbb{E}[\bar{s}_X] = 0$, so the term equals $\mathbb{E}[(w\Delta_{j,h} f(X) - \theta_h(f))\, \bar{s}_X(X)] = \mathbb{E}[(w\Delta_{j,h} f(X) - \theta_h(f))\, s(X,Y)]$, the last equality because $w\Delta_{j,h} f(X) - \theta_h(f)$ is X-measurable. For the second term, $\dot{f} \in L^1(P)$ (indeed L^2) since s is bounded, so Lemma 1 applies:

$$\mathbb{E}\left[w\, \Delta_{j,h} \dot{f}(X)\right] = \mathbb{E}\left[\alpha_h(X)\, \dot{f}(X)\right] = \mathbb{E}\left[\alpha_h(X)\, (Y - f(X))\, s(X,Y)\right] \tag{77}$$

by the tower property. Summing:

$$\frac{d\theta}{dt}\Big|_{t=0} = \mathbb{E}\left[\psi(X, Y; f, \alpha_h, \theta_h(f))\; s(X,Y)\right] \tag{78}$$

for every score s, so θ_h is pathwise differentiable with influence function ψ; since $\psi \in L_0^2(P)$ lies in the tangent space, it is the efficient influence function and the bound is $\mathbb{E}[\psi^2]$ (van der Vaart, 1998, Theorem 25.20). The cross term in $\mathbb{E}[\psi^2]$ vanishes by conditioning on X, yielding (30). Attainment follows from Theorem 1, whose expansion exhibits $\hat{\theta}_h$ as asymptotically linear with influence function ψ, hence regular and efficient (van der Vaart, 1998,

Lemma 25.23). □

G Proof of Lemma 3 (Derivative Existence, Sobolev)

G.1 What Needs to Be Shown

That $\hat{f}_n \in W^{2,p}(\Omega)$ for $p > d$ implies $\nabla \hat{f}_n \in W^{1,p}(\Omega)^d$ with the norm bound:

$$\left\|\nabla \hat{f}_n\right\|_{L^p} \leq \left\|\hat{f}_n\right\|_{W^{1,p}} \leq \left\|\hat{f}_n\right\|_{W^{2,p}} \tag{79}$$

Remark 7. *The condition $p > d$ is required by the proof. It is the condition under which Theorem 6.5 of Evans and Gariepy (2015) guarantees pointwise differentiability of $W^{1,p}_{loc}$ functions. This is slightly stronger than the condition $p > 1$ stated in the main text and is an explicit assumption of the lemma.*

G.2 Step 1: Existence of the Gradient as an L^p Function

Since $\hat{f}_n \in W^{2,p}(\Omega)$ and $p > d$, we have in particular that $\hat{f}_n \in W^{1,p}(\Omega)$. Extending $\hat{f}_n$ to a $W^{1,p}_{loc}(\mathbb{R}^d)$ function by a standard extension operator (Evans and Gariepy, 2015, Section 4.4), we may apply Theorem 6.5 of Evans and Gariepy (2015):

> *Theorem 6.5 (Evans and Gariepy, 2015).* Assume $f \in W^{1,p}_{loc}(\mathbb{R}^n)$ for some $n < p \leq \infty$. Then f is differentiable $\mathcal{L}^n$-a.e., and its classical derivative equals its weak derivative $\mathcal{L}^n$-a.e.

Applying this theorem to $\hat{f}_n$ with $n = d$ and $p > d$, we conclude that $\hat{f}_n$ is classically differentiable at $\mathcal{L}^d$-almost every $x \in \Omega$, and:

$$\nabla \hat{f}_n(x) = D\hat{f}_n(x) \quad \mathcal{L}^d\text{-a.e.} \tag{80}$$

where $D\hat{f}_n$ denotes the weak gradient of $\hat{f}_n$. Since $\hat{f}_n \in W^{1,p}(\Omega)$, the weak gradient $D\hat{f}_n \in L^p(\Omega)^d$ by definition of the Sobolev space. Therefore $\nabla \hat{f}_n \in L^p(\Omega)^d$.

Applying the same argument to each first-order weak derivative $\partial_j \hat{f}_n \in W^{1,p}(\Omega)$ — which belongs to $W^{1,p}(\Omega)$ because $\hat{f}_n \in W^{2,p}(\Omega)$ — we conclude that each $\partial_j \hat{f}_n$ is classically differentiable $\mathcal{L}^d$-a.e. with classical second derivatives in $L^p(\Omega)$. Therefore $\nabla \hat{f}_n \in W^{1,p}(\Omega)^d$.

G.3 Step 2: The Norm Bound

We establish the chain of inequalities:

$$\left\|\nabla \hat{f}_n\right\|_{L^p} \leq \left\|\hat{f}_n\right\|_{W^{1,p}} \leq \left\|\hat{f}_n\right\|_{W^{2,p}} \tag{81}$$

First Inequality

By definition of the $W^{1,p}(\Omega)$ norm:

$$\left\|\hat{f}_n\right\|_{W^{1,p}} = \left\|\hat{f}_n\right\|_{L^p} + \sum_{j=1}^{d} \left\|\partial_j \hat{f}_n\right\|_{L^p} = \left\|\hat{f}_n\right\|_{L^p} + \left\|\nabla \hat{f}_n\right\|_{L^p} \tag{82}$$

Since $\left\|\hat{f}_n\right\|_{L^p} \geq 0$:

$$\left\|\hat{f}_n\right\|_{W^{1,p}} = \left\|\hat{f}_n\right\|_{L^p} + \left\|\nabla \hat{f}_n\right\|_{L^p} \geq \left\|\nabla \hat{f}_n\right\|_{L^p} \tag{83}$$

Second Inequality

By definition of the $W^{2,p}(\Omega)$ norm using multi-index notation:

$$\left\|\hat{f}_n\right\|_{W^{2,p}} = \sum_{|\alpha| \leq 2} \left\|\partial^\alpha \hat{f}_n\right\|_{L^p} \tag{84}$$

The multi-indices with $|\alpha| \leq 1$ form a strict subset of those with $|\alpha| \leq 2$, and all terms are nonnegative, so:

$$\left\|\hat{f}_n\right\|_{W^{2,p}} = \sum_{|\alpha|\leq 2} \left\|\partial^\alpha \hat{f}_n\right\|_{L^p} \geq \sum_{|\alpha|\leq 1} \left\|\partial^\alpha \hat{f}_n\right\|_{L^p} = \left\|\hat{f}_n\right\|_{W^{1,p}} \tag{85}$$

G.4 Conclusion

For $p > d$, membership of $\hat{f}_n$ in $W^{2,p}(\Omega)$ implies via Theorem 6.5 of Evans and Gariepy (2015) that $\nabla \hat{f}_n$ exists $\mathcal{L}^d$-almost everywhere as a classical gradient, coincides with the weak gradient, and belongs to $W^{1,p}(\Omega)^d \subset L^p(\Omega)^d$. The gradient operator $\nabla : W^{2,p}(\Omega) \to W^{1,p}(\Omega)^d$ is bounded with:

$$\left\|\nabla \hat{f}_n\right\|_{L^p} \leq \left\|\hat{f}_n\right\|_{W^{1,p}} \leq \left\|\hat{f}_n\right\|_{W^{2,p}} \tag{86}$$

□

H Proof of Lemma 4 (Derivative Consistency, Sobolev)

H.1 What Needs to Be Shown

That well-behavedness in the Sobolev sense implies $\left\|\nabla \hat{f}_n - \nabla f\right\|_{L^p} \xrightarrow{p} 0$.

H.2 Proof Strategy

The proof uses a subsequence argument. We show that every subsequence of $\{\hat{f}_n\}$ has a further subsequence along which $\left\|\nabla \hat{f}_n - \nabla f\right\|_{L^p} \to 0$ almost surely. By the standard metric space characterization of convergence in probability — a sequence converges in probability if and only if every subsequence has an almost surely convergent further subsequence — this implies $\left\|\nabla \hat{f}_n - \nabla f\right\|_{L^p} \xrightarrow{p} 0$.

Suppose for contradiction that $\left\|\nabla \hat{f}_n - \nabla f\right\|_{L^p}$ does not converge to zero in probability. Then there exist $\epsilon > 0$ and $\delta > 0$ and a subsequence $\{n_k\}$ such that:

$$\mathbb{P}\Big(\left\|\nabla \hat{f}_{n_k} - \nabla f\right\|_{L^p} > \epsilon\Big) > \delta \quad \forall k \tag{87}$$

We derive a contradiction by extracting a further subsequence along which the gradient converges to zero almost surely.

H.3 Step 1: Extract a Strongly Convergent Subsequence

Take any subsequence $\{\hat{f}_{n_k}\}$. By the uniform $W^{2,p}(\Omega)$ boundedness condition:

$$\sup_k \left\|\hat{f}_{n_k}\right\|_{W^{2,p}} < \infty \quad \text{almost surely} \tag{88}$$

The Rellich–Kondrachov compact embedding theorem states that for bounded Lipschitz Ω, the embedding $W^{2,p}(\Omega) \hookrightarrow\hookrightarrow W^{1,p}(\Omega)$ is compact (Evans, 2010, Theorem 5, Section 5.7). Compactness of the embedding means that every sequence bounded in $W^{2,p}(\Omega)$ has a subsequence converging strongly in $W^{1,p}(\Omega)$. Therefore there exists a further subsequence $\{\hat{f}_{n_{k_j}}\}$ and a function $g \in W^{1,p}(\Omega)$ such that:

$$\left\|\hat{f}_{n_{k_j}} - g\right\|_{W^{1,p}} \to 0 \quad \text{almost surely as } j \to \infty \tag{89}$$

H.4 Step 2: Identify the Limit

Strong $W^{1,p}(\Omega)$ convergence implies strong L^p convergence, since by definition of the $W^{1,p}$ norm:

$$\left\|\hat{f}_{n_{k_j}} - g\right\|_{L^p} \leq \left\|\hat{f}_{n_{k_j}} - g\right\|_{W^{1,p}} \to 0 \quad \text{almost surely} \tag{90}$$

Therefore $\hat{f}_{n_{k_j}} \to g$ in L^p almost surely. By the L^p consistency assumption, $\left\|\hat{f}_n - f\right\|_{L^p} \xrightarrow{p} 0$, so in particular along the subsequence $\hat{f}_{n_{k_j}} \to f$ in L^p in probability, and hence almost surely

along a further subsequence if necessary. By uniqueness of limits in $L^p(\Omega)$:

$$g = f \quad \mathcal{L}^d\text{-a.e. on } \Omega \tag{91}$$

Therefore the strongly convergent subsequence satisfies:

$$\left\| \hat{f}_{n_{k_j}} - f \right\|_{W^{1,p}} \to 0 \quad \text{almost surely} \tag{92}$$

H.5 Step 3: Extract Gradient Convergence

Since $\hat{f}_{n_{k_j}} \to f$ strongly in $W^{1,p}(\Omega)$, expanding the $W^{1,p}$ norm:

$$\left\| \hat{f}_{n_{k_j}} - f \right\|_{W^{1,p}} = \left\| \hat{f}_{n_{k_j}} - f \right\|_{L^p} + \left\| \nabla \hat{f}_{n_{k_j}} - \nabla f \right\|_{L^p} \to 0 \quad \text{almost surely} \tag{93}$$

Since both terms are nonnegative and their sum goes to zero, each term goes to zero individually. In particular:

$$\left\| \nabla \hat{f}_{n_{k_j}} - \nabla f \right\|_{L^p} \to 0 \quad \text{almost surely as } j \to \infty \tag{94}$$

H.6 Conclusion

We have shown that from any subsequence $\{\hat{f}_{n_k}\}$ we can extract a further subsequence $\{\hat{f}_{n_{k_j}}\}$ along which $\left\| \nabla \hat{f}_{n_{k_j}} - \nabla f \right\|_{L^p} \to 0$ almost surely. This contradicts the assumption that $\mathbb{P}(\left\| \nabla \hat{f}_{n_k} - \nabla f \right\|_{L^p} > \epsilon) > \delta$ for all k. Therefore the assumption was false and:

$$\left\| \nabla \hat{f}_n - \nabla f \right\|_{L^p} \xrightarrow{p} 0 \tag{95}$$

which is the conclusion of Lemma 4. □

Remark 8. *The key step is the application of Rellich–Kondrachov in Step 1. The theorem requires Ω to be bounded and Lipschitz — both of which are assumed throughout — and*

$p > 1$, which is assumed in the lemma statement. The compact embedding $W^{2,p} \hookrightarrow\hookrightarrow W^{1,p}$ is strictly stronger than the continuous embedding $W^{2,p} \hookrightarrow W^{1,p}$: the latter follows from the norm bound in Lemma 3, while the former provides the compactness needed to extract convergent subsequences. It is this compactness, not merely the norm bound, that allows the interchange of limit and gradient in the proof.

I Proof of Lemma 5 (Derivative Existence, BV)

I.1 What Needs to Be Shown

Two things: that BV membership implies $D\hat{f}_n$ exists as a Radon measure with finite total variation, and the explicit formula for piecewise constant functions.

I.2 Step 1: Existence of $D\hat{f}_n$ as a Radon Measure

We establish existence in three steps, using the Riesz representation theorem and the Radon-Nikodym theorem.

Substep 1a: The Distributional Gradient as a Linear Functional

For any $\hat{f}_n \in BV(\Omega)$, the distributional gradient is defined as the linear functional $\Lambda : C_c^1(\Omega)^d \to \mathbb{R}$ given by:

$$\Lambda(\phi) = -\int_\Omega \hat{f}_n \operatorname{div} \phi \, dx \quad \forall \phi \in C_c^1(\Omega)^d \tag{96}$$

By the definition of $BV(\Omega)$, this functional has finite total variation:

$$|\Lambda|(\Omega) = \sup\left\{\Lambda(\phi) : \phi \in C_c^1(\Omega)^d,\, \|\phi\|_{L^\infty} \leq 1\right\} < \infty \tag{97}$$

This is precisely the condition that $\hat{f}_n \in BV(\Omega)$ imposes — the distributional gradient is a bounded linear functional on $C_c^1(\Omega)^d$ with respect to the supremum norm.

Substep 1b: Riesz Representation

Since Λ is a bounded linear functional on $C_c^1(\Omega)^d$ with finite total variation, the Riesz representation theorem (Evans and Gariepy, 2015, Theorem 1.38) guarantees the existence of a vector-valued Radon measure $D\hat{f}_n \in \mathcal{M}(\Omega)^d$ such that:

$$\Lambda(\phi) = \int_\Omega \phi \cdot d(D\hat{f}_n) \quad \forall \phi \in C_c^1(\Omega)^d \tag{98}$$

That is:

$$-\int_\Omega \hat{f}_n \operatorname{div} \phi \, dx = \int_\Omega \phi \cdot d(D\hat{f}_n) \quad \forall \phi \in C_c^1(\Omega)^d \tag{99}$$

This is the distributional gradient equation, and $D\hat{f}_n$ is the distributional gradient of $\hat{f}_n$ represented as a Radon measure.

Substep 1c: Finiteness of Total Variation

By the Radon-Nikodym theorem for Radon measures (Evans and Gariepy, 2015, Theorem 1.29), the derivative $D_{\mathcal{L}^d}(D\hat{f}_n)$ exists and is finite $\mathcal{L}^d$-almost everywhere on Ω. Since Ω is bounded and the total variation of Λ is finite by the BV assumption, we conclude:

$$|D\hat{f}_n|(\Omega) = |\Lambda|(\Omega) < \infty \tag{100}$$

Therefore $D\hat{f}_n \in \mathcal{M}(\Omega)^d$ with $|D\hat{f}_n|(\Omega) < \infty$, as claimed.

I.3 Step 2: Explicit Formula for Piecewise Constant Functions

Let $\hat{f}_n = \sum_j \bar{y}_j \mathbf{1}_{R_j}$ where $\{R_j\}$ is a finite partition of Ω into sets with Lipschitz boundaries, and $\bar{y}_j$ denotes the response mean on leaf R_j. We derive the explicit formula for $D\hat{f}_n$ by

applying the Gauss–Green theorem to each leaf.

Substep 2a: Applying the Definition

For any test function $\phi \in C_c^1(\Omega)^d$, the distributional gradient equation from Substep 1b gives:

$$\int_\Omega \phi \cdot d(D\hat{f}_n) = -\int_\Omega \hat{f}_n \operatorname{div} \phi \, dx = -\sum_j \bar{y}_j \int_{R_j} \operatorname{div} \phi \, dx \tag{101}$$

where the second equality uses the piecewise constant structure of $\hat{f}_n$ and linearity of the integral.

Substep 2b: Gauss–Green on Each Leaf

Since each R_j has Lipschitz boundary, the Gauss–Green theorem for BV functions (Evans and Gariepy, 2015, Theorem 5.16) applies to each leaf:

$$\int_{R_j} \operatorname{div} \phi \, dx = \int_{\partial R_j} \phi \cdot \nu_j \, d\mathcal{H}^{d-1} \tag{102}$$

where ν_j denotes the outward unit normal to ∂R_j and $\mathcal{H}^{d-1}$ is the $(d-1)$-dimensional Hausdorff measure on the boundary.

Substep 2c: Cancellation on Interior Boundaries

Substituting the Gauss–Green formula into the expression from Substep 2a:

$$\int_\Omega \phi \cdot d(D\hat{f}_n) = -\sum_j \bar{y}_j \int_{\partial R_j} \phi \cdot \nu_j \, d\mathcal{H}^{d-1} \tag{103}$$

Every interior boundary $\partial R_j \cap \partial R_k$ between adjacent leaves j and k appears twice in this sum — once with outward normal ν_j for leaf R_j and once with outward normal $\nu_k = -\nu_j$ for leaf R_k. Collecting these paired contributions and writing $\nu_{jk} = \nu_j$ for the unit normal

pointing from R_j into R_k:

$$\begin{aligned} -\sum_j \bar{y}_j \int_{\partial R_j} \phi \cdot \nu_j \, d\mathcal{H}^{d-1} &= -\sum_{j\sim k} \bar{y}_j \int_{\partial R_j \cap \partial R_k} \phi \cdot \nu_{jk} \, d\mathcal{H}^{d-1} \\ &\quad + \sum_{j\sim k} \bar{y}_k \int_{\partial R_j \cap \partial R_k} \phi \cdot \nu_{jk} \, d\mathcal{H}^{d-1} \\ &= \sum_{j\sim k} (\bar{y}_k - \bar{y}_j) \int_{\partial R_j \cap \partial R_k} \phi \cdot \nu_{jk} \, d\mathcal{H}^{d-1} \end{aligned} \tag{104}$$

Boundary terms on $\partial\Omega$ vanish because $\phi \in C_c^1(\Omega)^d$ has compact support in Ω.

Substep 2d: Identifying the Measure

Since the equation:

$$\int_\Omega \phi \cdot d(D\hat{f}_n) = \sum_{j\sim k} (\bar{y}_k - \bar{y}_j) \int_{\partial R_j \cap \partial R_k} \phi \cdot \nu_{jk} \, d\mathcal{H}^{d-1} \tag{105}$$

holds for all $\phi \in C_c^1(\Omega)^d$, the uniqueness part of the Riesz representation theorem (Evans and Gariepy, 2015, Theorem 1.38) identifies $D\hat{f}_n$ uniquely as the measure:

$$D\hat{f}_n = \sum_{j\sim k} (\bar{y}_k - \bar{y}_j) \cdot \nu_{jk} \cdot \mathcal{H}^{d-1} \lfloor_{\partial R_j \cap \partial R_k} \tag{106}$$

where $\mathcal{H}^{d-1} \lfloor_{\partial R_j \cap \partial R_k}$ denotes the $(d-1)$-dimensional Hausdorff measure restricted to the shared boundary between leaves j and k. This is the formula stated in the lemma. □

J Proof of Lemma 6 (Derivative Consistency, BV)

J.1 What Needs to Be Shown

That well-behavedness in the BV sense implies $D\hat{f}_n \xrightarrow{w*} Df$ in $\mathcal{M}(\Omega)^d$, and the strict convergence upgrade under the additional condition $\mathbb{E}[|D\hat{f}_n|(\Omega)] \to |Df|(\Omega)$.

J.2 Proof Strategy

The proof mirrors the subsequence argument of Lemma 4, replacing the Rellich–Kondrachov compact embedding with the BV compactness theorem and replacing strong L^p convergence with weak-* convergence of measures. We show that every subsequence of $\{D\hat{f}_n\}$ has a further subsequence converging weak-* to Df. By the metric space characterization of convergence in probability, this implies $D\hat{f}_n \xrightarrow{w*} Df$ in $\mathcal{M}(\Omega)^d$ in probability.

Suppose for contradiction that $D\hat{f}_n$ does not converge weak-* to Df in probability. Then there exist $\epsilon > 0$, $\delta > 0$, a test function $\phi \in C_0(\Omega)^d$ with $\|\phi\|_\infty \leq 1$, and a subsequence $\{n_k\}$ such that:

$$\mathbb{P}\left(\left|\int_\Omega \phi \cdot d(D\hat{f}_{n_k}) - \int_\Omega \phi \cdot d(Df)\right| > \epsilon\right) > \delta \quad \forall k \tag{107}$$

We derive a contradiction by extracting a further subsequence along which weak-* convergence holds almost surely.

J.3 Part I: Weak-* Convergence

Step 1: Bound the Full BV Norm

To apply the BV compactness theorem we require a uniform bound on the full BV norm:

$$\left\|\hat{f}_n\right\|_{BV} = \left\|\hat{f}_n\right\|_{L^1} + |D\hat{f}_n|(\Omega) \tag{108}$$

For the total variation term, the well-behaved condition gives:

$$\sup_n \mathbb{E}\Big[|D\hat{f}_n|(\Omega)\Big] < \infty \tag{109}$$

so $|D\hat{f}_n|(\Omega)$ is bounded in expectation and hence bounded almost surely along subsequences by Markov's inequality.

For the L^1 term, the consistency assumption $\left\|\hat{f}_n - f\right\|_{L^1} \xrightarrow{p} 0$ implies that $\left\|\hat{f}_n\right\|_{L^1}$ is

eventually bounded in probability. Since $f \in W^{1,p}(\Omega) \subset L^1(\Omega)$, we have $\|f\|_{L^1} < \infty$, and:

$$\left\|\hat{f}_n\right\|_{L^1} \leq \left\|\hat{f}_n - f\right\|_{L^1} + \|f\|_{L^1} \tag{110}$$

which is bounded in probability. Combining the two terms:

$$\sup_n \left\|\hat{f}_n\right\|_{BV} < \infty \quad \text{almost surely} \tag{111}$$

Step 2: Extract a Weakly-* Convergent Subsequence

Take any subsequence $\{\hat{f}_{n_k}\}$. By Step 1 it is bounded in $BV(\Omega)$ almost surely. The BV compactness theorem (Evans and Gariepy, 2015, Theorem 5.5) states that every sequence bounded in $BV(\Omega)$ for bounded Lipschitz Ω has a subsequence $\{\hat{f}_{n_{k_j}}\}$ and $g \in BV(\Omega)$ such that:

1. $\hat{f}_{n_{k_j}} \to g$ strongly in $L^1(\Omega)$
2. $D\hat{f}_{n_{k_j}} \xrightarrow{w*} Dg$ in $\mathcal{M}(\Omega)^d$

both holding almost surely. This is the BV analog of the Rellich–Kondrachov compact embedding: the BV compactness theorem gives simultaneously the L^1 convergence of the functions and the weak-* convergence of their gradient measures, in a single extraction.

Step 3: Identify the Limit

From item (1) of Step 2, $\hat{f}_{n_{k_j}} \to g$ strongly in $L^1(\Omega)$ almost surely. By the L^1 consistency assumption $\left\|\hat{f}_n - f\right\|_{L^1} \xrightarrow{p} 0$, along the subsequence $\hat{f}_{n_{k_j}} \to f$ in L^1 in probability, hence almost surely along a further subsequence. By uniqueness of L^1 limits:

$$g = f \quad \mathcal{L}^d\text{-a.e. on } \Omega \tag{112}$$

Therefore $Dg = Df$ as Radon measures, and from item (2) of Step 2:

$$D\hat{f}_{n_{k_j}} \xrightarrow{w*} Df \quad \text{in } \mathcal{M}(\Omega)^d \quad \text{almost surely} \tag{113}$$

In particular, for the test function ϕ from the contradiction assumption:

$$\int_\Omega \phi \cdot d(D\hat{f}_{n_{k_j}}) \to \int_\Omega \phi \cdot d(Df) \quad \text{almost surely} \tag{114}$$

Step 4: Derive the Contradiction

The almost sure convergence in Step 3 implies:

$$\mathbb{P}\left(\left|\int_\Omega \phi \cdot d(D\hat{f}_{n_{k_j}}) - \int_\Omega \phi \cdot d(Df)\right| > \epsilon\right) \to 0 \tag{115}$$

This contradicts the assumption that the probability exceeds $\delta > 0$ for all k. Therefore the assumption was false.

Conclusion of Part I

Since every subsequence $\{D\hat{f}_{n_k}\}$ has a further subsequence converging weak-* to Df almost surely, the full sequence satisfies:

$$D\hat{f}_n \xrightarrow{w*} Df \quad \text{in } \mathcal{M}(\Omega)^d \quad \text{in probability} \tag{116}$$

J.4 Part II: Strict Convergence Upgrade

We now prove the additional claim: if $\mathbb{E}[|D\hat{f}_n|(\Omega)] \to |Df|(\Omega)$ then $D\hat{f}_n \to Df$ strictly in $\mathcal{M}(\Omega)^d$, meaning weak-* convergence holds and $|D\hat{f}_n|(\Omega) \to |Df|(\Omega)$.

Step 1: Lower Semicontinuity Gives One Direction

The total variation functional $\mu \mapsto |\mu|(\Omega)$ is lower semicontinuous with respect to weak-* convergence of Radon measures (Evans and Gariepy, 2015, Theorem 5.2). Since $D\hat{f}_n \xrightarrow{w*} Df$ in $\mathcal{M}(\Omega)^d$ by Part I:

$$|Df|(\Omega) \leq \liminf_{n\to\infty} |D\hat{f}_n|(\Omega) \tag{117}$$

This gives the lower bound on the total variation of the limit.

Step 2: The Additional Condition Gives the Other Direction

By the additional assumption $\mathbb{E}[|D\hat{f}_n|(\Omega)] \to |Df|(\Omega)$, we have:

$$\limsup_{n\to\infty} |D\hat{f}_n|(\Omega) \leq |Df|(\Omega) \tag{118}$$

in expectation, hence in probability along subsequences.

Step 3: Pin the Total Variation

Step 1 gives $|Df|(\Omega) \leq \liminf_{n\to\infty} |D\hat{f}_n|(\Omega)$ by lower semicontinuity of total variation under weak-* convergence. Step 2 gives $\limsup_{n\to\infty} |D\hat{f}_n|(\Omega) \leq |Df|(\Omega)$ from the additional assumption $\mathbb{E}[|D\hat{f}_n|(\Omega)] \to |Df|(\Omega)$. Since $\liminf \leq \limsup$ always, combining gives:

$$|Df|(\Omega) \leq \liminf_{n\to\infty} |D\hat{f}_n|(\Omega) \leq \limsup_{n\to\infty} |D\hat{f}_n|(\Omega) \leq |Df|(\Omega) \tag{119}$$

All inequalities are therefore equalities and $|D\hat{f}_n|(\Omega) \to |Df|(\Omega)$ in probability. Combined with the weak-* convergence $D\hat{f}_n \xrightarrow{w*} Df$ established in Part I, this is precisely the definition of strict convergence in $\mathcal{M}(\Omega)^d$ (Ambrosio et al., 2000, Definition 3.13). □

Remark 9. *The parallel between Lemma 4 and Lemma 6 is exact at the structural level. In both cases: uniform boundedness in a strong space gives compactness via a compact embedding or compactness theorem; the consistency condition identifies the limit; and the subsequence*

argument promotes deterministic convergence to convergence in probability. The difference is entirely in the function spaces and the topology of convergence — strong L^p in the Sobolev case, weak- of measures in the BV case — reflecting the fundamental difference in the inductive biases of smooth and piecewise constant estimators.*

Remark 10. *The strict convergence upgrade requires the additional assumption* $\mathbb{E}[|D\hat{f}_n|(\Omega)] \to |Df|(\Omega)$*, which is not implied by weak-* convergence alone. This is because weak-* convergence of measures does not in general imply convergence of total masses — mass can escape to the boundary or concentrate in ways invisible to continuous test functions. The lower semicontinuity of total variation (Evans and Gariepy, 2015, Theorem 5.2) gives the liminf inequality for free; pinning the limsup requires the additional assumption. In applications where only weak-* convergence is needed — as in the bootstrap corollaries — the additional assumption is not required.*

K Proof of Theorem 3 (Pairing Identity)

Proof. Write points as $x = (x_{-j}, x_j)$ and for fixed x_{-j} let $\Omega^{x_{-j}} = \{\tau \in \mathbb{R} : (x_{-j}, \tau) \in \Omega\}$ denote the one-dimensional section of the domain and $g^{x_{-j}}(\tau) = g(x_{-j}, \tau)$ the section of g. By the one-dimensional section theory of BV functions (Ambrosio et al., 2000, Section 3.11), for $\mathcal{H}^{d-1}$-almost every x_{-j} the section $g^{x_{-j}}$ of the precise representative is a good representative of a function of bounded variation on $\Omega^{x_{-j}}$, its distributional derivative $Dg^{x_{-j}}$ is the corresponding section of $D_j g$, and the total variations disintegrate:

$$\int_\Omega \phi \, dD_j g = \int \left(\int_{\Omega^{x_{-j}}} \phi(x_{-j}, \tau) \, dDg^{x_{-j}}(\tau) \right) d\mathcal{H}^{d-1}(x_{-j}) \tag{120}$$

for every bounded Borel ϕ. For a good representative of a BV function of one variable, the fundamental theorem of calculus holds in the form $g^{x_{-j}}(b) - g^{x_{-j}}(a) = Dg^{x_{-j}}\big((a, b]\big)$ for all $a < b$ in $\Omega^{x_{-j}}$ that are not atoms of $Dg^{x_{-j}}$; since each section has at most countably many atoms, for Lebesgue-almost every x_j the pair $(x_j - h, x_j + h)$ avoids them. Hence for almost

every $x \in \mathrm{supp}(w)$ (where the whole segment lies in Ω by the trimming condition):

$$g(x + he_j) - g(x - he_j) = Dg^{x_{-j}}\big((x_j - h,\ x_j + h]\big) \tag{121}$$

Multiplying by $q(x)/2h$, integrating, and applying Tonelli on each section (justified since $|Dg^{x_{-j}}|$ is finite for a.e. section and q is bounded):

$$\begin{aligned}
\mathbb{E}\left[w(X)\,\Delta_{j,h}g(X)\right] &= \int d\mathcal{H}^{d-1}(x_{-j}) \int dx_j\ \frac{q(x_{-j}, x_j)}{2h} \int \mathbf{1}\{\tau \in (x_j - h, x_j + h]\}\ dDg^{x_{-j}}(\tau) \\
&= \int d\mathcal{H}^{d-1}(x_{-j}) \int dDg^{x_{-j}}(\tau)\ \frac{1}{2h} \int \mathbf{1}\{x_j \in [\tau - h, \tau + h)\}\, q(x_{-j}, x_j)\ dx_j \\
&= \int d\mathcal{H}^{d-1}(x_{-j}) \int (K_h \star q)(x_{-j}, \tau)\ dDg^{x_{-j}}(\tau) = \int_\Omega (K_h \star q)\ dD_j g
\end{aligned} \tag{122}$$

using $\tau \in (x_j - h, x_j + h] \Leftrightarrow x_j \in [\tau - h, \tau + h)$ in the second line, the definition of $K_h \star q$ (half-open versus closed windows agree up to Lebesgue-null sets) in the third, and (120) in the last step. The piecewise constant case follows by substituting the explicit jump formula of Lemma 5 for $D_j g$.

For the corollary: if q extends to a uniformly continuous function of compact support on $\mathbb{R}^d$, then $K_h \star q \to q$ uniformly as $h \downarrow 0$, so:

$$\left| \int (K_h \star q)\ dD_j g - \int q\ dD_j g \right| \le \|K_h \star q - q\|_\infty \cdot |D_j g|\,(\Omega) \to 0 \tag{123}$$

since $|D_j g|\,(\Omega) < \infty$. When $g \in W^{1,1}(\Omega)$, the measure $D_j g$ is absolutely continuous with density $\partial_j g$, so $\int q\, dD_j g = \int w\, \partial_j g\, p\, dx = \mathbb{E}[w\, \partial_j g(X)]$. □

L Proof of Theorem 4 (Neural Networks)

L.1 Setup and Notation

Let $\hat{f}_n : \Omega \to \mathbb{R}$ be a feedforward neural network with L layers, weight matrices $W_\ell^{(n)} \in \mathbb{R}^{d_\ell \times d_{\ell-1}}$, bias vectors $b_\ell^{(n)} \in \mathbb{R}^{d_\ell}$, and activation function $\sigma \in C^2(\mathbb{R})$. The network computes:

$$\hat{f}_n(x) = W_L^{(n)} z_{L-1} + b_L^{(n)} \tag{124}$$

where the intermediate representations are defined recursively by:

$$z_\ell = \sigma\Big(W_\ell^{(n)} z_{\ell-1} + b_\ell^{(n)}\Big), \quad z_0 = x \tag{125}$$

with σ applied elementwise. Denote the pre-activation at layer ℓ by $a_\ell = W_\ell^{(n)} z_{\ell-1} + b_\ell^{(n)}$ and the diagonal matrix of activation derivatives by $\Sigma'_\ell = \mathrm{diag}(\sigma'(a_\ell^{(1)}), \ldots, \sigma'(a_\ell^{(d_\ell)}))$ and similarly $\Sigma''_\ell = \mathrm{diag}(\sigma''(a_\ell^{(1)}), \ldots, \sigma''(a_\ell^{(d_\ell)}))$.

Define the product of weight matrices from layer k to layer ℓ as:

$$M_{\ell:k} = W_\ell^{(n)} \Sigma'_{\ell-1} W_{\ell-1}^{(n)} \Sigma'_{\ell-2} \cdots W_{k+1}^{(n)} \Sigma'_k \quad \text{for } \ell > k \tag{126}$$

with the convention $M_{\ell:\ell} = \Sigma'_\ell$. This notation captures the Jacobian of the composition of layers from k to ℓ.

The proof proceeds in two steps. Step 1 establishes uniform $W^{2,p}(\Omega)$ boundedness. Step 2 applies Lemma 4 to conclude.

L.2 Step 1: Uniform $W^{2,p}$ Boundedness

Claim: $\sup_n \left\| \hat{f}_n \right\|_{W^{2,p}(\Omega)} < \infty$.

Substep 1a: Bound on $\hat{f}_n$ in L^p

Since $\left\|\hat{f}_n - f\right\|_{L^p} \xrightarrow{p} 0$ and $f \in W^{1,p}(\Omega) \subset L^p(\Omega)$, the triangle inequality gives:

$$\left\|\hat{f}_n\right\|_{L^p} \leq \left\|\hat{f}_n - f\right\|_{L^p} + \|f\|_{L^p} \tag{127}$$

The first term converges to zero in probability and the second is finite since $f \in L^p(\Omega)$. Therefore $\left\|\hat{f}_n\right\|_{L^p}$ is eventually bounded in probability and this bound can be made uniform in n by passing to a subsequence or imposing a mild boundedness condition on the network output.

Substep 1b: First Derivative Bound

Differentiating $\hat{f}_n$ with respect to the input x via the chain rule through all L layers, the gradient at $x \in \Omega$ is:

$$\nabla \hat{f}_n(x) = \left(W_L^{(n)} M_{L-1:1}\right)^\top \tag{128}$$

where $M_{L-1:1} = W_{L-1}^{(n)} \Sigma'_{L-2} W_{L-2}^{(n)} \cdots \Sigma'_1 W_1^{(n)}$ is the product of weight-activation Jacobians from layer 1 to layer $L-1$. More explicitly, for the j-th input coordinate:

$$\frac{\partial \hat{f}_n}{\partial x_j}(x) = \left(W_L^{(n)}\right)_{:} \cdot \prod_{\ell=L-1}^{1} \left(W_\ell^{(n)} \Sigma'_{\ell-1}\right) \cdot e_j \tag{129}$$

where e_j is the j-th standard basis vector.

Taking the operator norm of this product and applying submultiplicativity:

$$\left|\frac{\partial \hat{f}_n}{\partial x_j}(x)\right| \leq \left\|W_L^{(n)}\right\| \cdot \prod_{\ell=1}^{L-1} \left\|W_\ell^{(n)}\right\| \cdot \|\Sigma'_\ell\| \leq \prod_{\ell=1}^{L} \left\|W_\ell^{(n)}\right\| \cdot \|\sigma'\|_\infty^{L-1} \tag{130}$$

using that $\|\Sigma'_\ell\| = \max_i |\sigma'(a_\ell^{(i)})| \leq \|\sigma'\|_\infty < \infty$ since $\sigma \in C^2$. This bound holds pointwise

for every $x \in \Omega$, so integrating over Ω:

$$\left\|\nabla \hat{f}_n\right\|_{L^p} \leq \left\|\nabla \hat{f}_n\right\|_{L^\infty} \cdot |\Omega|^{1/p} \leq \prod_{\ell} \left\|W_\ell^{(n)}\right\| \cdot \|\sigma'\|_\infty^{L-1} \cdot |\Omega|^{1/p} \tag{131}$$

Substep 1c: Second Derivative Bound

Differentiating the gradient formula once more with respect to x_i using the product rule and chain rule, the (i,j) entry of the Hessian $\nabla^2 \hat{f}_n(x)$ is:

$$\frac{\partial^2 \hat{f}_n}{\partial x_i \partial x_j}(x) = \sum_{\ell=1}^{L-1} \left(W_L^{(n)} M_{L-1:\ell+1}\right)_{:} \cdot \Sigma_\ell'' \cdot (M_{\ell:1} e_j) \cdot (M_{\ell:1} e_i)^\top \cdot \left(W_L^{(n)} M_{L-1:\ell+1}\right)_{:}^\top \tag{132}$$

where the sum runs over all layers ℓ at which the second derivative of σ is applied. The structure is: the forward pass from the input to layer ℓ contributes $M_{\ell:1} e_j$ for the j-th coordinate direction, the activation second derivative at layer ℓ contributes Σ_ℓ'', and the backward pass from layer ℓ to the output contributes $W_L^{(n)} M_{L-1:\ell+1}$.

Taking absolute values and applying submultiplicativity of operator norms:

$$\begin{aligned}
\left|\frac{\partial^2 \hat{f}_n}{\partial x_i \partial x_j}(x)\right| &\leq \sum_{\ell=1}^{L-1} \left\|W_L^{(n)} M_{L-1:\ell+1}\right\|^2 \cdot \|\Sigma_\ell''\| \cdot \|M_{\ell:1}\|^2 \\
&\leq \sum_{\ell=1}^{L-1} \left(\prod_{k=\ell+1}^{L} \left\|W_k^{(n)}\right\|\right)^2 \cdot \|\sigma''\|_\infty \cdot \left(\prod_{k=1}^{\ell} \left\|W_k^{(n)}\right\|\right)^2 \cdot \|\sigma'\|_\infty^{2(L-2)} \\
&\leq (L-1) \cdot \left(\prod_{\ell=1}^{L} \left\|W_\ell^{(n)}\right\|\right)^2 \cdot \|\sigma''\|_\infty \cdot \|\sigma'\|_\infty^{2(L-2)}
\end{aligned} \tag{133}$$

using $\|\Sigma_\ell''\| \leq \|\sigma''\|_\infty < \infty$ since $\sigma \in C^2$, and collecting all weight norm factors into a single product. This bound holds pointwise for every $x \in \Omega$, so:

$$\left\|\nabla^2 \hat{f}_n\right\|_{L^p} \leq \left\|\nabla^2 \hat{f}_n\right\|_{L^\infty} \cdot |\Omega|^{1/p} \leq (L-1) \cdot \left(\prod_{\ell} \left\|W_\ell^{(n)}\right\|\right)^2 \cdot \|\sigma''\|_\infty \cdot \|\sigma'\|_\infty^{2(L-2)} \cdot |\Omega|^{1/p} \tag{134}$$

Combining Substeps 1a–1c

Summing the three bounds:

$$\left\|\hat{f}_n\right\|_{W^{2,p}} = \left\|\hat{f}_n\right\|_{L^p} + \left\|\nabla \hat{f}_n\right\|_{L^p} + \left\|\nabla^2 \hat{f}_n\right\|_{L^p} \leq C\left(\prod_{\ell} \left\|W_\ell^{(n)}\right\|, \, \|\sigma'\|_\infty, \, \|\sigma''\|_\infty, \, L, \, |\Omega|\right) \tag{135}$$

where C is a finite constant depending on the indicated quantities but not on n or x. By the assumption $\sup_n \prod_\ell \left\|W_\ell^{(n)}\right\| < \infty$ — which formalizes the effect of L^2 regularization during training (Bartlett et al., 2017) — and the finiteness of $\|\sigma'\|_\infty$ and $\|\sigma''\|_\infty$ from $\sigma \in C^2$, the right-hand side is uniformly bounded in n. Therefore:

$$\sup_n \left\|\hat{f}_n\right\|_{W^{2,p}} < \infty \tag{136}$$

This completes Step 1.

L.3 Step 2: L^p Consistency

The L^p consistency condition $\left\|\hat{f}_n - f\right\|_{L^p} \xrightarrow{p} 0$ is the standard universal approximation result for smooth neural networks. Hornik et al. (1990) establish that multilayer feedforward networks with smooth activation functions are dense in $W^{1,p}(\Omega)$ with respect to the $W^{1,p}$ norm, implying in particular density in $L^p(\Omega)$. Combined with standard empirical risk minimization results establishing that the training procedure converges to the best approximation in the function class, this gives $\left\|\hat{f}_n - f\right\|_{L^p} \xrightarrow{p} 0$.

L.4 Step 3: Applying Lemma 4

By Step 1, $\sup_n \left\|\hat{f}_n\right\|_{W^{2,p}} < \infty$. By Step 2, $\left\|\hat{f}_n - f\right\|_{L^p} \xrightarrow{p} 0$. These are precisely the two conditions of the well-behaved definition in the Sobolev sense (Definition 1). Lemma 4 then gives:

$$\left\|\nabla \hat{f}_n - \nabla f\right\|_{L^p} \xrightarrow{p} 0 \tag{137}$$

which is the conclusion of Theorem 4. □

Remark 11. *The weight norm assumption* $\sup_n \prod_\ell \left\|W_\ell^{(n)}\right\| < \infty$ *is the key regularity condition on the estimator. It formalizes the effect of* L^2 *weight decay regularization during training, which penalizes large weights and keeps the product of operator norms bounded. Under standard conditions on the regularization parameter and optimization procedure, this bound holds with high probability over the training randomness (Bartlett et al., 2017). The condition is stated as an assumption of the theorem rather than derived from first principles because its verification depends on details of the training procedure that are outside the scope of the present analysis.*

Remark 12. *Theorem 4 is stated for smooth activations* $\sigma \in C^2$*. For ReLU networks* $\sigma \notin C^2$ *and the Hessian calculation in Substep 1c does not apply directly. The first derivative exists almost everywhere with* $\sigma' = \mathbf{1}_{(0,\infty)}$ *bounded, so Substep 1b goes through unchanged. The distributional second derivative is* $\sigma'' = \delta_0$*, a Dirac mass at zero, placing* $\hat{f}_n$ *in a mixed* $W^{1,p} \cap BV$ *space rather than* $W^{2,p}(\Omega)$*. A modified version of Lemma 4 applicable in this mixed space, or a restriction to input distributions placing zero mass on the kink set* $\{x : a_\ell(x) = 0\}$*, would be required to handle this case. We leave this extension for future work.*

M Proof of Theorem 5 (Gradient Boosting)

M.1 Setup and Notation

By Proposition 1 of Bühlmann and Yu (2003), the L_2Boost estimator after m iterations can be represented as:

$$\hat{F}_m = (I - (I - S)^{m+1})Y \tag{138}$$

where $S : \mathbb{R}^n \to \mathbb{R}^n$ is the base learner operator. The residual at step m is:

$$u_m = Y - \hat{F}_m = (I - S)^m Y \tag{139}$$

so that each base learner fit is $\hat{f}_m = Su_m$ and the ensemble updates as $\hat{F}_m = \hat{F}_{m-1} + \hat{f}_m$.

By Proposition 2 of Bühlmann and Yu (2003), for a symmetric linear learner S with eigenvalues $\{\lambda_k; k = 1, \ldots, n\}$, the eigenvalues of the boosting operator $\mathcal{B}_m = I - (I - S)^{m+1}$ are $\{1 - (1 - \lambda_k)^{m+1}; k = 1, \ldots, n\}$.

Under the eigenvalue condition $0 < \lambda_k \leq 1$ for all k (Bühlmann and Yu, 2003, Theorem 1), let:

$$\rho = \max_k (1 - \lambda_k) \in [0, 1) \tag{140}$$

Since $\lambda_k > 0$ for all k, we have $\rho < 1$. The bias decays exponentially fast: by Theorem 1(a) of Bühlmann and Yu (2003):

$$\|u_m\|_{L^2}^2 = \|(I - S)^m Y\|_{L^2}^2 \leq \rho^{2m} \|Y\|_{L^2}^2 \tag{141}$$

In particular:

$$\|u_m\|_{L^2} \leq C\rho^m, \quad C = \|Y\|_{L^2} \tag{142}$$

This geometric decay is the key engine for both cases of the proof.

The two cases of the theorem are handled separately.

M.2 Case (i): Smooth Base Learners

Step 1: $W^{2,p}$ Bound for a Single Base Learner

Each $\hat{f}_m = Su_m \in W^{2,p}(\Omega)$ by assumption on the base learner class. For shallow neural net base learners this follows from Appendix L. For spline base learners it follows from standard spline approximation theory (Wahba, 1990). We require a uniform bound:

$$\sup_m \left\| \hat{f}_m \right\|_{W^{2,p}} \leq C' \|u_m\|_{L^2} < \infty \tag{143}$$

where the bound on $\left\|\hat{f}_m\right\|_{W^{2,p}}$ in terms of $\|u_m\|_{L^2}$ follows from the boundedness of S as an operator from L^2 to $W^{2,p}$.

Step 2: Ensemble Bound via Geometric Decay

By linearity of the $W^{2,p}$ norm and the triangle inequality:

$$\left\|\hat{F}_m\right\|_{W^{2,p}} \leq \sum_{j=1}^{m} \left\|\hat{f}_j\right\|_{W^{2,p}} \leq C' \sum_{j=1}^{m} \|u_j\|_{L^2} \leq C'C \sum_{j=1}^{m} \rho^j \leq \frac{C'C\rho}{1-\rho} < \infty \tag{144}$$

using the geometric decay (142) and the fact that $\sum_{j=1}^{\infty} \rho^j = \rho/(1-\rho) < \infty$ for $\rho < 1$. The bound is uniform in m.

Step 3: Apply Lemma 4

L^p consistency is assumed. Uniform $W^{2,p}(\Omega)$ boundedness follows from Step 2. Lemma 4 gives $\left\|\nabla \hat{F}_m - \nabla f\right\|_{L^p} \xrightarrow{p} 0$. □

M.3 Case (ii): Tree Stump Base Learners

Each base learner $\hat{f}_m = Su_m$ is a depth-1 tree (stump) of the form:

$$\hat{f}_m(x) = c_m^L \, \mathbf{1}[x_j \leq t_m] + c_m^R \, \mathbf{1}[x_j > t_m] \tag{145}$$

for some split variable j, threshold t_m, and leaf constants:

$$c_m^L = \frac{1}{|I_L|} \sum_{i \in I_L} u_m(x_i), \quad c_m^R = \frac{1}{|I_R|} \sum_{i \in I_R} u_m(x_i) \tag{146}$$

where $I_L = \{i : x_{ij} \leq t_m\}$ and $I_R = \{i : x_{ij} > t_m\}$ are the left and right leaf index sets.

Step 1: BV Norm of a Single Stump

By Lemma 5 the distributional gradient of $\hat{f}_m$ is:

$$D\hat{f}_m = (c_m^R - c_m^L) \cdot \nu_m \cdot \mathcal{H}^{d-1}\lfloor_{\{x_j = t_m\} \cap \Omega} \tag{147}$$

so the total variation is:

$$|D\hat{f}_m|(\Omega) = |c_m^R - c_m^L| \cdot \mathcal{H}^{d-1}(\{x_j = t_m\} \cap \Omega) \tag{148}$$

The geometric term satisfies $\mathcal{H}^{d-1}(\{x_j = t_m\} \cap \Omega) \leq \operatorname{diam}(\Omega)^{d-1}$, a constant depending only on Ω. For the response term, by Cauchy–Schwarz applied to the leaf means:

$$|c_m^R - c_m^L| \leq \frac{1}{|I_R|} \sum_{i \in I_R} |u_m(x_i)| + \frac{1}{|I_L|} \sum_{i \in I_L} |u_m(x_i)| \leq 2 \|u_m\|_{L^2} \tag{149}$$

Therefore:

$$|D\hat{f}_m|(\Omega) \leq 2 \operatorname{diam}(\Omega)^{d-1} \cdot \|u_m\|_{L^2} \leq 2 \operatorname{diam}(\Omega)^{d-1} \cdot C\rho^m \tag{150}$$

using the geometric decay (142).

Step 2: Ensemble BV Bound via Geometric Series

By subadditivity of total variation:

$$|D\hat{F}_m|(\Omega) \leq \sum_{j=1}^{m} |D\hat{f}_j|(\Omega) \leq 2 \operatorname{diam}(\Omega)^{d-1} \cdot C \sum_{j=1}^{m} \rho^j \leq \frac{2 \operatorname{diam}(\Omega)^{d-1} \cdot C\rho}{1-\rho} < \infty \tag{151}$$

The geometric series $\sum_{j=1}^{\infty} \rho^j = \rho/(1-\rho)$ converges because $\rho < 1$, which follows from the eigenvalue condition $\lambda_k > 0$ for all k. Taking expectations:

$$\mathbb{E}[|D\hat{F}_m|(\Omega)] \leq \frac{2 \operatorname{diam}(\Omega)^{d-1} \cdot \mathbb{E}[C]\rho}{1-\rho} < \infty \tag{152}$$

uniformly in m, where $\mathbb{E}[C] = \mathbb{E}[\|Y\|_{L^2}] < \infty$ by assumption.

Step 3: Apply Lemma 6

L^1 consistency is assumed. Uniform BV boundedness follows from Step 2. Lemma 6 gives $D\hat{F}_m \xrightarrow{w*} Df$ in $\mathcal{M}(\Omega)^d$. □

Remark 13. *The key step in both cases is the geometric decay $\|u_m\|_{L^2} \leq C\rho^m$, which follows from the eigenvalue condition $0 < \lambda_k \leq 1$ via Proposition 1 and Theorem 1 of Bühlmann and Yu (2003). This decay is much faster than any polynomial rate and implies that the geometric series $\sum_j \rho^j$ converges without any condition on step sizes. The eigenvalue condition is the operative regularity assumption for both cases of the theorem, replacing the step size conditions common in the boosting literature.*

Remark 14. *When $\lambda_k \in \{0, 1\}$ for all k — as occurs for linear projection operators — Corollary 1 of Bühlmann and Yu (2003) shows that $\mathcal{B}_m \equiv S$ for all m, meaning boosting does not improve on the base learner. In this degenerate case $\rho = \max_k(1-\lambda_k)$ takes the value 1 for any k with $\lambda_k = 0$, and the geometric series does not converge. The strict positivity condition $\lambda_k > 0$ excludes this case and corresponds to the requirement that the base learner has positive but incomplete explanatory power — the standard "weak learner" assumption in the boosting literature.*

N Proof of Theorem 6 (Mondrian Forests)

N.1 Setup

Let $\hat{f}_n = \frac{1}{B}\sum_{b=1}^{B} T_b$ where T_b are individual Mondrian trees. Each T_b is piecewise constant on a random partition $\mathcal{P}_b$ of $\Omega = [0, 1]^d$ generated by the Mondrian process $MP(\lambda_n, [0, 1]^d)$ independently of the data, with budget $\lambda_n \asymp n^{1/(d+2)}$ (Mourtada et al., 2017). The partition $\mathcal{P}_b$ consists of $K_{\lambda_n} + 1$ hyperrectangular cells, where K_{λ_n} is the number of splits (Roy and Teh, 2009).

N.2 Step 1: Each Tree is in BV

Each T_b is piecewise constant on a finite partition and therefore in $BV(\Omega)$ by Lemma 5. The forest average $\hat{f}_n = \frac{1}{B}\sum_b T_b$ is a finite convex combination of BV functions, hence $\hat{f}_n \in BV(\Omega)$ with:

$$|D\hat{f}_n|(\Omega) \leq \frac{1}{B}\sum_{b=1}^{B} |DT_b|(\Omega) \tag{153}$$

by convexity of the total variation functional (Ambrosio et al., 2000, Proposition 3.6).

N.3 Step 2: Uniform BV Boundedness via the Second Moment Argument

Claim: $\sup_n \mathbb{E}[|D\hat{f}_n|(\Omega)] < \infty$.

The triangle inequality applied to the forest average is too crude because individual tree BV norms grow with n. Instead we work with the second moment and apply Cauchy–Schwarz:

$$\mathbb{E}\Big[|D\hat{f}_n|(\Omega)\Big] \leq \sqrt{\mathbb{E}\Big[|D\hat{f}_n|(\Omega)^2\Big]} \tag{154}$$

It therefore suffices to show $\mathbb{E}[|D\hat{f}_n|(\Omega)^2] < \infty$ uniformly in n.

Substep 2a: Expand the Second Moment

Writing $D\hat{f}_n = \frac{1}{B}\sum_b DT_b$ and expanding:

$$\mathbb{E}\Big[|D\hat{f}_n|(\Omega)^2\Big] = \frac{1}{B^2}\mathbb{E}\left[\left|\sum_b DT_b\right|(\Omega)^2\right] = \frac{1}{B^2}\sum_{b,b'}\mathbb{E}\left[\int \phi \cdot d(DT_b)\cdot\int \phi \cdot d(DT_{b'})\right] \tag{155}$$

for a suitable test function ϕ with $\|\phi\|_\infty \leq 1$. The double sum splits into diagonal terms $b = b'$ and cross terms $b \neq b'$.

Substep 2b: Cross Terms Vanish

For $b \neq b'$ the partitions $\mathcal{P}_b$ and $\mathcal{P}_{b'}$ are independent of each other and of the data. By independence:

$$\mathbb{E}\left[\int \phi \cdot d(DT_b) \cdot \int \phi \cdot d(DT_{b'})\right] = \mathbb{E}\left[\int \phi \cdot d(DT_b)\right] \cdot \mathbb{E}\left[\int \phi \cdot d(DT_{b'})\right] \tag{156}$$

We claim each factor is zero. By Lemma 5 and the explicit formula for DT_b:

$$\mathbb{E}\left[\int \phi \cdot d(DT_b)\right] = \mathbb{E}\left[\sum_{j\sim k} (\bar{y}_k^b - \bar{y}_j^b) \int_{\partial R_j^b \cap \partial R_k^b} \phi \cdot \nu_{jk}\, d\mathcal{H}^{d-1}\right] \tag{157}$$

Since $\mathcal{P}_b$ is independent of the data, we condition on the partition and take the expectation over the data first:

$$= \mathbb{E}_{\mathcal{P}_b}\left[\sum_{j\sim k} \mathbb{E}_{\text{data}}\left[\bar{y}_k^b - \bar{y}_j^b\right] \int_{\partial R_j^b \cap \partial R_k^b} \phi \cdot \nu_{jk}\, d\mathcal{H}^{d-1}\right] \tag{158}$$

The leaf means satisfy $\mathbb{E}_{\text{data}}[\bar{y}_j^b] \to f(\bar{x}_j^b)$ as $n \to \infty$, where $\bar{x}_j^b$ is the center of leaf R_j^b. For adjacent leaves j and k, by the Lipschitz continuity of f:

$$\mathbb{E}_{\text{data}}\left[\bar{y}_k^b - \bar{y}_j^b\right] \approx f(\bar{x}_k^b) - f(\bar{x}_j^b) = O(D_{\lambda_n}) \tag{159}$$

where D_{λ_n} is the cell diameter. As $\lambda_n \to \infty$ the cell diameter $D_{\lambda_n} \to 0$ and each term vanishes. More precisely, since $f \in W^{1,p}(\Omega)$ with $p > d$, the Sobolev embedding $W^{1,p} \hookrightarrow C^{0,\alpha}$ for $\alpha = 1 - d/p$ gives:

$$\left|\mathbb{E}_{\text{data}}\left[\bar{y}_k^b - \bar{y}_j^b\right]\right| \leq C \cdot \text{diam}(R_j^b \cup R_k^b)^\alpha \leq C \cdot D_{\lambda_n}^\alpha \to 0 \tag{160}$$

Therefore $\mathbb{E}[\int \phi \cdot d(DT_b)] \to 0$ and the cross terms vanish asymptotically. For n large enough the cross terms are negligible and the second moment is dominated by the diagonal terms.

Substep 2c: Bound the Diagonal Terms

For the diagonal terms $b = b'$:

$$\mathbb{E}\big[|DT_b|(\Omega)^2\big] = \mathbb{E}\left[\left(\sum_{j\sim k} |\bar{y}_k^b - \bar{y}_j^b| \cdot \mathcal{H}^{d-1}(\partial R_j^b \cap \partial R_k^b)\right)^2\right] \tag{161}$$

Expanding the square and separating geometric and response terms using the independence of $\mathcal{P}_b$ from the data:

$$\mathbb{E}\big[|DT_b|(\Omega)^2\big] \leq \mathbb{E}_{\mathcal{P}_b}\big[K_{\lambda_n}^2\big] \cdot \sup_{j\sim k} \mathbb{E}_{\text{data}}\big[|\bar{y}_k^b - \bar{y}_j^b|^2\big] \tag{162}$$

where K_{λ_n} is the number of adjacent pairs, which equals the number of splits.

Geometric factor. The Mondrian process generates cuts according to a recursive tree structure. By the conditional independence of subtrees (Roy and Teh, 2009), the number of splits satisfies the distributional recursion:

$$K_{\lambda_n} = 1 + K_{\lambda_n,L} + K_{\lambda_n,R} \tag{163}$$

where $K_{\lambda_n,L}$ and $K_{\lambda_n,R}$ are the split counts in the left and right subtrees, conditionally independent given the first cut. From Mourtada et al. (2017):

$$\mathbb{E}[K_{\lambda_n}] \leq (e(\lambda_n + 1))^d = O(n^{d/(d+2)}) \tag{164}$$

Taking the second moment via the recursion and using conditional independence:

$$\mathbb{E}[K_{\lambda_n}^2] = \mathbb{E}[K_{\lambda_n}] + \mathbb{E}[K_{\lambda_n}]^2 = O(n^{d/(d+2)}) + O(n^{2d/(d+2)}) = O(n^{2d/(d+2)}) \tag{165}$$

where the first equality uses the Poisson-like variance structure of the Mondrian process implied by the conditional independence recursion.

Response factor. For adjacent leaves j and k, by Cauchy–Schwarz on the sample means:

$$\mathbb{E}_{\text{data}}\big[|\bar{y}_k^b - \bar{y}_j^b|^2\big] \le 4\,\mathbb{E}_{\text{data}}\big[\|r^{(b)}\|_{L^2}^2\big] \tag{166}$$

where $r^{(b)}$ denotes the residual within the relevant cells. Using the cell diameter bound of Roy and Teh (2009):

$$\mathbb{E}[D_{\lambda_n}(x)^2] \le \frac{4d}{\lambda_n^2} = O(n^{-2/(d+2)}) \tag{167}$$

and the Hölder continuity of $f \in W^{1,p}$ with $p > d$ giving $\alpha = 1 - d/p$:

$$\mathbb{E}_{\text{data}}\big[|\bar{y}_k^b - \bar{y}_j^b|^2\big] \le C^2\left(\mathbb{E}[D_{\lambda_n}(x)^2]\right)^{\alpha} \le C^2\left(\frac{4d}{\lambda_n^2}\right)^{\alpha} = O(n^{-2\alpha/(d+2)}) \tag{168}$$

Combining.

$$\mathbb{E}\big[|DT_b|(\Omega)^2\big] \le O(n^{2d/(d+2)}) \cdot O(n^{-2\alpha/(d+2)}) = O(n^{2(d-\alpha)/(d+2)}) \tag{169}$$

Therefore:

$$\mathbb{E}\Big[|D\hat{f}_n|(\Omega)^2\Big] = \frac{1}{B^2}\sum_b \mathbb{E}\big[|DT_b|(\Omega)^2\big] = \frac{1}{B}O(n^{2(d-\alpha)/(d+2)}) \tag{170}$$

For this to be bounded uniformly in n we require:

$$B \gg n^{2(d-\alpha)/(d+2)} \tag{171}$$

This is a condition on the number of trees relative to the dimension and smoothness of f. Under this condition:

$$\mathbb{E}\Big[|D\hat{f}_n|(\Omega)\Big] \le \sqrt{\mathbb{E}\Big[|D\hat{f}_n|(\Omega)^2\Big]} = \sqrt{\frac{1}{B}O(n^{2(d-\alpha)/(d+2)})} < \infty \tag{172}$$

uniformly in n, completing the uniform BV boundedness claim.

Remark 15. *The condition $B \gg n^{2(d-\alpha)/(d+2)}$ requires the number of trees to grow with n*

at a rate depending on dimension d and Hölder exponent $\alpha = 1 - d/p$. For α close to d — achieved when $p \gg d$, meaning f is highly smooth — the condition is mild and a slowly growing number of trees suffices. For α small — meaning f has limited smoothness or d is large — more trees are required. In practice random forests with $B = 500$ or $B = 1000$ trees satisfy this condition for moderate d and n.

N.4 Step 3: Apply Lemma 6

The L^1 consistency condition follows from the L^2 consistency result of Mourtada et al. (2017) under the budget $\lambda_n \asymp n^{1/(d+2)}$ by the Cauchy–Schwarz inequality on the bounded domain Ω:

$$\left\|\hat{f}_n - f\right\|_{L^1} \leq |\Omega|^{1/2} \left\|\hat{f}_n - f\right\|_{L^2} \xrightarrow{p} 0 \tag{173}$$

Uniform BV boundedness follows from Step 2 under the condition $B \gg n^{2(d-\alpha)/(d+2)}$. Lemma 6 then gives $D\hat{f}_n \xrightarrow{w*} Df$ in $\mathcal{M}(\Omega)^d$. □

Remark 16. *Theorem 6 is stated for Mondrian forests where the partition is generated independently of the data. The independence is used critically in the decomposition of Substep 2a and in the cross term cancellation of Substep 2b. For Breiman forests with data-dependent splits the partition geometry and response variation are coupled — splits are chosen to maximize response differences across boundaries, which is precisely the worst case for the BV bound, and the cross term cancellation argument fails because the jump directions are not independent across trees. Extending the result to Breiman forests requires additional control over this coupling and is left as future work.*

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